MONASH University

Buildings and Property Division

BUILDING AUTOMATION SYSTEM (BAS) STANDARD

NIAGARA 4 GRAPHICS STANDARD AND GUIDE

Version 1.4

November 2024

CONTENTS

DOCUMENT CONTROL

Owner: BAS Administrator

Location: \BAS Team Drive\Forms & Sheets\BAS Standards & Checklist\Graphic Standards

Headings and content that are highlighted in yellow indicate that content has been modified from the previous version.

Version	Date	Action
V0.21	06/08/2021	Drafted document.
V0.22	30/08/2021	Drafted document.
V0.23	10/09/2021	Drafted document.
V1.0	22/04/2022	Document format changed to Monash template including heading numbering. Updated to include the clarifications sent throughout the MSI1 project. Added 'Note' icon background requirement.
V1.1	12/01/2023	Minor amendment to important information section
V1.2	15/03/2023	Amendments to facets, alarms and multi-trends sections
V1.3	24/11/2023	Added point folder structure standard
V1.4	26/11/2024	Amendments to Controlled Environments graphics and features. Removal of reference to SMS, replacement with AIMS features. Update to offline device configuration and network health pages

PURPOSE

This document encompasses Monash University's graphic specifications for the user-interface on the Building Automation System and contains information required for the configuration and implementation on Tridium Niagara N4. The standard is intended to be used as a reference for the minimum requirements in undertaking graphics work unless otherwise specified. In addition, the document offers procedures in order to produce a consistent Tridium Niagara N4 Building Automation System that meet the requirements.

IMPORTANT INFORMATION

- Take graphic and controller backups before and after any graphic or control work. Before and after backups are to be submitted to the BAS Administrator at the completion of the project.
- Only JACE offline AIMS alarms are to be sent from the Supervisor. All other AIMS messages are to be sent from the relevant JACE (not field controller or Supervisor).
- New projects must not configure new SMS alarms, as the SMSBroadcast system is deprecated. Where SMS alarms were in use, these alarms must route to AIMS.
- JACE must have FOXS connection only – Disable FOX.
- Standard passphrase, platform username and platform passwords to be used as per Monash University BAS Administrator.
- A 'Building Alarm List' must be submitted to the Monash University BAS Administrator, in the approved format, with suggested priorities for approval and modification before alarms are implemented.
- The Monash Supervisor Port Forward module is to be updated prior to handover. This is to include all JACE secondary IP devices and all ports required to fully interface with it. For example, each EasyIO FW controller would have port TCP 80 (http), TCP 443 (https) and UPD 1876 (sox) in the Port Forward table. It is important that each secondary IP device has the JACE secondary port as its gateway address.
- Users for any JACE must be generated via network sync with the Monash N4 Supervisor
- JACE time synchronization must be configured to Network Time Protocol Server provided by Monash BAS Administrator
- JACE backups to be automatically configured via the Supervisor's Provisioning Network Extension, with no more than 10 JACEs to be scheduled at any one time. Liaise with BAS Administrator for further details if required.

GRAPHICS STANDARD

The following section outlines Monash University's graphics requirements. Implementation and instructions are detailed below, in order to produce consistent graphics.

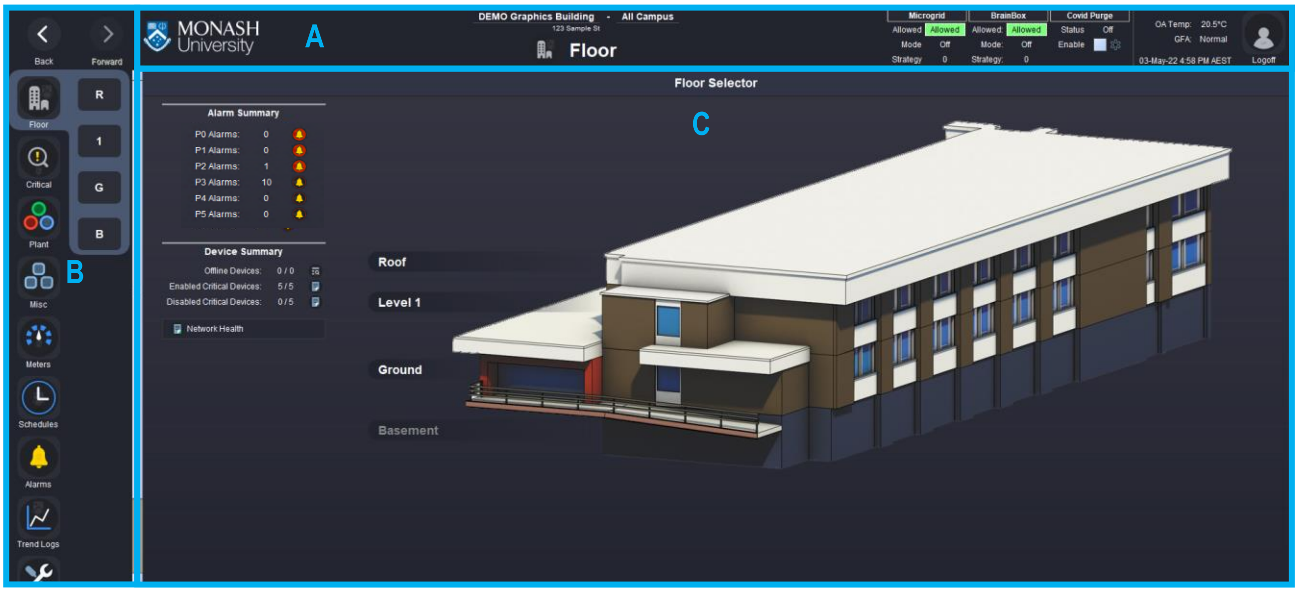
All colours used must be identical to the sample provided.

Monash Supervisor, sample graphics information:

Item	Location
Sample JACE	station:|slot:/Drivers/NiagaraNetwork/DEMO_GraphicsAndTaggingStandards
Graphics Folder	station:|slot:/Drivers/Graphics/DEMO_GraphicsSamples
Graphics File	file:^DEMO_GraphicsSamples
Icon Library / Templates	file:^muGraphicsStandards

1 PAGE LAYOUT

The following image illustrates the basic layout for the BMS user-interface at Monash University. Standard graphic pages are comprised of three key components: Navigation Menu Pane, Header Pane and Main Content Pane. All pages must adhere to this layout.



Building Homepage –Header Pane (A), Navigation Menu Pane (B), Main Content Pane (C)

MONASH University

2 HEADER PANE

The header specifies the building and page that is currently being viewed. The header can be constructed using a template found in the *muGraphicLibrary* palette.

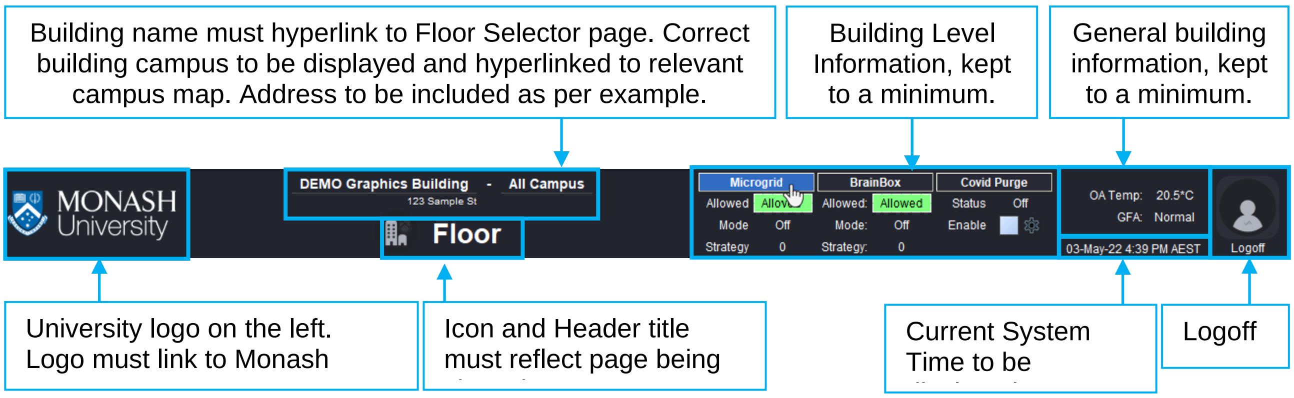
It must display the building name, campus location, building address and page title information, time as well as useful information where available. This information can include: supervisory control information, weather station data, general fire alarm status, outside air temp and the current time.

Additional information can be displayed above the system time, such as current weather, but must not alter the overall header height where possible. Information in the header must be kept to a minimum. Building level information such as Covid Purge, Microgrid and Brainbox may be added to the header per building requirement. Where there is no campus available, delete the dash and campus BoundLabels from the header Px Include for that building. The header must look identical to the example image below, with relevant building specific information updated as needed.

Contact BAS Administrator where additional information cannot be displayed in the header effectively.

Label Guide:

Building Name	Campus Location Name	Building Address
Building 1	Clayton Campus	42 Scenic Boulevard
Building 69	Clayton Campus	22 Alliance Lane
Building A	Caulfield Campus	(Street name of entrance optional)
30 Collins St	(Delete label – not applicable)	30 Collins St, VIC 3000

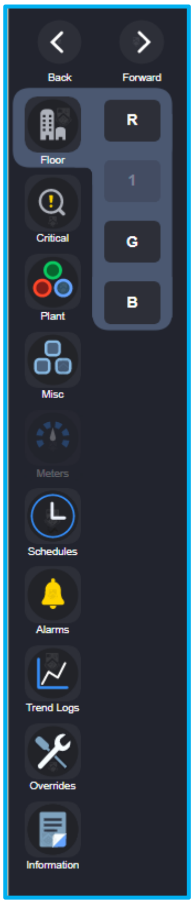


(Example of header)

3 NAVIGATION MENU PANE

The navigation menu is to provide a logical method of navigation that offers users fast button recognition through the use of icons and shortened page names. The menu uses standard navigation buttons in the location shown. Menus are unique to the building and are positioned on the left-hand side of all pages, contained within a scroll pane, then a constrained pane (See *Graphical Application Guide*, *Navigation Menu)*. The main and submenu can be constructed as per the relevant *Graphical Application Guide* sections using templates found in the *muGraphicLibrary* palette.

3.1 Main Menu



The navigation menu is constructed in two parts: main menu (left column) and submenu (right column, with a highlighted background that encompasses the main and submenu buttons). Main menu navigation should consist of the following buttons, icons and names in this order and location:

- Back/Forward
- Floor
- Critical
- Plant
- Misc
- Meters (if applicable)
- Schedule
- Alarm
- Overrides
- Information

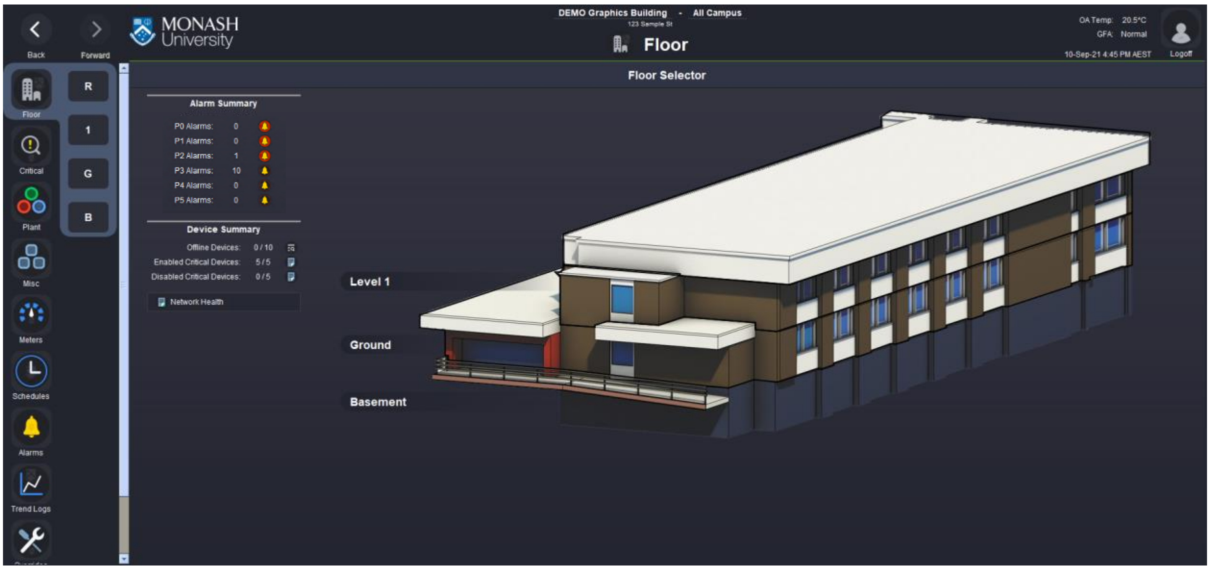
Additional main menu buttons can be added in cases where pages are unsuitable within the categories outlined above, but if a page can be categorised, it must reside in the relevant location. Any additional buttons must be organised in a logical order, with the existing menu and its information kept in mind.

A building's submenu navigation must be organised logically within the main menu framework outlined above. Each main menu button must have a unique submenu. This submenu must appear to the right-hand side of the main menu. When a main menu button is selected, a highlight shape must encompass the relevant submenu buttons and the associated main menu button. The highlight allows the user to identify what section of the graphics they are viewing. The first submenu button should always be located at the upper most slot, while ensuring there is no disconnect in the highlight shape between the main and submenu buttons. In cases where corners would be formed where the main and submenu highlight connect, curved inner corner shapes must be used to avoid the appearance of right-angles. Where no submenu buttons are required, the right-hand column will be empty and only the main menu button will be highlighted with a shape.

(Example of ghosted meter and floor button)

3.2 Floor Submenu

The Floor submenu is a navigational tool to access floorplan pages. The menu order must be in descending order, where roof is the first button and basement the last. The order mimics the 3D floor selector view where roof is the top floor and basement is the bottom. The menu style must be identical to the example provided, using text style buttons, minimalist button names and a highlighted menu background. All levels in the building must be shown in the Floor's submenu, regardless of the scope of works. Where no information is required or provided for particular levels, the button may be disabled and ghosted, appearing transparent, as needed (See *Graphics Application Guide, Navigation Menu*). The floorplan submenu button must be visible on all floor selector and floorplan pages, as well as area selector pages where implemented.



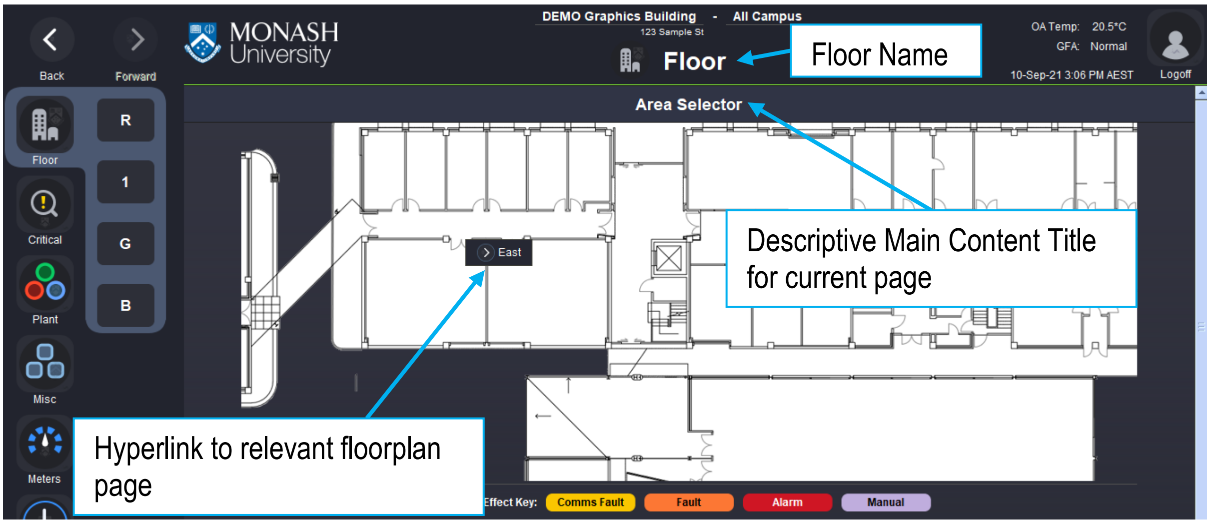
(Building Home page. Example indicating the Floor Selector content only)

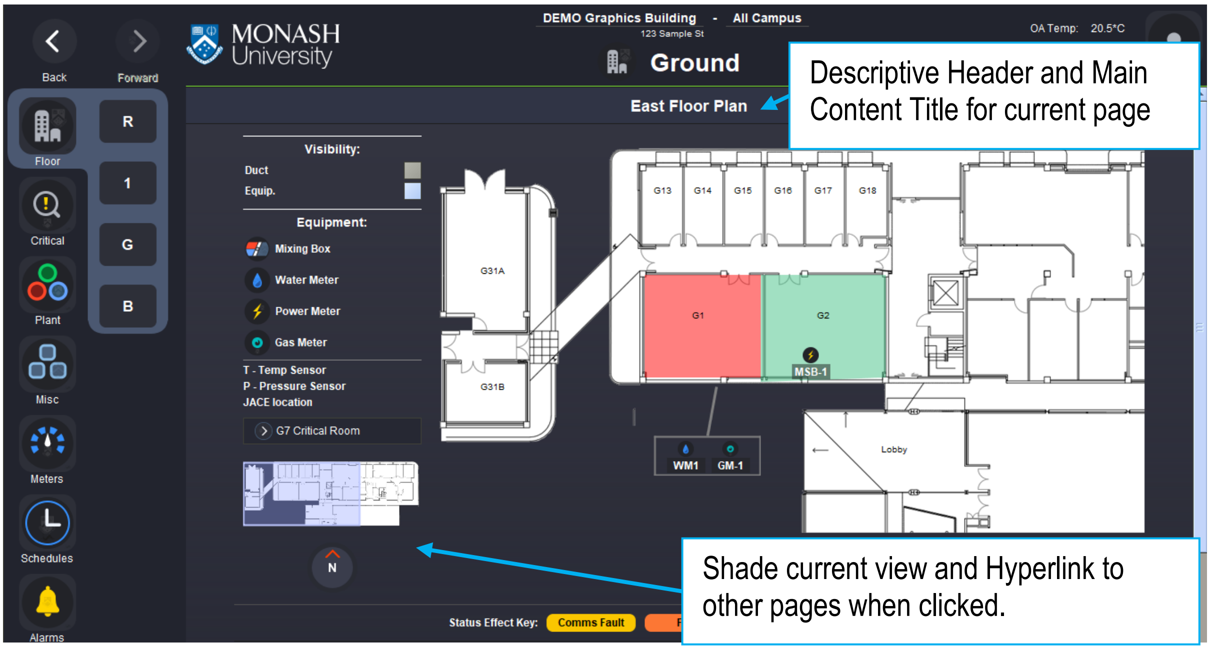
(Floor submenu)

Label Guide:

Floor	Nav Menu (Short Name)	Header Title	Content Title
Roof	R	Roof	Floor Plan
Level 3	3	Level 3	Floor Plan
Level 2	2	Level 2	Floor Plan
Level 1	1	Level 1	Floor Plan
Mezzanine	M	Mezzanine	Floor Plan
Ground	G	Ground	Floor Plan
Basement	B	Basement	Floor Plan
Basement 1	B1	Basement 1	Floor Plan
Basement 2	B2	Basement 2	Floor Plan
Lower Basement	LB	Lower Basement	Floor Plan

For large floorplans where multiple pages are required to view the level, the Floor's submenu button will direct you to a Zone or Area Selector page. It may also direct you to the first logical page for that level. For floorplans that are split across multiple pages, a minimap image should displayed. It is to be a scaled down version of the Zone or Area Selector image. The current area being viewed should be shaded in minimap. The minimap hyperlink allows the user to navigate to other floorplan pages on the same floor. Zone navigation is not handled by the submenu. For more information on the minimap and floorplan page layout, see *Main Content – Floorplan* section.





In special circumstances where a critical area may require a quick link button to a dedicated page, a button may be added to the Floor's submenu but should be avoided. A direct button can link to the critical page via the floorplan page, in order to keep the submenu orderly.

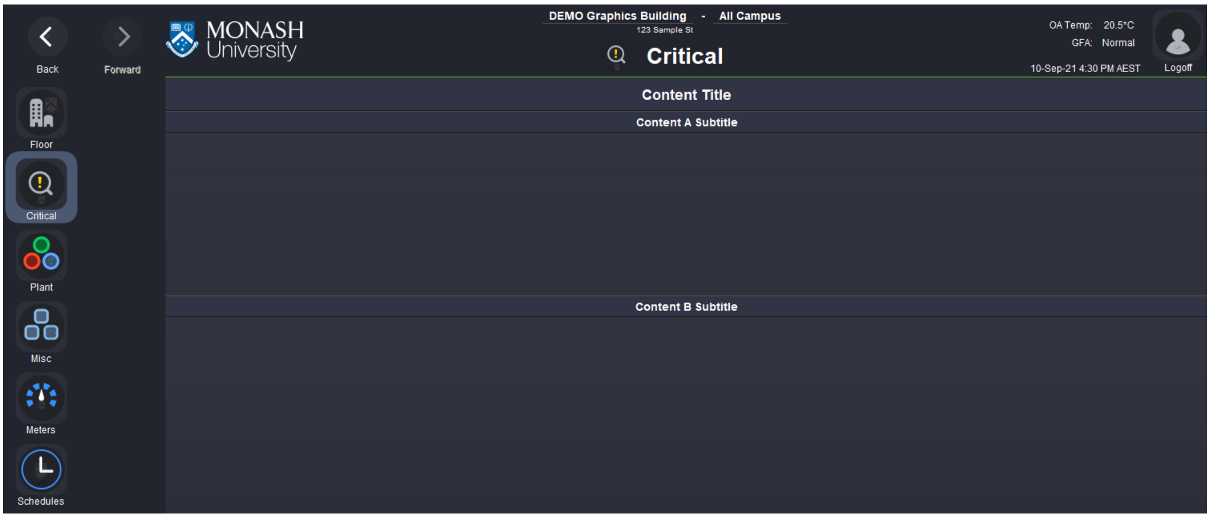
(Example link on a floorplan page, typically used for dedicated pages for critical rooms/systems)

3.3 Critical Submenu

The Critical submenu is a navigational tool to access a building specific overview page that displays critical systems. The intent is to provide a quick, centralised overview page that users can navigate to in order to monitor and control important areas and equipment. On selection, the Critical button is to be highlighted, with no submenu icons required. All information is to be located beneath the header, in the Content component of the page. It must be organised logically, separating by Main Content Titles or Content Subtitles in a cascading manner. The content must facilitate vertical scrolling and avoid horizontal scrolling where possible.

Label Guide:

Nav Menu (Short Name)	Header Title	Content Title
Critical	Critical	Critical Overview

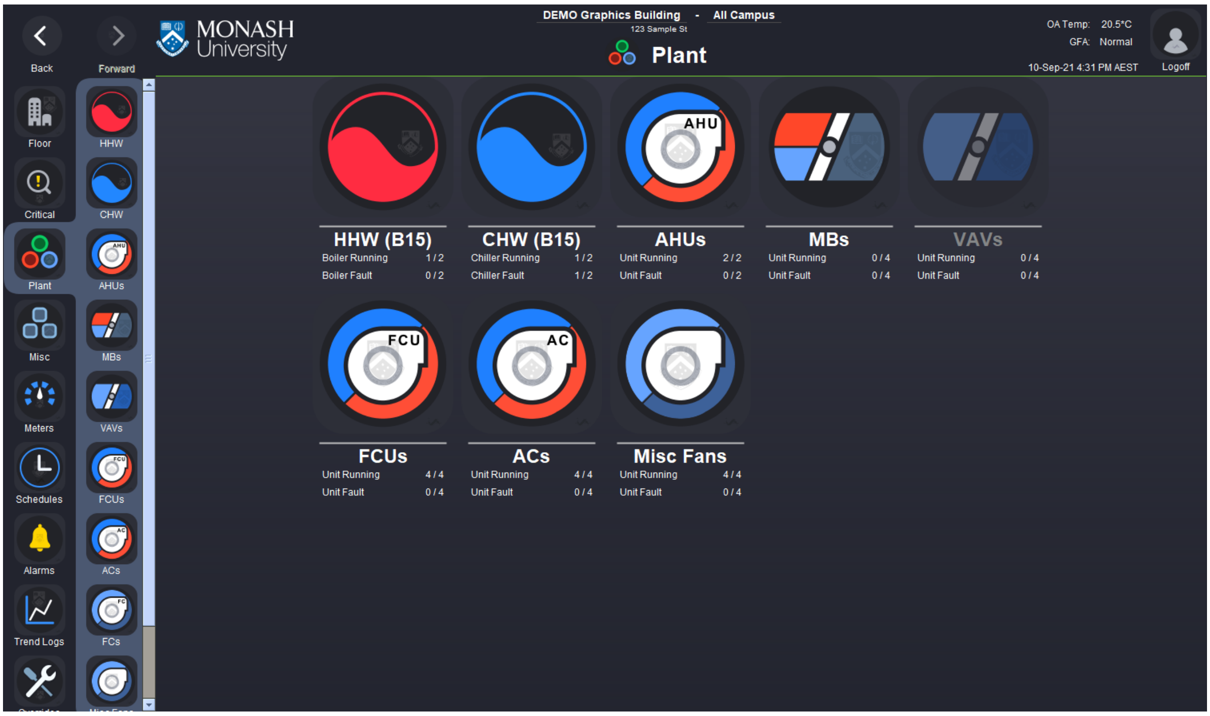


(Critical button highlighted. Cascading information sections)

3.4 Plant Submenu

The Plant submenu is a navigational tool that must display what system schematics or equipment summary pages are accessible via the graphics. On selection, the button must link to the Plant's icon summary landing page, see image below. The submenu is to be sorted vertically by size of plant, units or fans. By selecting the Plant button, the associated menu buttons must highlight as per the standard. On selection, the page must navigate to a Plant landing page, reflecting the same icons, names and order as the building Plant's sub menu. Information such as equipment running, equipment in alarm etc. may be added beneath the respective landing page icons. The Plant submenu must be visible on all pages associated with the building's Plant navigation.

In addition, the number of columns must show icons clearly, without horizontal scrolling. The example below shows icons in a GridPane set to five columns. A landing page should not exceed five columns. Where there are less than five icon summary objects being displayed, the GridPane column count should reflect this. For example, a building displaying a single boiler, will have a HHW icon summary button displayed on the landing page, with a GridPane column count set to 1 column.



(Plant landing page)

The table below is to be used as a reference when sorting the sub menu order for the building Plant submenu. As shown in the table, most equipment will hyperlink to an equipment summary table. The summary table can be divided by equipment type or by level. Specific pages will link directly to schematic pages, for example, heating, chilled and condenser water systems.

Label Guide:

Nav Menu (Short Name)	Header Title	Content Title	Navigates to…
HHW	HHW	Heating Hot Water System	Pipework schematic page
CHW	CHW	Chilled Water System	Pipework schematic page
CCW	CCW	Condenser Water System	Pipework schematic page
AHUs	AHUs	Air Handling Units	Table summary for the building
MBs	MBs	Mixing Box Units	Table summary by level
VAVs	VAVs	Variable Air Volume Units	Table summary by level
FCUs	FCUs	Fan Coil Units	Table summary by level
ACs	ACs	Air Conditioner Units	Table summary by level
Air Curtains	Air Curtains	Air Curtains	Table summary by level
Misc Fans	Misc Fans	Miscellaneous Fans	Table summary by fan type

3.5 Misc Submenu

The Misc submenu is a navigational tool that displays what miscellaneous system schematics or equipment summary pages can be viewed via the graphics. On selection, the button must link to the Misc icon summary landing page, see image below. The submenu is to be sorted vertically by importance, where possible. By selecting the Misc button, the associated menu buttons must highlight as per the standard. On selection the page must navigate to a Misc landing page. Information such as equipment running or equipment in alarm, may be added beneath the respective landing page icons where it is beneficial. The Misc submenu must be visible on all pages associated with the building's Plant navigation. This page content style must be identical to the one below, displaying the same icons, names and order as the building Misc's sub menu.

(*Misc plant landing page)*

Label Guide:

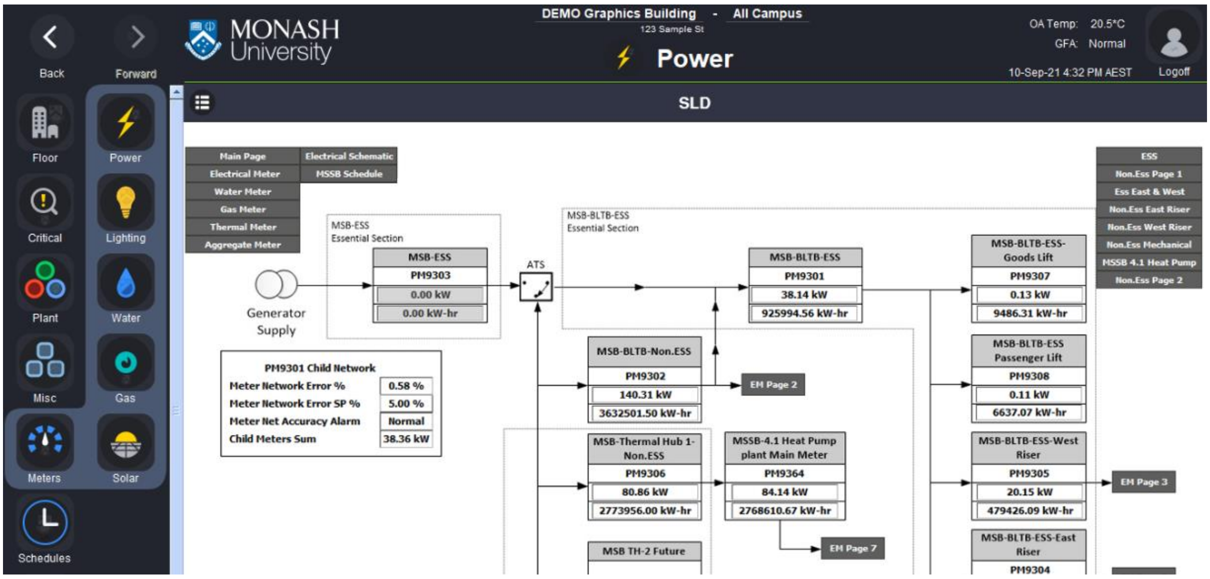
Nav Menu (Short Name)	Header Title	Content Title	Navigates to…
Lift	Lift	Lift Monitoring	Lift monitoring page
Pumps	Pumps	Pump Systems	Pump page grouped by type or separated by content subtitles
Radiator	Radiator	Radiator Equipment	Pipework schematic page
Weather	Weather	Weather Station	Weather monitoring page

3.6 Meters Submenu

The Meters submenu is a navigational tool to access single line diagrams (SLD) for each meter type. The meter summary pages can be accessed via the Detailed View button in the main content title. The individual meter pages can be accessed via hyperlink buttons on the SLD or summary table of the respective meter type. The Meter button in the main menu must navigate to a landing page displaying the available meter types associated with the building. The exact order, name and icons used in the submenu must be reflected on landing page. These meter types include: power, water, gas, thermal energy and solar meters. Where there are no meters to be displayed in the building, the main menu button may be removed. Alternatively, the button can be disabled and ghosted. See *Graphical Application Guide – Navigation Menu*. The Meter submenu must be visible on all pages associated with the building's meter navigation. Each icon summary button must link to the meter single line diagram page of that type. Where a SLD is not available, the icon summary button must link directly to the detailed view, displaying the summary table view instead. Individual meter pages can be accessed via hyperlink buttons on the diagram. The respective summary table pages can be accessed using the Detailed View button, which will be visible on pages such as the power meter SLD page, as well as individual power meter pages.

Format Guide:

Nav Menu (Short Name)	Header Title	Content Title	Navigates to…
Power	Power	Power Meters	Power meter SLD
Water	Water	Water Meters	Water meter SLD
Gas	Gas	Gas Meters	Gas meter SLD
Thermal	Thermal	Thermal Energy Meters	Thermal energy meter SLD
Solar	Solar	Solar Meter	Solar meter SLD

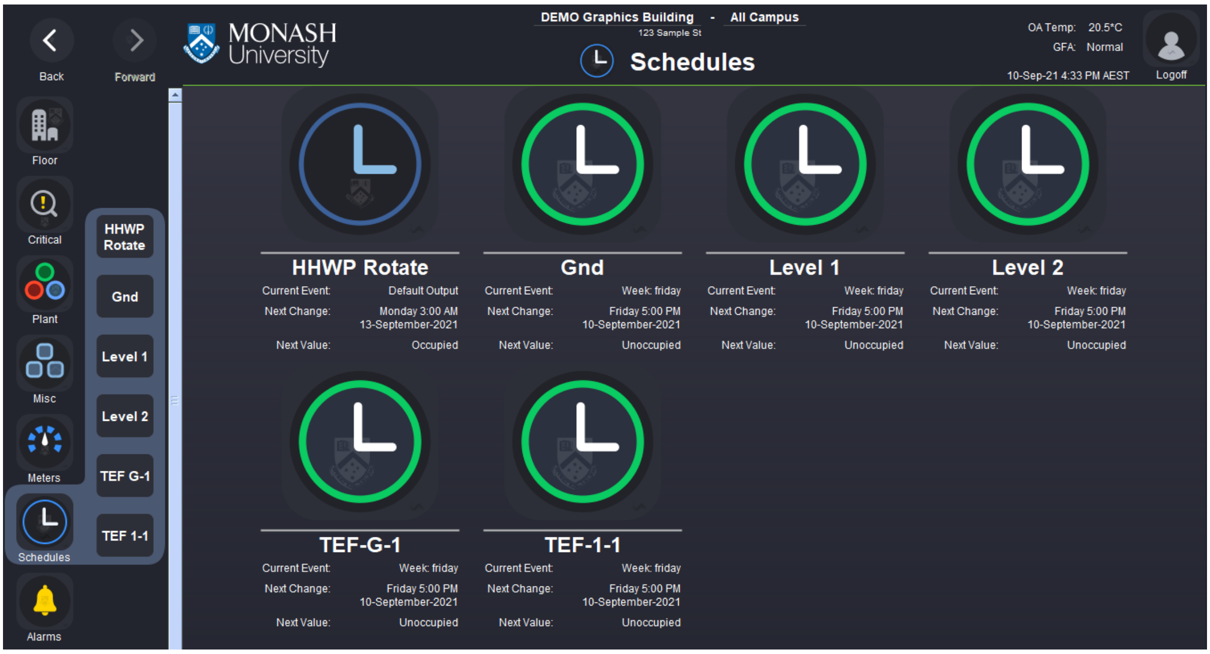


(SLD with 'Detailed View' button in the title bar hyperlinking to the summary page)

(Power meter summary with 'Detailed View' button in the title bar hyperlinking to the SLD page)

3.7 Schedule Submenu

The Schedule submenu is a navigational tool to access the Schedule's icon summary landing page. The submenu indicates the available Schedule pages that can be accessed and updated via the graphics. Unlike the Floor Submenu that mimics the floor selector, the submenu order must be organised by equipment types or in ascending order. For example, basement to roof. The exact order in the submenu must be reflected on landing page. 'Current Event', 'Next Schedule Change' and 'Next Value' are to be shown beneath the respective landing page icons, as per the example. A template for this schedule icon summary widget is available. See *Graphical Application Guide - Template*. A suitable column count must be used, for example 4 columns, in order to clearly show the icon summaries without horizontal page scrolling, see example below. Vertical scroll is utilised for buildings with many schedules. The menu style must use text style buttons for the submenu, minimalist button names and a highlighted menu background. Submenu button text must not exceed 2 lines. The Schedule submenu must be visible on all Scheduler pages.

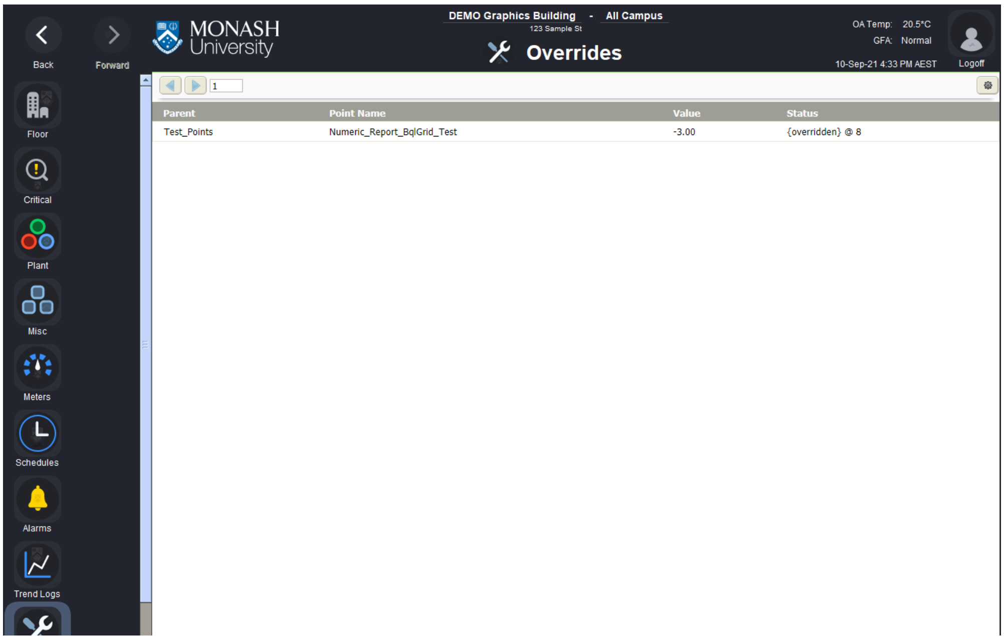


3.8 Alarm Submenu

The Alarm submenu is a navigational tool to access the building specific alarm console pages. The submenu must be sorted vertically from most critical to least critical alarm class. By selecting the Alarm button, the associated menu buttons must highlight as per the standard. A main content title is not required. The Alarm submenu must be visible on all pages associated with the building's alarm navigation.

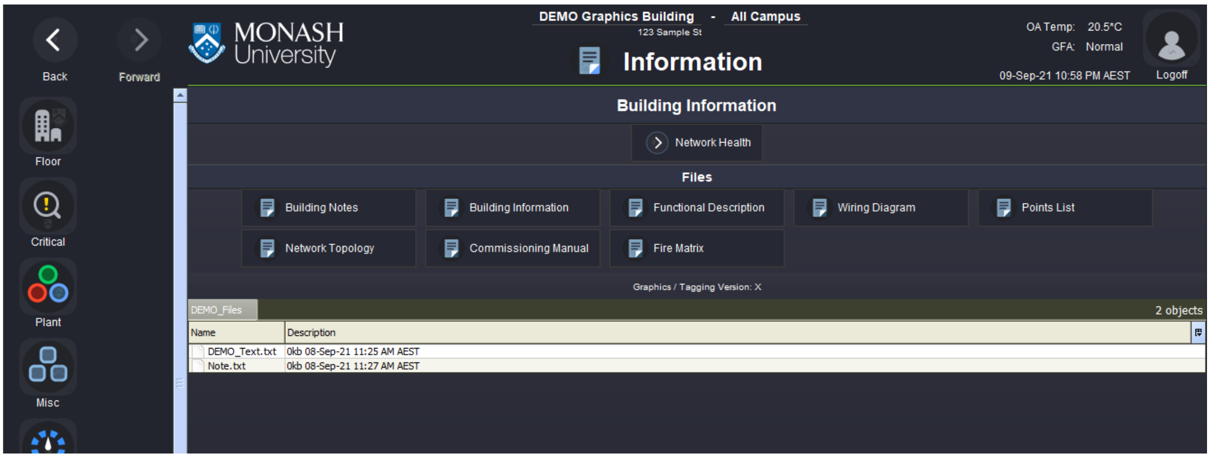
3.9 Overrides Submenu

The Overrides button navigates to a page containing a BQL report table (report:BqlGrid) that must display all overridden points and their current value for the building. There are no submenu buttons associated with this page. By selecting Overrides, a highlight fill must appear behind the Overrides button. A content title is not required. See *Graphics Application Guide – BQL Grid Table* for how to setup a BQL query.



3.10 Information Submenu

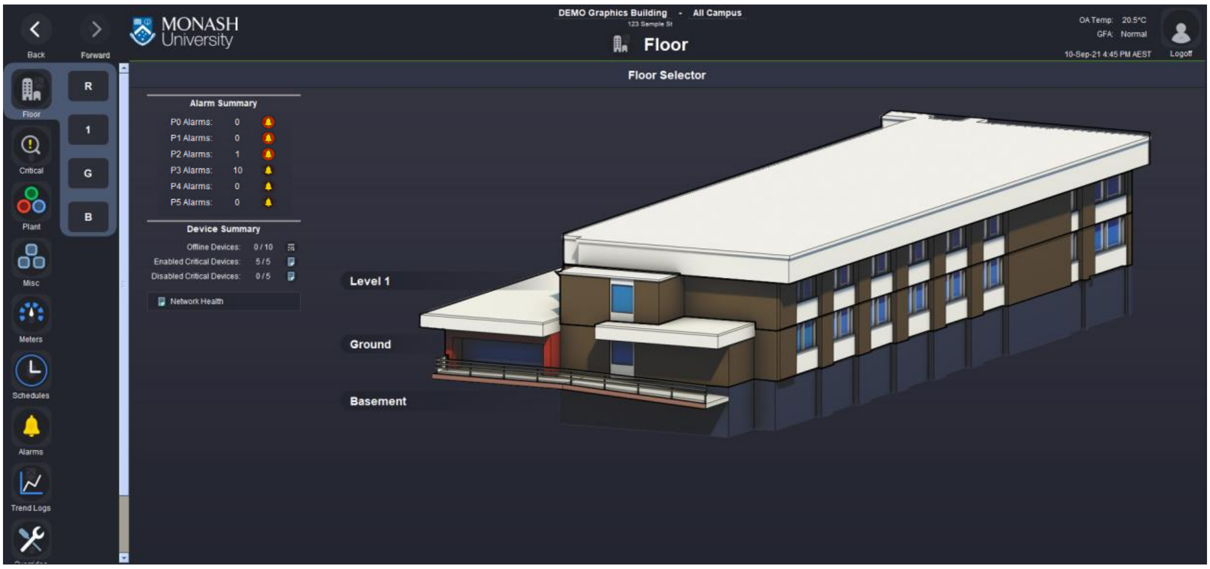
The Information button navigates to a page containing documentation via a NavContainerView. There are no submenu buttons associated with this page. By selecting Information, a highlight fill must appear behind the Information button. See *Main Content – Information*, for further details.



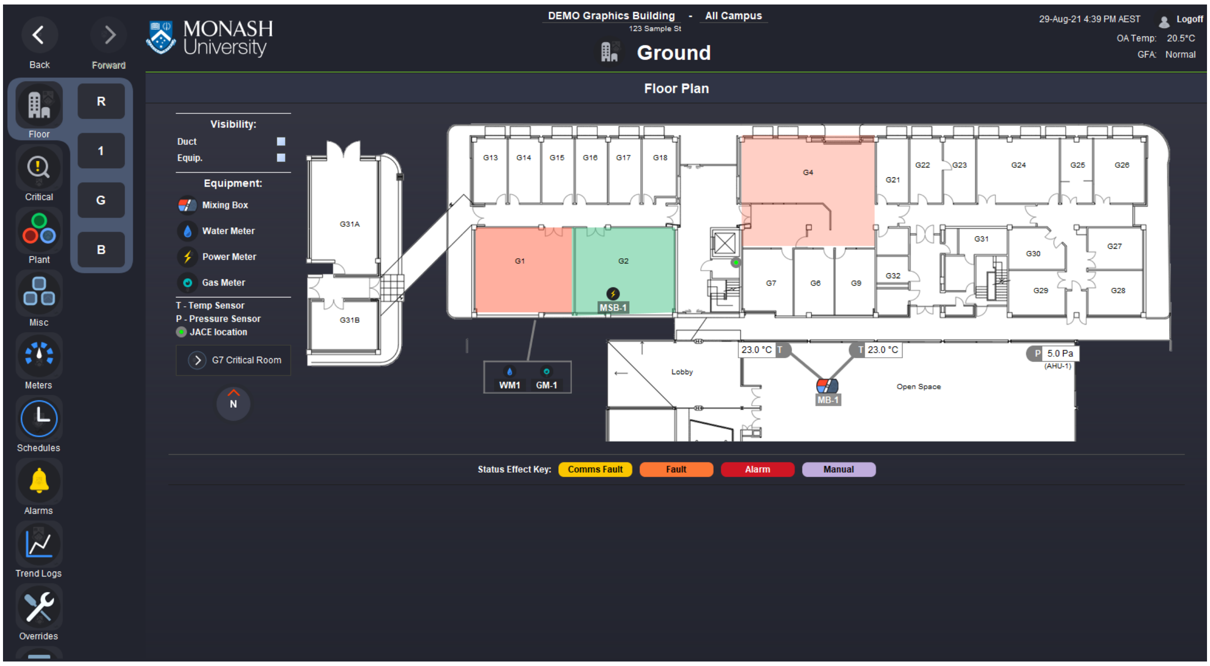
(Building information page, displaying building specific NavContainerView and files)

4 MAIN CONTENT PANE

The main content section is where information relevant to the page is displayed. Where available, specific pages may have content title buttons enabled. These buttons navigate to relevant floorplan, detailed view, settings and notes pages. All pages displaying control and monitoring points must have a status effect key beneath the content. The following sections outline specific page requirements and provide examples.



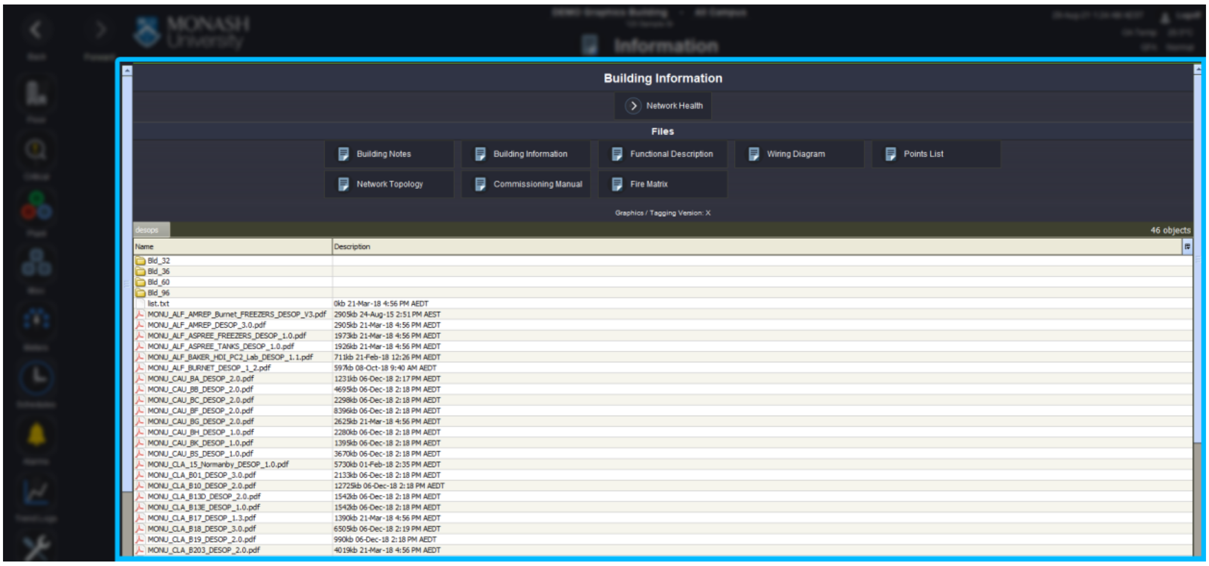
(Floor Selector page displaying Bql Query information)



(Floorplan displaying controller points and status effect key)

Monash University

(Equipment page displaying controller points and status effect key)

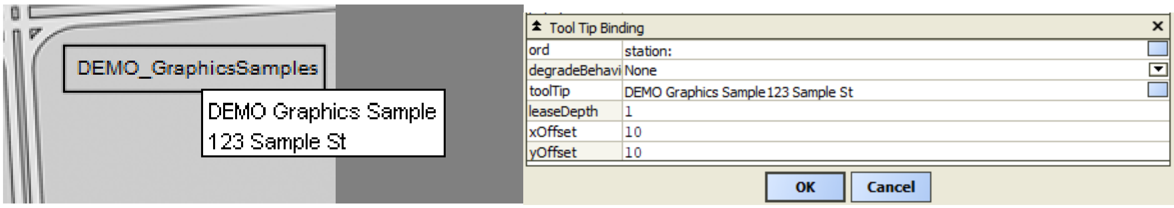


(Generic information page displaying buttons and a general file directory location)

4.1 Campus Map

A campus map is used as a navigational tool and visual representation of building location. This map uses existing graphics standards. For this page, use the appropriate standard for the project's scope of works, but you must add the addition of a tool tip binding. The tooltip binding must display 2 lines of text. The first line must be the building name. The second line must be the street number and name. Where information is not applicable, just display the building name. The building will have a path that outlines the building, where the fill colour indicates the highest active alarm in a building as per the alarm standard. The colour is determined by alarm class chooser logic.

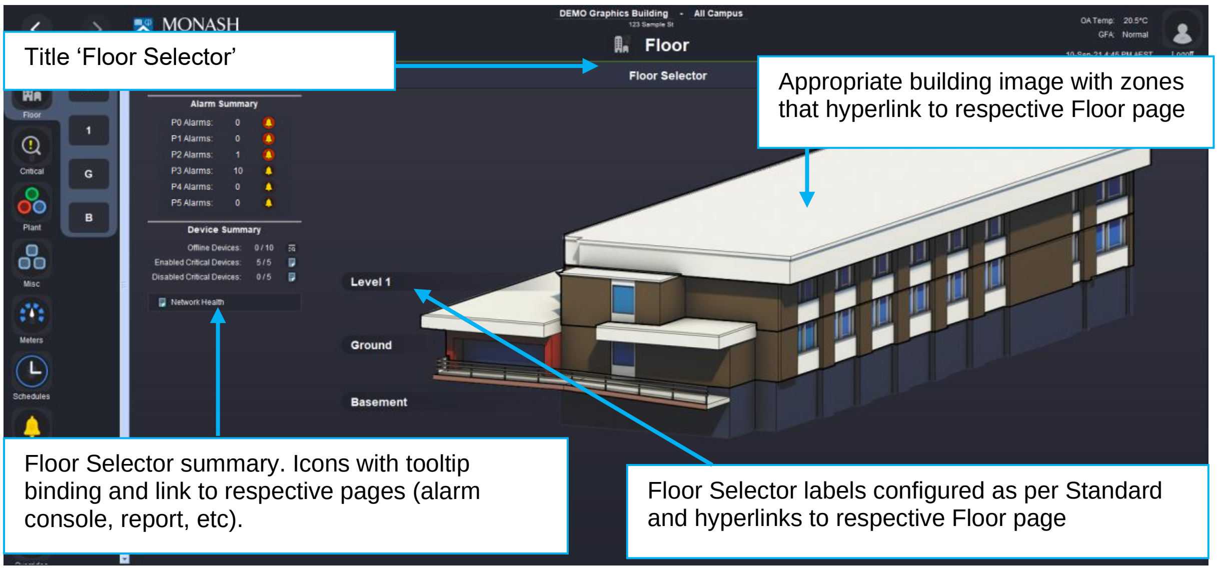
(Campus map with DEMO_GraphicsSample indicated)



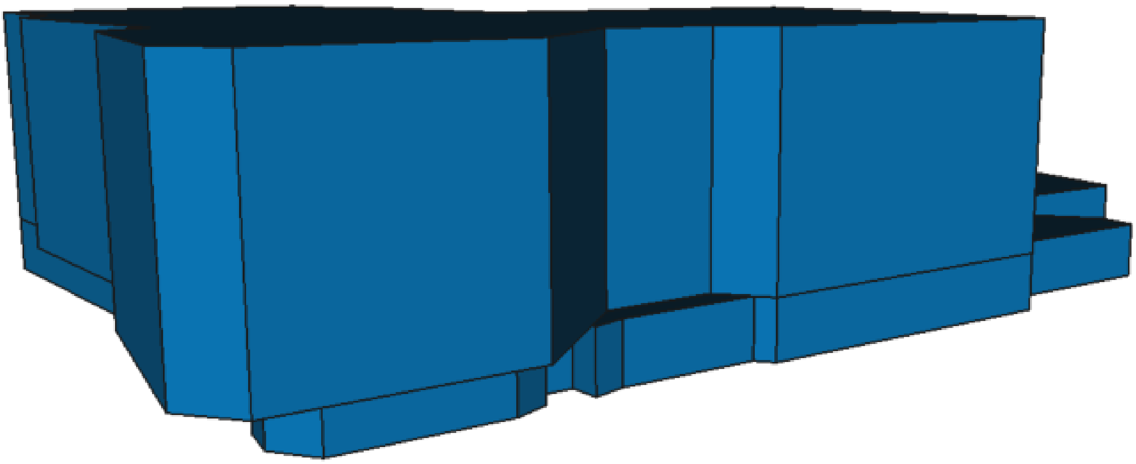
(Close up of a zone to be linked to building Floor Selector page with tool tip binding)

4.2 Floor Selector

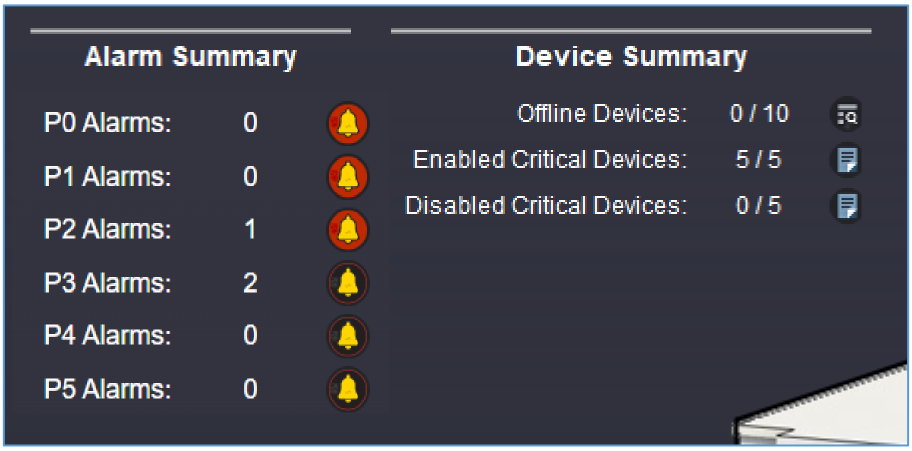
The Floor Selector page is the building home page and allows the user to view building summary information and navigate to floorplan pages via a button or zone. The page should not navigate to a building photo, that may exist in previous standards. The two images below highlight the type of 3D building images accepted. The following section outlines the requirements for the Floor Selector page.



(Floor page – Floor Selector with preferred type of 3D building image. Other examples in buildings 16, 31 and 96)



(Other acceptable 3D building image using FreeCAD – Request instructions from BAS Administrator if required)

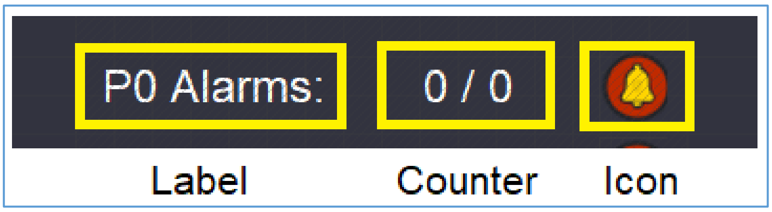


(Close up of the summary values)

Label Guide:

Item	Icon	Tool Tip	Hyperlink to…
P0 Alarms	alarmFill_20x.svg	P0 Alarm Console	Alarm Console
P1 Alarms	alarmFill_20x.svg	P1 Alarm Console	Alarm Console
P2 Alarms	alarmFill_20x.svg	P2 Alarm Console	Alarm Console
P3 Alarms	alarmStroke_20x.svg	P3 Alarm Console	Alarm Console
P4 Alarms	alarmStroke_20x.svg	P4 Alarm Console	Alarm Console
P5 Alarms	alarmStroke_20x.svg	P5 Alarm Console	Alarm Console
P6 Alarms	alarmStroke_20x.svg	P6 Alarm Console	Alarm Console
Offline Devices:	infoLook_20x.svg	Offline Devices	Network Health Page
Disabled Offline Devices	report_20x.svg	Disabled Offline Devices	To Report
Disabled Critical Devices	report_20x.svg	Disabled Critical Devices	To Report

The floor selector summaries are a pair of lists that contains counters, one for alarms and the other for devices. Each list must have a row that contains the following three elements in the listed order: A label, a counter and an icon.



(Breakdown of a summary *row)*

The labels describe the what information is being displayed in its counter.

The counter is to display:

- For devices: The current amount and the total amount for the data that it is displaying.
- For alarms: The current amount.

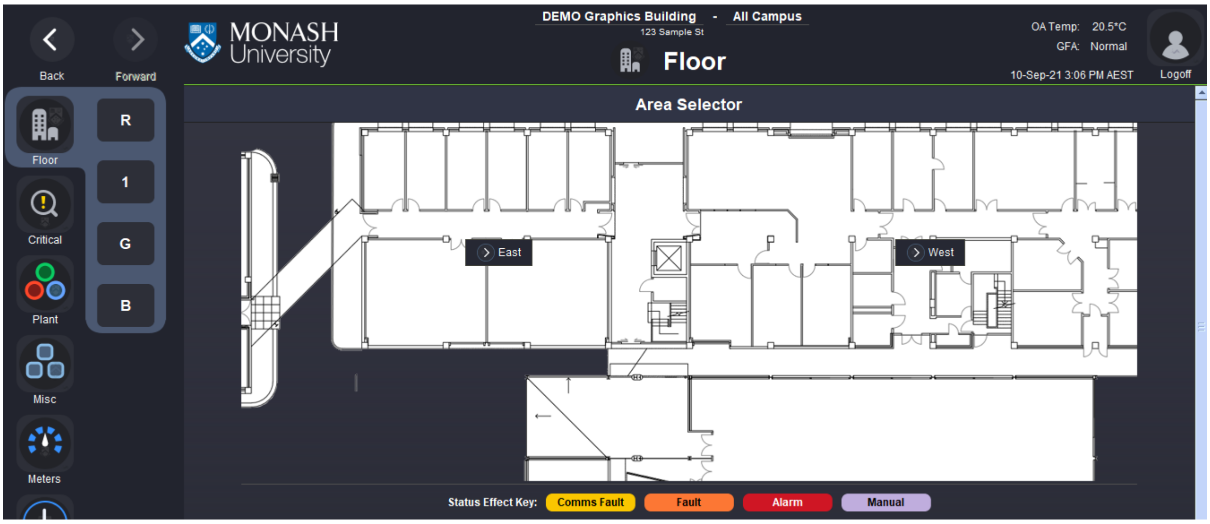
The icon is an image that when clicked will navigate the user to the relevant page that displays the data in greater detail. The image used for the icon should match the icon used on the target destinations page. However, if the destination page has no icon assigned to it, an icon appropriate to the information displayed in the row can be selected at the user's discretion.

The alarm summary contains a list of all of the different alarm groups ordered from most important to least important. The label must state the name of the group that the row is representing. The counter must show the number of active alarms for that alarm group. The icon must also have a hyperlink which directs the user to the relevant building's alarm console page.

The device summary contains a list of information about the status of device in the station. The first row must display the total amount of devices that are offline. This rows icon must hyperlink to the network health page. The second row must display the total amount of devices marked as critical that are enabled. This rows icon must hyperlink to the critical devices report. The third row is similar to the second, except it must instead show the critical devices that are disabled.

4.3 Area Selector

The Area Selector page is a floorplan page that allows the user to view the entire level, as well as navigate to sections of a floor via a button or zone selectable zone. There should be no floor icons or labels on this page. The area selector is used where floorplan sections need to be enlarged in order to clearly display the required information. Where a floorplan page can adequately show all required information without scrolling, no area selector is required.

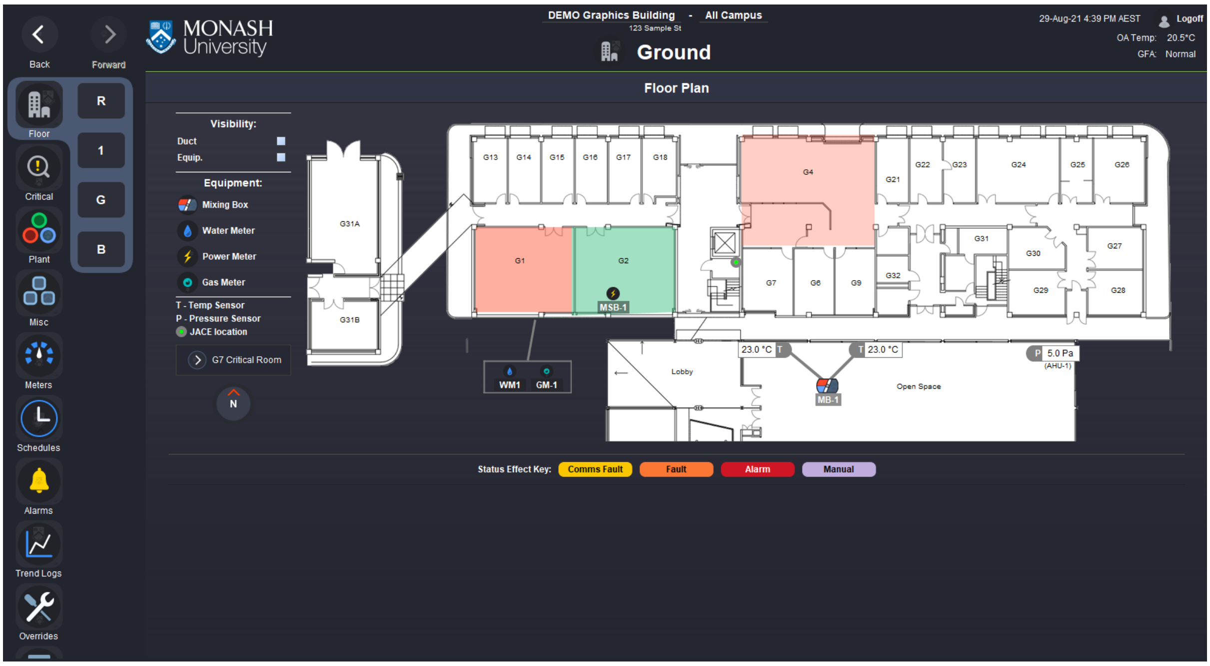


(Area Selector page with buttons)

(Area Selector page with buttons and highlight, in this example the button is below the zone in the hierarchy)

4.4 Floor Plan

The Floor Plan page shows the relationship between rooms, spaces and equipment location viewed from above. The user can view thermal comfort information, navigate to equipment pages and toggle specific visual elements in this view. The section below will outline the requirements for the Floor Plan page and provide examples.



(Floor Plan page)

The Floor Plan must use display room names and use an appropriately sized floorplan image with minimal horizontal and vertical scrolling. Where some scrolling is necessary, vertical scrolling is preferred. If a floorplan is too large and scaling the image down will hinder the usability, sectioning the floorplan in a logical and consistent way throughout the building may be necessary. An example may be dissecting larger floors, having an east and west floorplan page allowing a closer view of the rooms and ample space for placing equipment icons. See *Navigation Menu - Floor Submenu* for examples.

The view must display all equipment controlled and monitored on the floor plan using an icon with a text label. For example, this includes icons for physical locations of air handling units (AHU) shown in the plant room. If too many items are located in the one area, a fly out box with an indicator can be used, as shown below. Where equipment associated with a system are located in one area, for example heating hot water systems (HHW), a boiler icon with a 'HHW' label can be used hyperlinking to the HHW System schematic page. The original size icon should be retained, but the smaller version can be used if required. Where the option of a smaller unit or meter icon is needed in a space where no other layout choices are available, a smaller meter icon is preferred.

(Flyout box and indicator line)

4.4.1 Floor Plan Image

The Floor Plan image must be suitable and consistent across the building. A floorplan should not change orientation and sizing when navigating through the pages. For example, if a floorplan is orientated north, it should remain that way throughout the building. The floorplan orientation must be practical and minimise unnecessary scrolling. The floorplan image must be sourced from Monash University's Sisfm database:

https://space.bpd.monash.edu/sisfm-enquiry/monash/home/

After providing your login credentials, navigate to Map Enquiry and select the required campus and building. Use the right-side menu to locate the Floor Documents section. Download an appropriate floorplan file. Remove all layers except for walls and doors and save image in a format of your choice, for example Portable Network Graphics (.PNG) file format.

4.4.2 Floor Icons and Labels

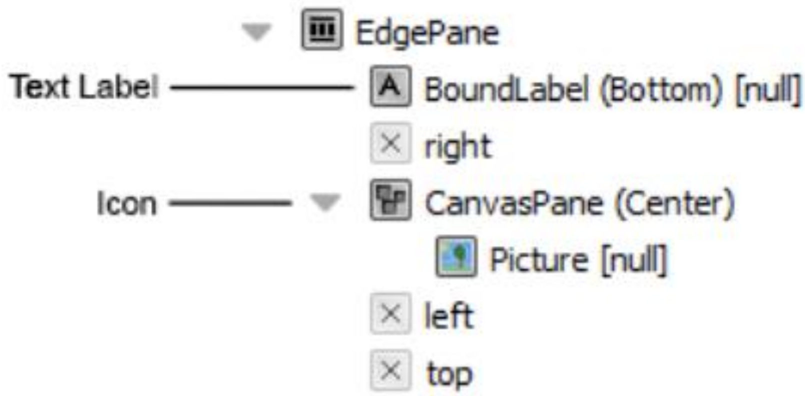
Floor Icons are used to indicate the location of controlled and monitored equipment on the Floor Plan page. These are also used as a navigational tool to view specific pages. The section below will outline the requirements and provide examples.

A standard Floor Icon must look identical to the table below. The icon and text label should reflect the page it links to, in line with the icon and name used across the building's graphic pages. The hyperlink resides on the image only, not the text button. When using the template, only the image, the image hyperlink and the text need to be updated. A template for all 4 sizes is available in the muGraphicsLibrary palette.

Example Size Guide:

Floor Plan Icon	Image	Canvas and Picture Dimension
	lightning.svg	32 x 32px
	lightning.svg	20 x 20px
	mbSml.svg	32 x 20px
	mbSml.svg	24x 16px

If you are not using the template, note that in order for the icon to scale, the canvas (view size) and picture (layout) need to be set to the same width and height dimensions. The image below indicated the text label and the icon canvas and subsequent image that must share the same dimensions.



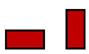
(Floorplan icon root, indicating text label and icon widgets)

The table below provide suggestions for floorplan icons and labels. When implementing these visual elements on the floorplan, some customisation can be used as required. Where a floorplan has a lot of objects on a canvas pane or has not been sized appropriately for the information that is to be displayed, various techniques can be used to convey the necessary information.

These include:

- Repositioning the floor icon's text label to the left, right or above the icon.
- Resizing the floor icon's canvas and image size (but it must be consistent across the page)
- Using clear, abbreviated names. (For example, FCU 1-1, may be use the label 1-1)
- Where 2 units are next to each other, work together and share the same space, a single icon may be used with a descriptive name.

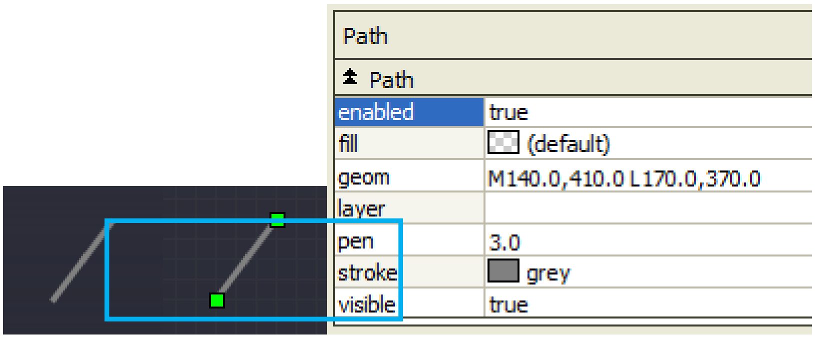
Examples of floorplan icons and labels:

Floor Icon	Description
	*Temperature Sensor*
	*Temperature Sensor 1 of 2 or more*
	*After Hours Button (Location Only) – If required, a black stroke (eg. 1.0pt thickness) can be added to aid visibility on floorplan.*
	*After Hours Button*
	*Pressure Sensor*
	*JACE/Controller/Panel Location (jace_x15.svg) – Use floorplan legend and/or tool tip bindings where additional information is necessary.*
	*Power Meter (lightning.svg).*
	*Smoke Dampers (label or path can be used).* *Vertical or horizontal shapes can be used indicating the orientation of the damper. Where status is to be displayed, flyouts and BoundLabel standards are to be used appropriately for the specific floorplan and information to be displayed. This must be clearly indicated in the equipment key.*

Floor Icon	Description
	*CHW System (chiller.svg) – where all equipment is located in one area and displayed on one page.*
	*Chiller (chiller.svg) – where a standalone unit location is indicated, eg. A chiller called CH-3.*
	*Pump 1 and 2 (hhwp.svg) – where a unit is located together. The hyperlink may hyperlink to a dedicated pump page, or a HHW System page if the pumps reside on that page.*
	*Pump 1 and 2 (greyPump.svg) – where a unit will benefit being identified in the label and are located together and displayed on the same graphics page.*
	*Fridge (fridge.svg) -*
	*Freezer*
	*BAS*

4.4.3 Indicators

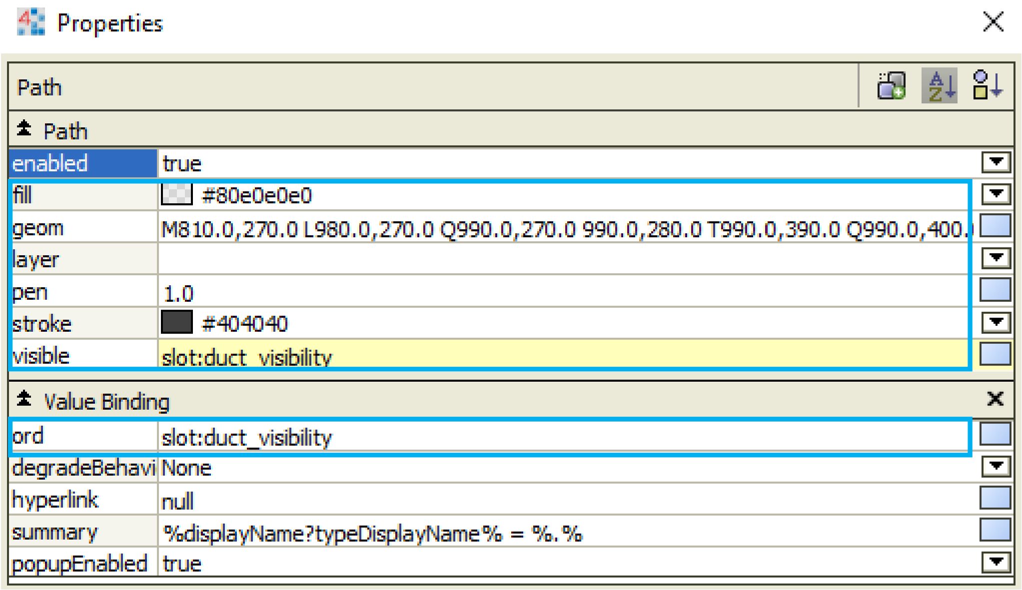
The follow image shows the minimum requirements for an indicator. Where a graphic may benefit with a thicker pen value, it can be increased. All indicator paths must be consistent across the page it is on.



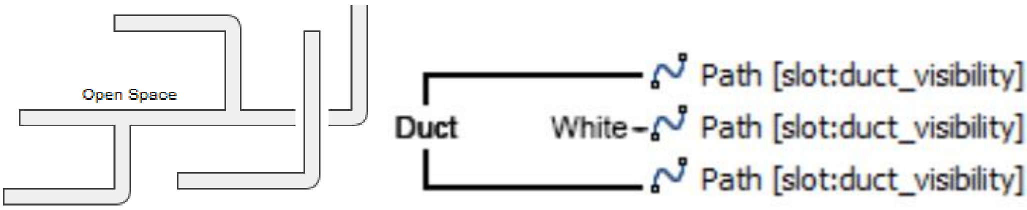
4.4.4 Duct

Ducting is represented using paths. Below are the minimum requirements for ductwork on a floorplan. Where thicker lines are more suitable, the pen value can be increased. The ducting must be consistent across the floor page. Where ducting may overlap, a short white path or shape, wider than the ductwork, can but positioned where the two ducts intersect. This can infer that a duct continues above or below another duct depending on the widget tree hierarchy. To create the path geometry, the 'Add Path' tool can be used.

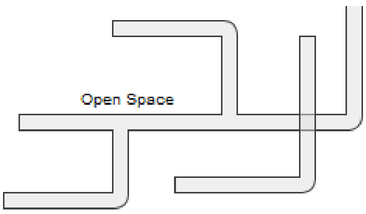
Having a lot of widgets on a page may become hard to edit. Layers may be used to group elements on the page and lock or hide them for ease of editing.



(Duct path properties)



(Example of wide, white path position between the ductwork)

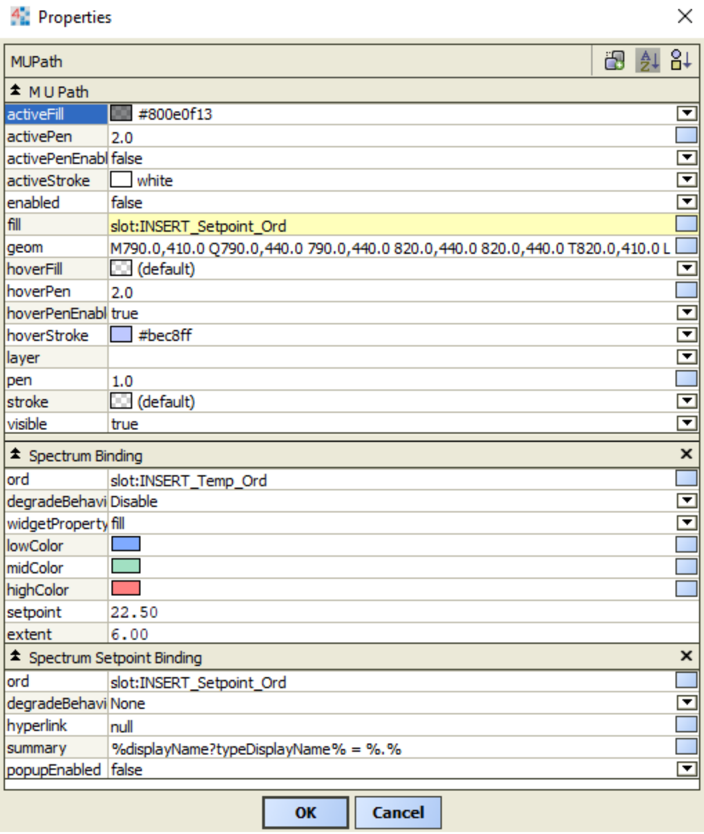


(Duct path without white path)

Alternatively, an image can be used as a layer for ductwork. A sample is required to be submitted to the BAS Administrator for approval.

4.4.5 Zones

The Floor Plan zones are a navigational tool that also indicates the thermal comfort of a space. An area is indicated using a path. A path can be created using bajaui Path or MUPaths, which has additional features. These features include web friendly hover and hyperlink capabilities. Both Path objects use a spectrum binding to indicate how far a temperature is from setpoint, as well as a spectrum setpoint binding. The setpoint binding is used when a specific unit setpoint is to be used to affect the colour fill. The zone area is to represent the temperature of a space being served by specific units.



(MUPath spectrum binding properties)

The default colour spectrum is;

- No colour = no unit/s start call
- Green = equal to SP
- Orange = within +3degC from SP
- Cyan = within -3degC from SP
- Red = greater than or equal to 3degC from SP
- Blue = less that or equal to 3degC from SP

Colour values;

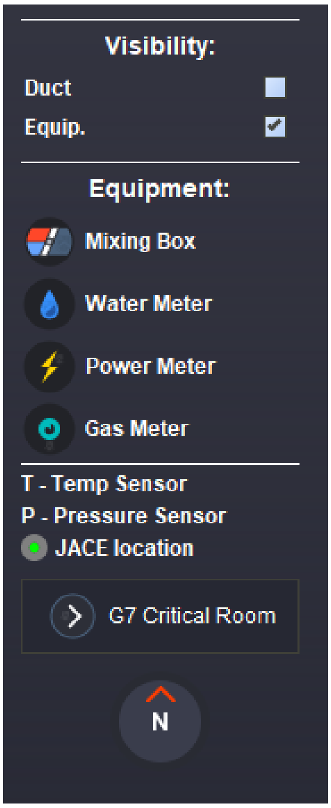
- Green = #8044c185
- Orange = Uses Spectrum Binding
- Cyan = Uses Spectrum Binding
- Red = #80ff0000
- Blue = #800055ff

Note, if there is a temperature alarm configured for the space, the degrees from setpoint are to be adjusted accordingly. For example, a lab with 22+-0.2 for an alarm.

A zone is required to show no colour, when a unit call is inactive, the point can be animated to the zone visibility.

4.4.6 Side Panel

The side panel is located to the left of the floorplan image in the main content section. The panel consists of a visibility, equipment, legend, additional links and a compass image section. Where a minimap is required, one may be included after the compass in the side panel. The side panel should be constructed in this order and updated with relevant information relating to the floorplan. The following section will outline the requirements for the side panel sections.



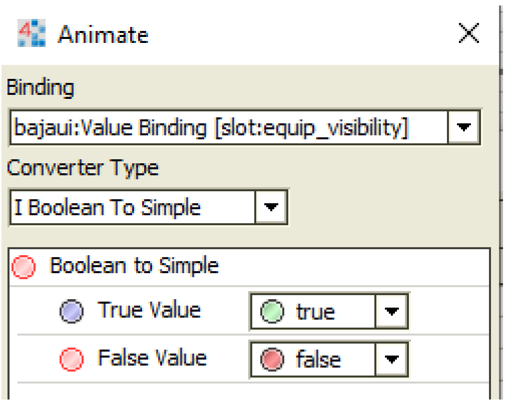
4.4.7 Side Panel: Visibility

Visibility check boxes allow the user to toggle on and off ducting or equipment icons, along with their associated indicators. This is done by adding a baja:Boolean slot to the graphics folder via the AX Slot Sheet view, see image below.

If using the template, the icon and indicator objects will have a value binding pre-configured with 'slot:equip_visibility' on the widget root. The 'visible' property will need to be animated to this ord in order for it to show and hide when the checkbox is toggled.

equip_visibility	equip_visibility	Dynamic		baja:Boolean
duct_visibility	duct_visibility	Dynamic		baja:Boolean

(Icon and indicator widget root)



(Animate example)

4.4.8 Side Panel: Equipment

All controlled and monitored equipment shown on the floorplan must be accounted for in the Equipment section of the side panel. This is to define the icons for the user. This is to be displayed in a single column list as shown below. Beneath the equipment is additional information related to points and JACE location. Where a sensor label uses an abbreviation, this section must inform the user what it stands for.

4.4.9 Side Panel: Additional Links

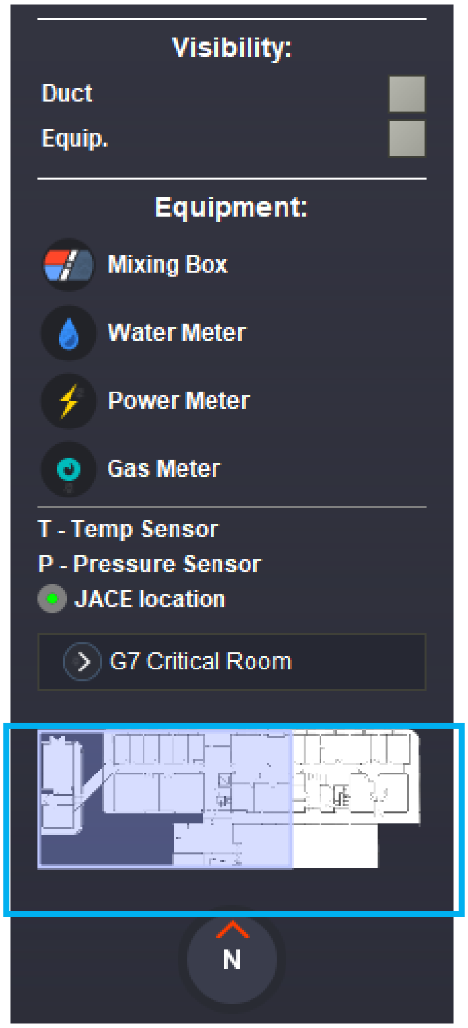
Any useful and additional links will be located beneath the equipment list. This may include critical rooms or information relating to the page.

4.4.10 Side Panel: Compass

A compass must accurately represent the real-world orientation of the floorplan. The floorplan orientation may be rotated and scaled to utilised screen dimensions more effectively, limiting scrolling.

4.4.11 Side Panel: Minimap

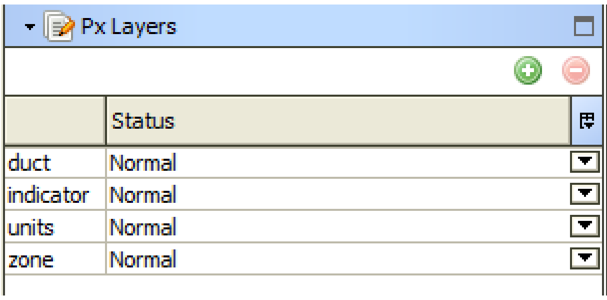
For floors that require numerous pages to be viewed, a minimap must be implemented. The minimap must be a scaled down version of the image used on the Zone or Area Selector page. The minimap is located below all the visibility, equipment and additional links info. The sizing should be practical and appropriate for the information to be displayed. Where no labels are required, the minimap can be small. Where complex floors with specific zones are labelled, a larger minimap must be utilised to display the information and hyperlinks. Shading must be used to indicate the current view. A minimap is not to be used where an entire floor is visible on a page.



(Minimap in floorplan side panel)

4.4.12 Px layers

Px Layers may be used to assist in editing and grouping floorplan elements. It allow you to group, select, hide and lock elements on a page assigned to a particular layer. Px Layer names should be consistent across all floor pages. Below are example page layers to be used where necessary. Note that if an element from another page is copied, and the destination page does not have the exact Px Layer name, you will not be able to interact with that object. You must add the correct layer the object it is expecting and was already assigned to.



4.5 Critical Overview

The Critical overview page displays critical campus systems. A suitable graphic must be constructed to be displayed the relevant building information, for example a summary table style or a button hyperlink to a specific page. See *Summary Table* section for examples for displaying information.

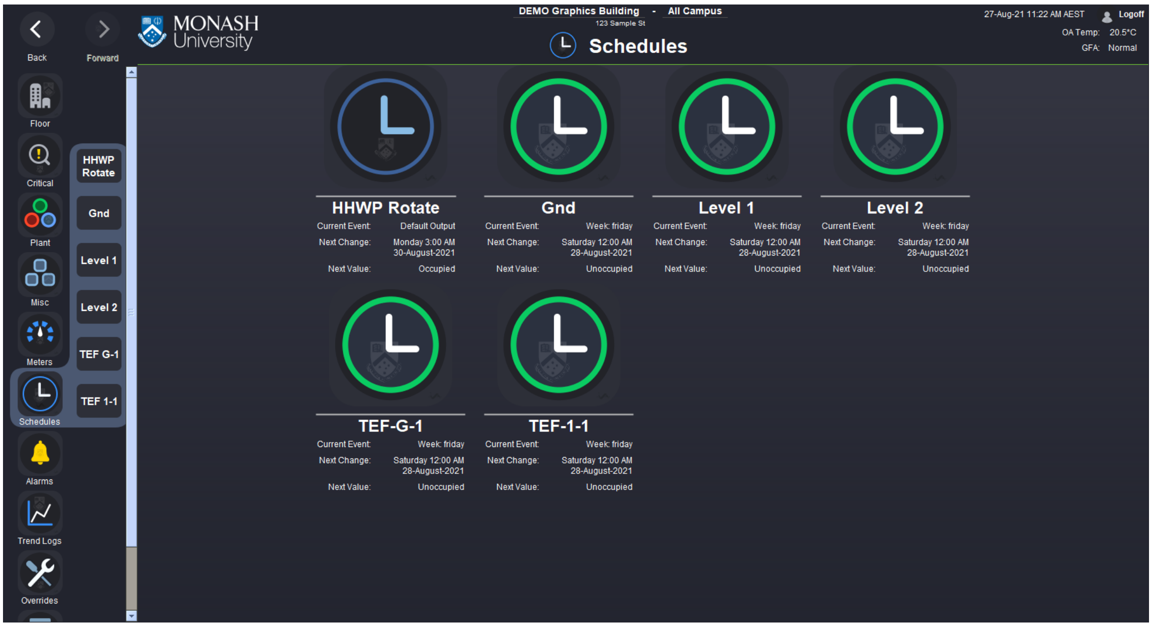
Critical campus systems include;

- Campus Clocks
- Freezer Alarms
- Constant Temperature Room
- Compressed Air Systems
- HV Substation Alarms
- Generators
- Power Monitoring
- Water Monitoring
- Gas Monitoring
- Service Tunnel
- Sump Pumps
- Tunnel ACS System
- Waste Pit pH Levels
- Sewer Pits
- Stormwater Harvesting (edited)

4.6 Landing Page

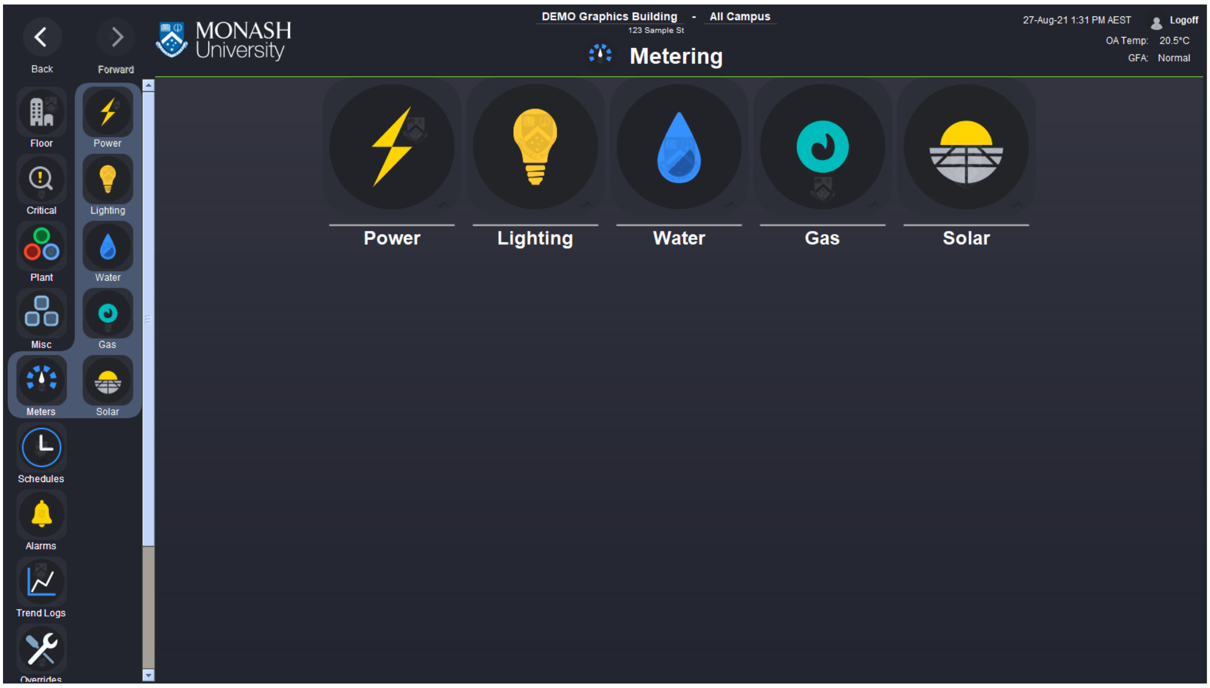
A landing page must display icons, a title and plant specific information, where practical. Buttons must be consistent across the landing page and submenu, sharing the same name, hyperlinks, imagery and order listed in the submenu. The Schedule submenu and landing page is the exception. The Schedule submenu uses a text style submenu instead of an icon style. Regarding the icon summary widgets on the landing page, any additional information is to be displayed beneath the icon summary title, as shown in the example below. The information must be practical for the related plant and use queries or points in order for information to update dynamically. In instances where plant is located in a different location, the text label must denote the building name. For example, a building 15 HHW system would have a label, 'HHW (B15)' and hyperlink to the relevant page. A button and hyperlink between the two buildings must be clear and accessible.

(Plant landing page with additional information)

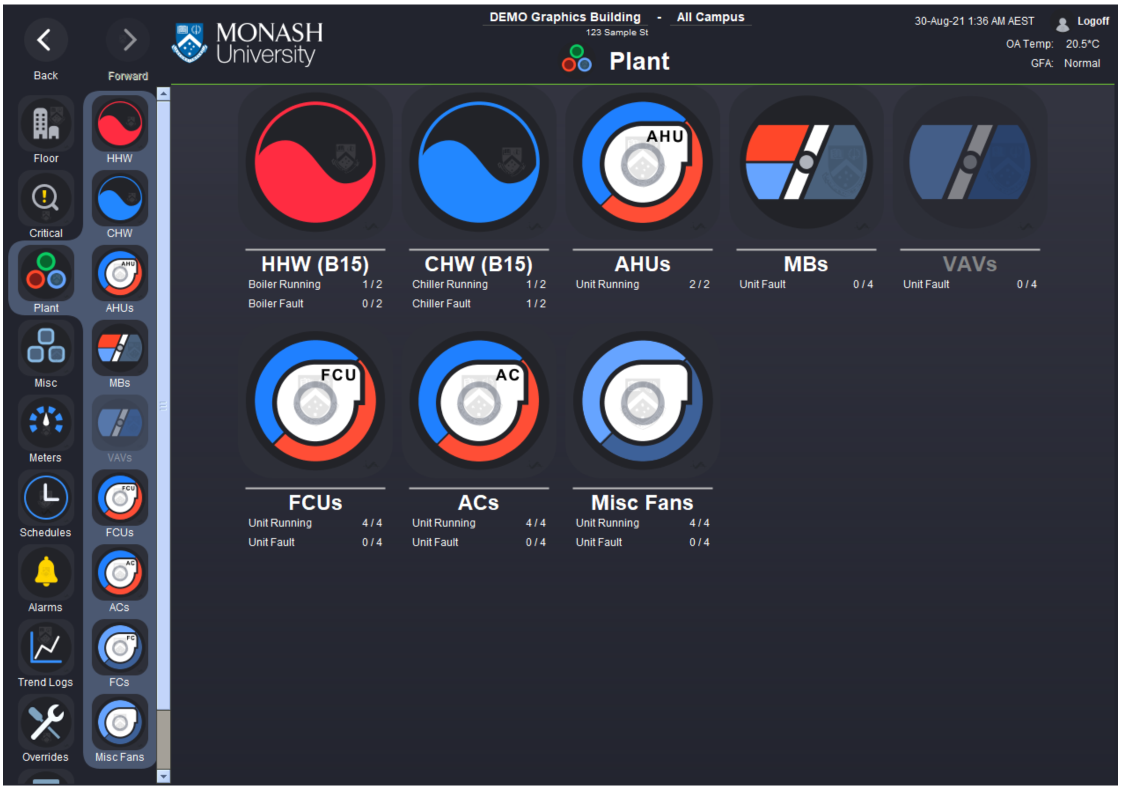


(Schedule landing page with additional information)

(Alarm landing page with additional information)



(Meter landing page, example of no information available)



(Example of HHW and CHW system coming from another building)

4.7 Summary Table

The summary table is made up a widget at the base called the Table Root which is used to hold a number of child widgets that represent the rows of the table.

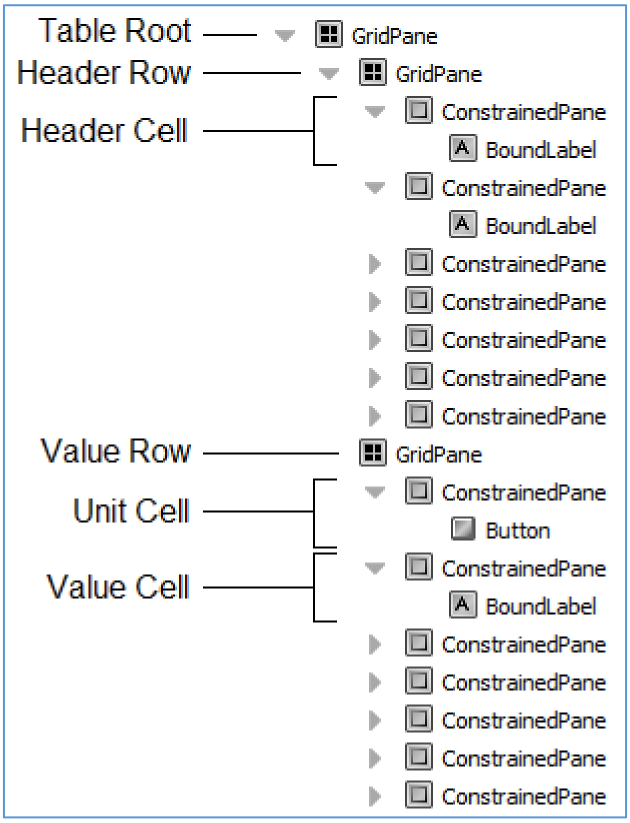
These row widgets have two different types: Header Rows & Value Rows.

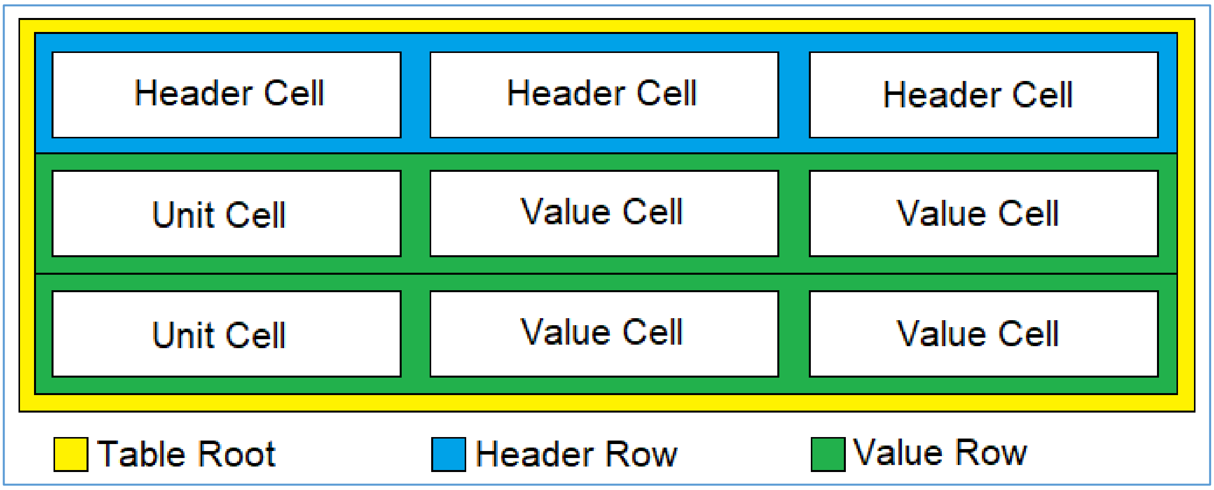
The row widgets hold a number of cell widgets, the types of which is determined by a combination of the type of row they sit in as well as their position in the table.

Header Cells must reside under the Header Row. Unit Rows and Value Rows must reside under a Value Row. Each Value Row is used to represent a unit, where the Unit Cell is used represent the unit itself and contains a hyperlink to the unit's info page if available. Each Value Cell displays the value of a point that belongs to the unit.

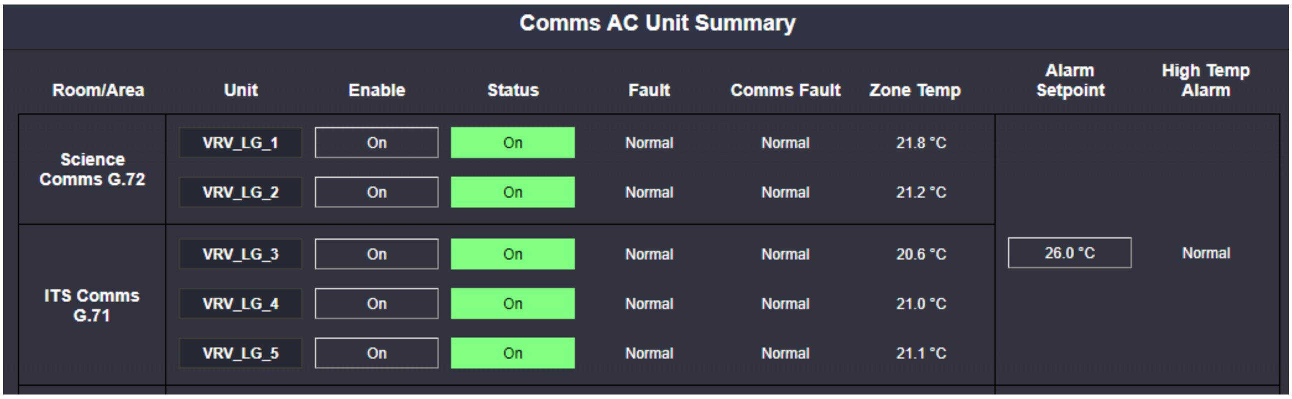
The ordering of the columns for the points should be by importance to the operation of the unit. For example, Enable and Status points should appear toward the left of the table.

"Area served" must be the first column of a summary table. Examples: Rooms 1.23 & 1.24, Level 1 West, AHU-1/2/3 Preconditioned Air.





(Diagram depicting of how table widgets are arranged)



(Example of an actual summary table graphic)

4.8 Summary Page By Level

A variation of the summary table is use, usually separated by level, for units where a single summary page would not be practical. Where there are many units and a floor-by-floor summary is beneficial, the summary table pages displayed, as shown in the image below. Unit may include mixing box (MB) and variable air volume units (VAV). Templates for the content title bars with hyperlinks and floorplan button are available.

The order of the VAV titles should be in ascending order from ground floor upward. Unlike the Floor Submenu that mimics the floor selector, the submenu order must be organised by equipment types or in ascending order. For example, basement to roof. The Content Header Titles that have an arrow hyperlink to the corresponding VAV summary page. The current level summary does not have an arrow and does not have a hyperlink, this is because this is the page that is currently being viewed. When the user navigates to the different pages, the effect is an expanding style navigation. A floorplan icon is accessible and is to be configured as per the Content Title Button standard.

“Area served” must be the first column of a summary table. Examples: Rooms 1.23 & 1.24, Level 1 West, AHU-1/2/3 Preconditioned Air.

Ground VAVs

Level 1 VAVs

Level 2 VAVs

Level 3 VAVs

VAVs	Control Override	Mode	SA Temp	Flow	Damper	Zone Temp SP	Zone Temp	Zone CO2	Pressure	Occupancy Override	Occupancy	Blacklist
VAV 3-1 & 3-2	Normal	Heat	20.4 °C	0.0 L/s	100 %	22.0 °C	20.8 °C	465 ppm	0.0 Pa	Off	Off	✓
VAV-3-3	Normal	Heat	20.4 °C	0.0 L/s	100 %	22.0 °C	20.9 °C	425 ppm	0.0 Pa	Off	Off	✓
VAV 3-4 & 3-5	Normal	Heat	20.4 °C	0.0 L/s	100 %	22.0 °C	20.9 °C	430 ppm	0.0 Pa	Off	Off	✓
VAV 3-6 & 3-7	Normal	Heat	21.5 °C	112.4 L/s	100 %	22.0 °C	21.8 °C	422 ppm	0.0 Pa	Off	Off	✓
VAV-3-9	Normal	Heat	20.7 °C	0.0 L/s	100 %	22.5 °C	21.7 °C	424 ppm	0.0 Pa	Off	Off	✓
VAV-3-10	Normal	Heat	21.5 °C	0.0 L/s	100 %	22.0 °C	21.5 °C	416 ppm	0.0 Pa	Off	Off	✓
VAV-3-13	Normal	Heat	22.1 °C	0.0 L/s	100 %	23.5 °C	22.6 °C	393 ppm	0.0 Pa	Off	Off	✓
VAV-3-14	Normal	Cool	22.7 °C	0.0 L/s	100 %	22.0 °C	22.9 °C	432 ppm	0.0 Pa	Off	Off	✓
VAV-3-15	Normal	Cool	21.9 °C	0.0 L/s	100 %	21.5 °C	22.5 °C	394 ppm	0.0 Pa	Off	Off	✓
VAV-3-16	Normal	Cool	21.1 °C	0.0 L/s	100 %	21.5 °C	22.2 °C	430 ppm	0.0 Pa	Off	Off	✓
VAV-LS-3-16	Normal	Heat	21.2 °C	0.0 L/s	100 %	22.0 °C	21.6 °C	430 ppm	0.0 Pa	Off	Off	✓
VAV-3-18	Normal	Idle	22.4 °C	0.0 L/s	100 %	22.0 °C	22.1 °C	416 ppm	0.0 Pa	Off	Off	✓
VAV-3-22	Normal	Cool	23.1 °C	0.0 L/s	100 %	22.0 °C	22.3 °C	427 ppm	0.0 Pa	Off	Off	✓
VAV-LS (3-23 to 3-25)	Normal	Cool	22.1 °C	0.0 L/s	100 %	22.0 °C	22.8 °C	434 ppm	0.0 Pa	Off	Off	✓
VAV-3-30	Normal	Heat	20.8 °C	0.0 L/s	100 %	22.0 °C	21.5 °C	416 ppm	0.0 Pa	Off	Off	✓

Status Effect Key: Comms Fault | Fault | Alarm | Manual

(Summary by Level – First Column Should Have Area Served)

Level 1 VAVs

Level 2 VAVs

Level 3 VAVs

VAVs	Zone Temp SP	Zone Temp	SA Temp	RHWV	Air Flow	Requested Flow	Damper	VAV Mode	Blacklist
L3 Internal	22.5 °C	20.4 °C	20.7 °C	0 %	50.6 L/s	0.0 L/s	0 %	Unoccupied	✓
L3 North	22.5 °C	20.2 °C	21.0 °C	0 %	0.0 L/s	0.0 L/s	0 %	Unoccupied	✓
L3 South	22.5 °C	18.8 °C	21.0 °C	0 %	53.3 L/s	0.0 L/s	0 %	Unoccupied	✓

Status Effect Key: Comms Fault | Fault | Alarm | Manual

(Close up– First Column Should Have Area Served)

Level 1 VAVs

Level 2 VAVs

VAVs	Zone Temp SP	Zone Temp	SA Temp	RHWV	Air Flow	Requested Flow	Damper	VAV Mode	Blacklist
L2 Internal	22.5 °C	21.8 °C	22.2 °C	18 %	726.9 L/s	741.0 L/s	74 %	Occupied	
L2 North	22.5 °C	22.0 °C	24.7 °C	3 %	653.8 L/s	467.0 L/s	32 %	Occupied	
L2 South	22.5 °C	19.9 °C	35.6 °C	100 %	287.3 L/s	300.0 L/s	72 %	Occupied	

Status Effect Key: Comms Fault | Fault | Alarm | Manual

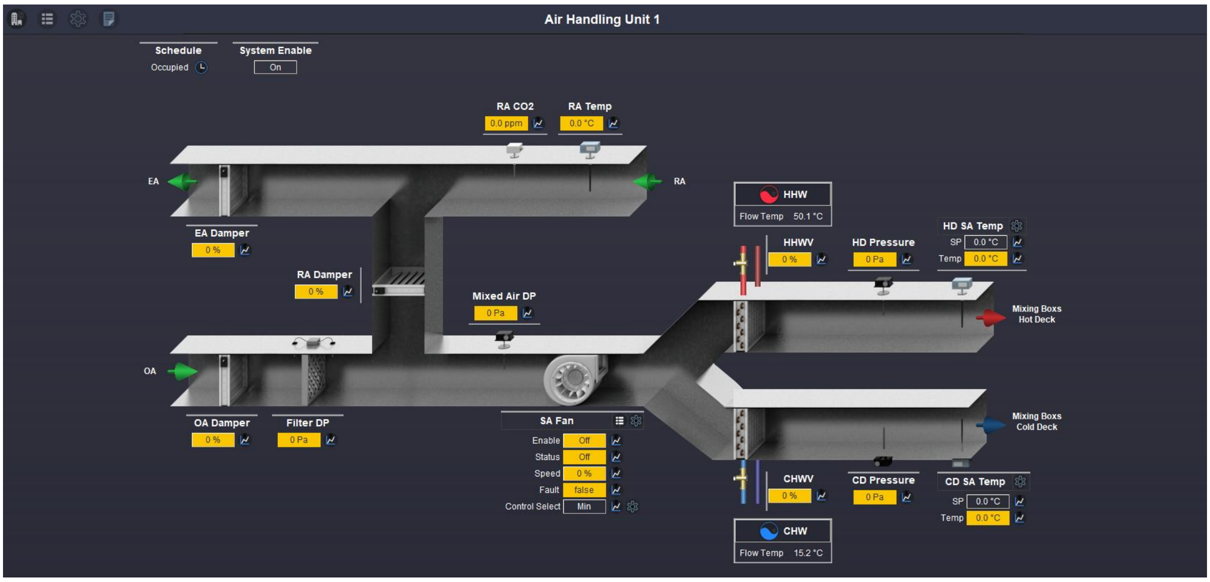
Level 3 VAVs

(Example of the expanding effect – First Column Should Have Area Served)

4.9 System Graphics

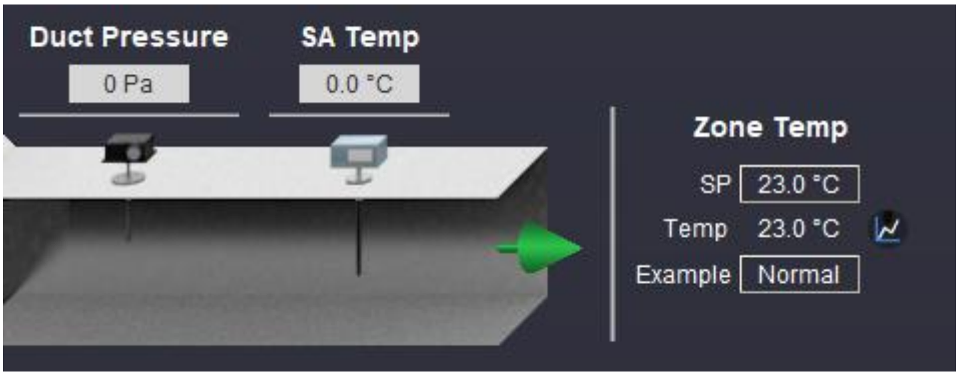
4.9.1 Schematic Page / Diagram

The Schematic Diagram page is a representation of the elements of a system using abstract, graphic symbols and visual language. The schematic page illustrates ductwork, for example air handling units (AHU), or pipework, such as heating hot water systems (HHW). The schematic must use elements from the kitPxGraphics module and have arrows indication the direction of flow. The schematic must not use elements from IAGraphicsLibrary or the IAN4GraphicsLibrary. The following section will outline graphics standard and summarise key elements that specific pages may require.



(AHU schematic)

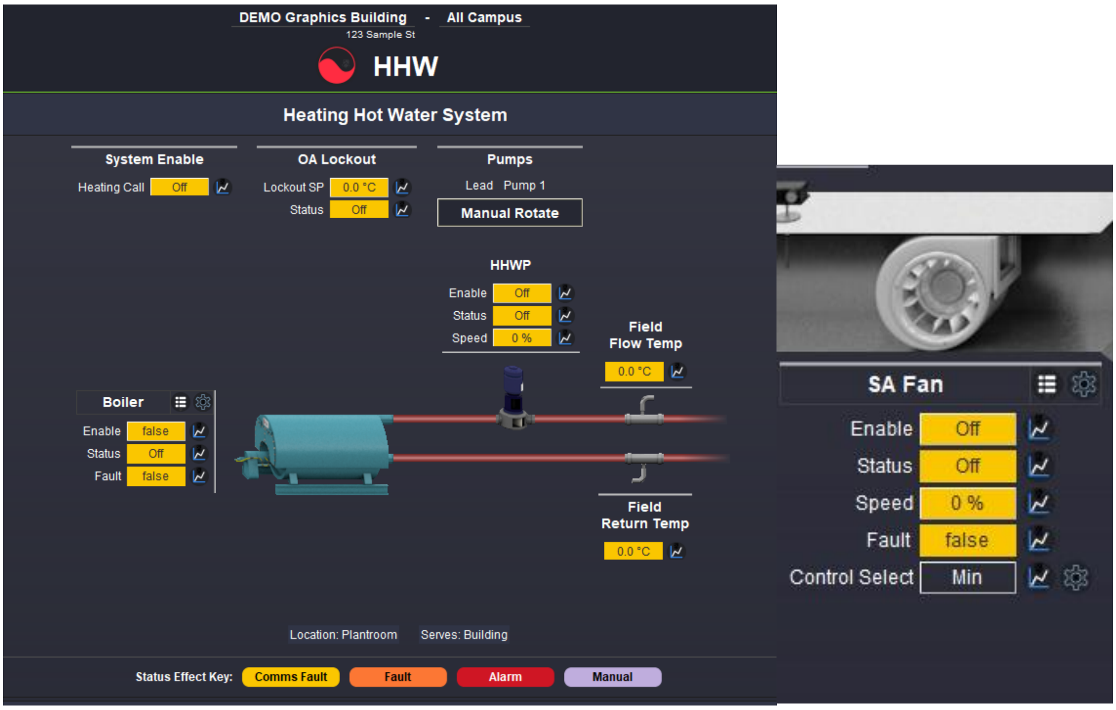
Schematics are to flow left to right, representing the transfer of thermal energy. Ductwork and pipework are to accurately reflect physical installation, for example a cooling coil before or after a heating coil. Equipment point values must be positioned next to the image representation. Depending on the location and orientation of the image representation, a GridPane will be positioned with a line indicating the associated equipment. In the example below, the duct sensor values are positioned above the probe. The line indicates the probe which is situated below the values. Note how the zone temp values use a left-hand side line, towards the arrow and discharge air. For more information about buttons and displaying points, see *Displaying Values, BoundLabels and Buttons.*



(GridPane examples properly indicating the appropriate visual elements in relation to the values)

4.9.2 High Level Interface (HLI)

High level interface (HLI) pages are to be accessed via a button in the title label. The HLI page must display points in a coherent manner following the graphics standards outlined in this document. For more information see *Displaying Values: BoundLabels and Buttons* section in this document.



(GridPane Header button. Button to Boiler HLI page - Left. Button to SA Fan HLI - Right)

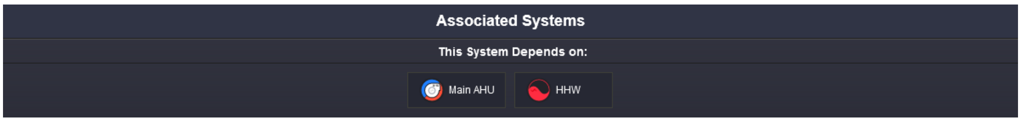
4.9.3 Associated / Dependent Units

Associated units or Zone Group must be positioned below the Status Effect Key. This section should display a button hyperlinked to the relevant unit, as well as indicate the room number, room name and asset number. At a minimum, the room number must be displayed. See examples below.

(FCU Zone Group section)

(AHU Associated Systems section)

MONASH University



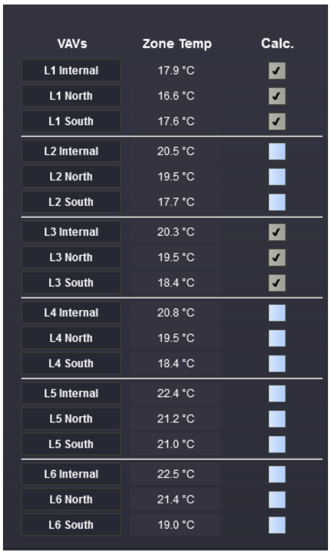
(VAV Reheat Associated Systems section)

4.9.4 Key Elements: Control Mode (Blacklists)

A Control Mode pop-up page is required where systems control using feedback from multiple points. For example, VAV temperatures for AHU supply air control or CHW valve positions for cooling calls.

A button linking to the Control Mode page must be accessible from the schematic and the settings page. The button name must be consistent across the pages. The control page must have an option to set ‘minimum’, ‘maximum’ or ‘average’ for control and calculations. The graphic must show point names, values and checkboxes. The check boxes must be configured where no check denotes that a value is blacklisted.

Note, the settings page is accessed via the ‘Settings’ main content title button. The document covers the content title buttons and settings page in a separate section.

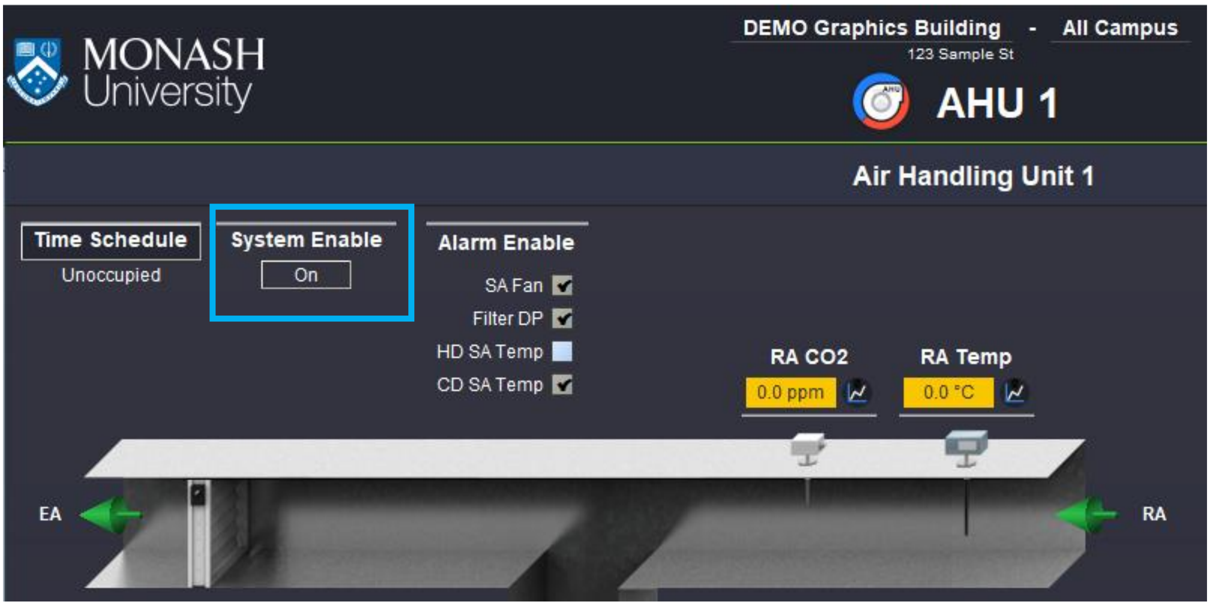


(Blacklist example)

4.9.5 Key Elements: System Enable

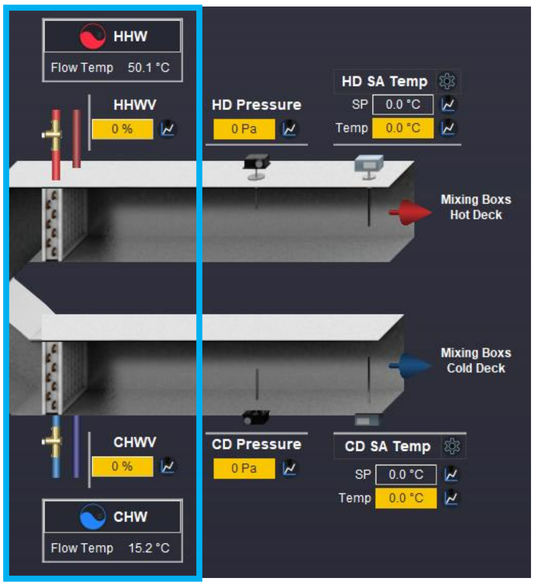
Central systems or ones that have a particular start process required a ‘System Enable’ point to be displayed on the graphics. For example, a CHW system enable 1 point would turn off or on pumps, isolate valves and initiate any other required items to safely turn on and off chiller 1. Another example is an AHU system enable point, which would open any isolation dampers before turning on. If there is doubt, the ‘System Enable’ should be added unless clarified with the BAS Administrator.

The system enable point is to be displayed in the top left of the schematic canvas pane, in the style illustrated below. Note, all systems related points are displayed above the duct or pipework with a silver GridPane line on top.



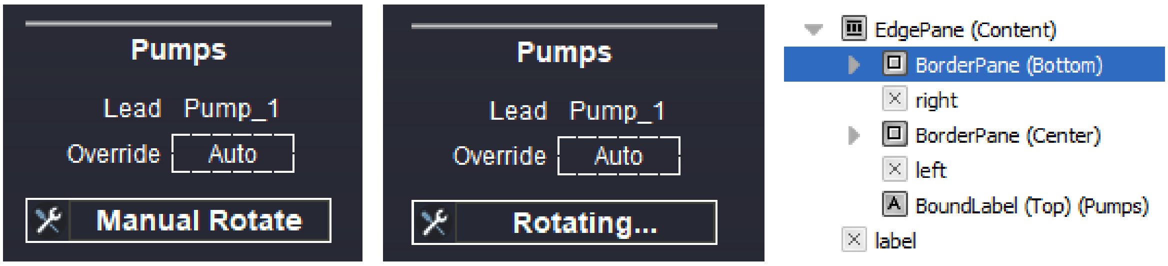
4.9.6 Key Elements: Coils

For all graphics that display coils, the graphic must indicate whether it is a 2 or 3-way valve. The graphic must also display the associated system, HHW, CHW, as well as the supply temp. See example below. Note the HHW and CHW GridPane title is a button style, which hyperlinks to the relevant system pages. In addition, the information is displayed with a silver border. This is because the system is located in a different location to the elements in the schematic.



4.9.7 Key Elements: Manual Rotate

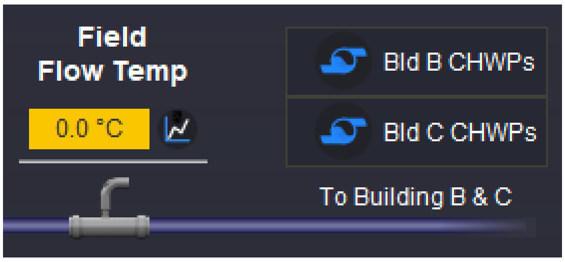
All lead-lag and duty-standby systems must have a manual rotate option, as well as auto and override options; For example, Auto, System 1 Lead, System [x] Lead. Below is an example of a GridPane with a Manual Rotate trigger. When the trigger is fired, the button text will animate.



(Manual Rotate button - Left. Animated text - Centre. Location in GridPane - Right)

4.9.8 Key Elements: Equipment Serving Multiple Buildings

All systems or equipment that serve other locations must allow for navigation between the two pages. The button must be named clearly, be consistent across pages and be placed in an appropriate and logical location.



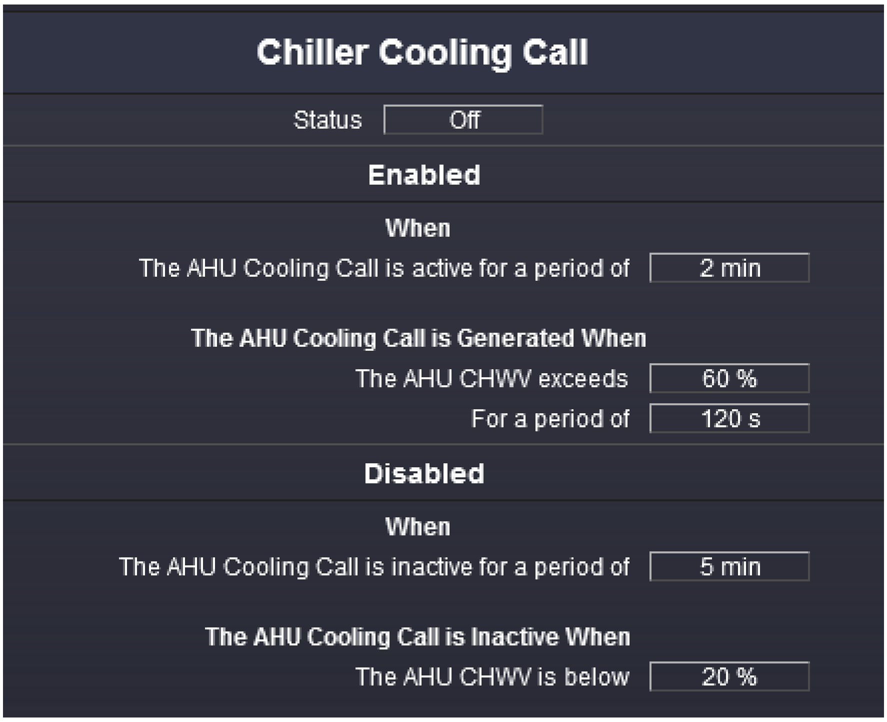
(CHW System supply. Bld B and C CHWP hyperlink buttons are visible)

4.10 Settings

The settings page is a pop-up with information a BAS engineer or mechanical technician may need access in order to troubleshoot and verify a units function. A settings page is indicated with a grey cog icon. Depending on where the icon is placed on the graphics, will infer the type of information that is to be displayed. For example, information pertaining to the system would be viewed via a button on the main content title. Information pertaining to a particular piece of equipment will be in the GridPane title. Information pertaining to a particular point may be found to the right of the BoundLabel.

All of the examples mentioned must use an icon in the appropriate locations for the information that is being displayed. Information may include, tuning values, setpoints, time delays, manual changeovers, economy settings, outside air lockouts, unit enable parameters, blacklists etc. The information on the settings page must be laid out in a logical and clear manner. For instance, if a unit enable is off, it should be clear why, as well as what conditions need to be met for it the unit to start.

(Example control settings information)

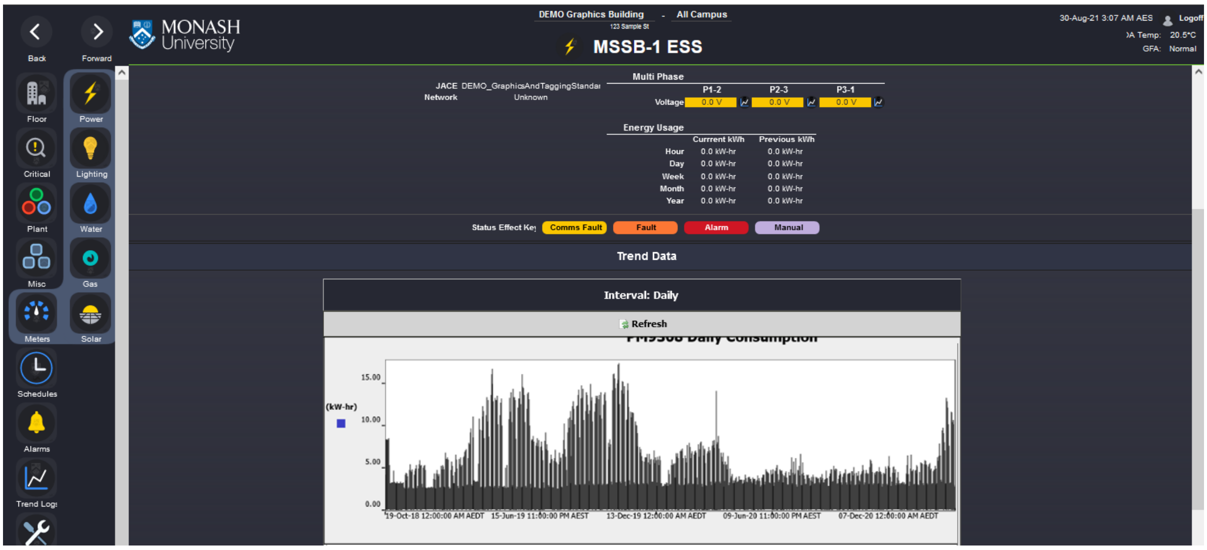


(Example cooling call information)

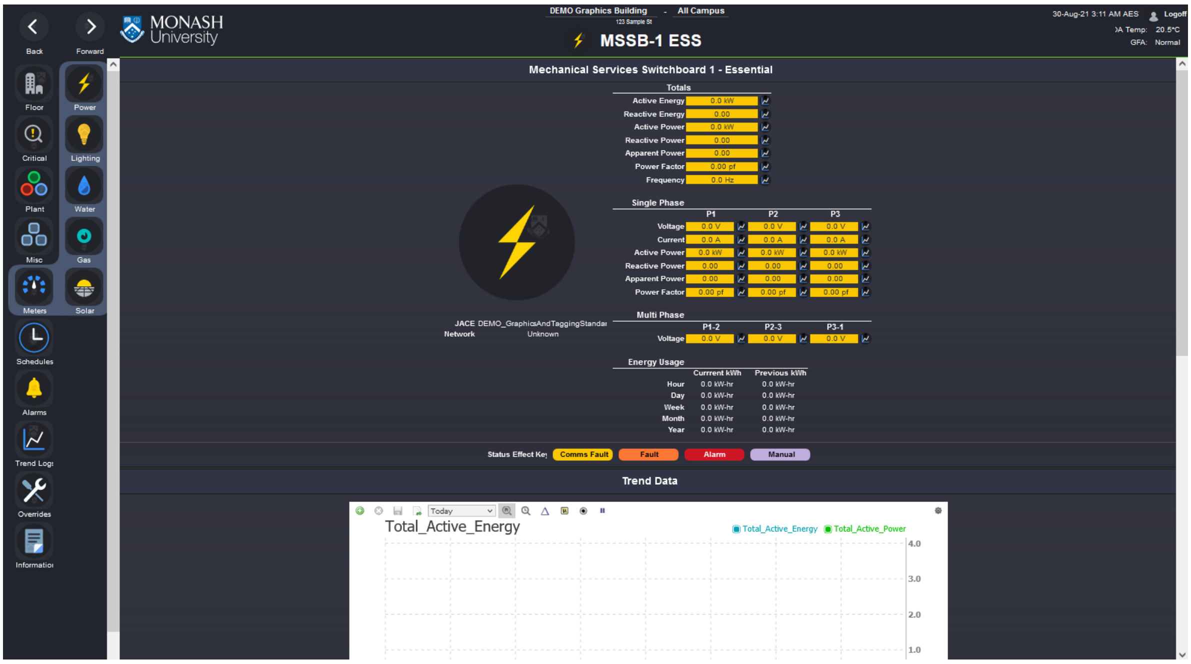
4.11 Meters

The Meter page displays meter values and histories for all logged points. The page has a ‘Trend Data’ section where the graphic time period can be adjusted.

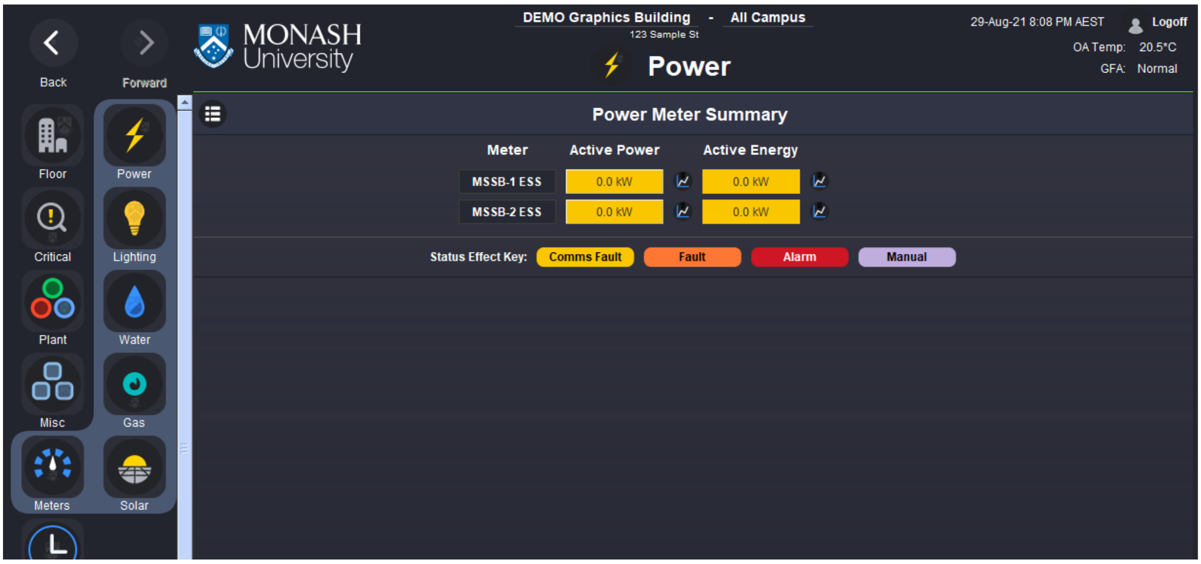
(Meter page)



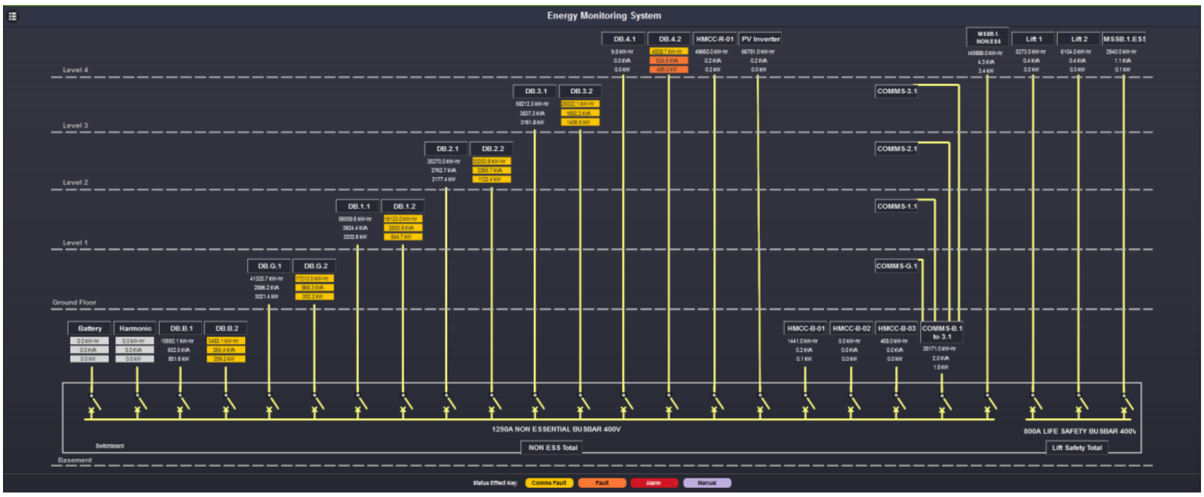
(Trend Data showing adjustable interval and time period chart)



(Trend Data showing standard chart)



(Power meter tabular summary view)

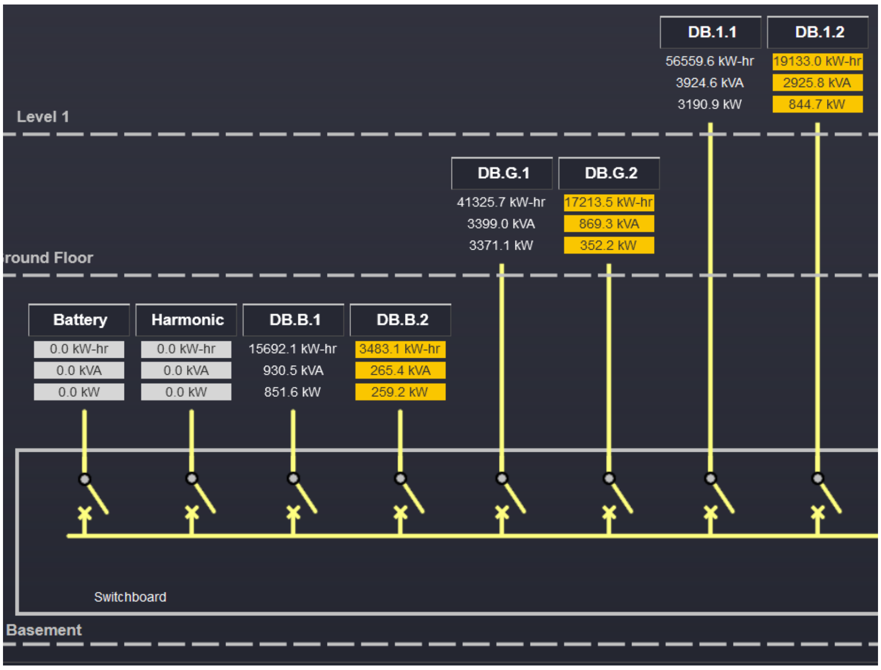


(Single-line diagram - SLD)

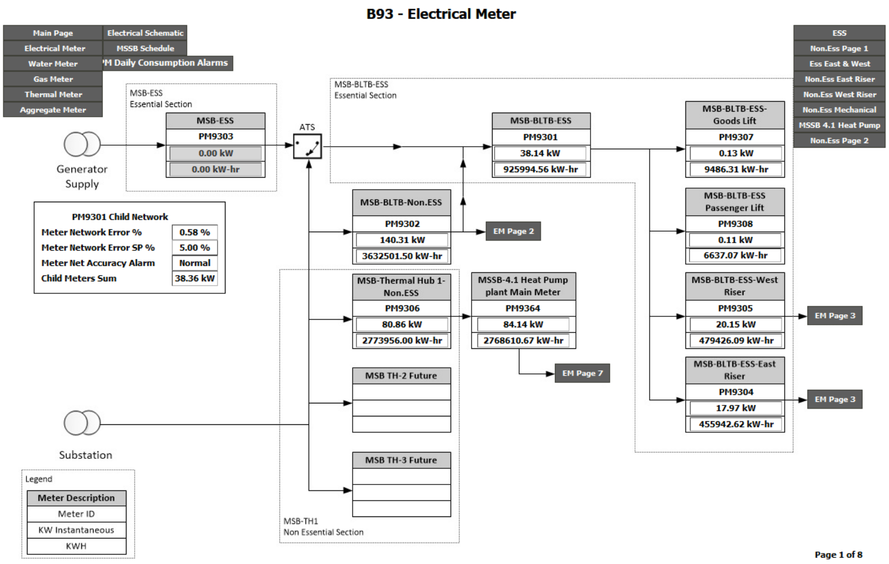
4.11.1 Content: Single Line Diagrams (SLD)

Below is are examples of single line diagrams (SLD). The first example is a page that is built using shapes, paths, labels and buttons. The second example is an existing page that uses an image as a background and places labels and buttons on top. Both SLD methods are accepted so long as they meet the page requirements and is as consistent as possible to the graphic standard.

The SLD page must show instantaneous value with history button. If the instantaneous value is not available, the current day usage should be displayed. The SLD must accurately represent the electrical power system.



(SLD Close up – does not display chart buttons in example)



(Example of existing SLD)

4.12 Building Information

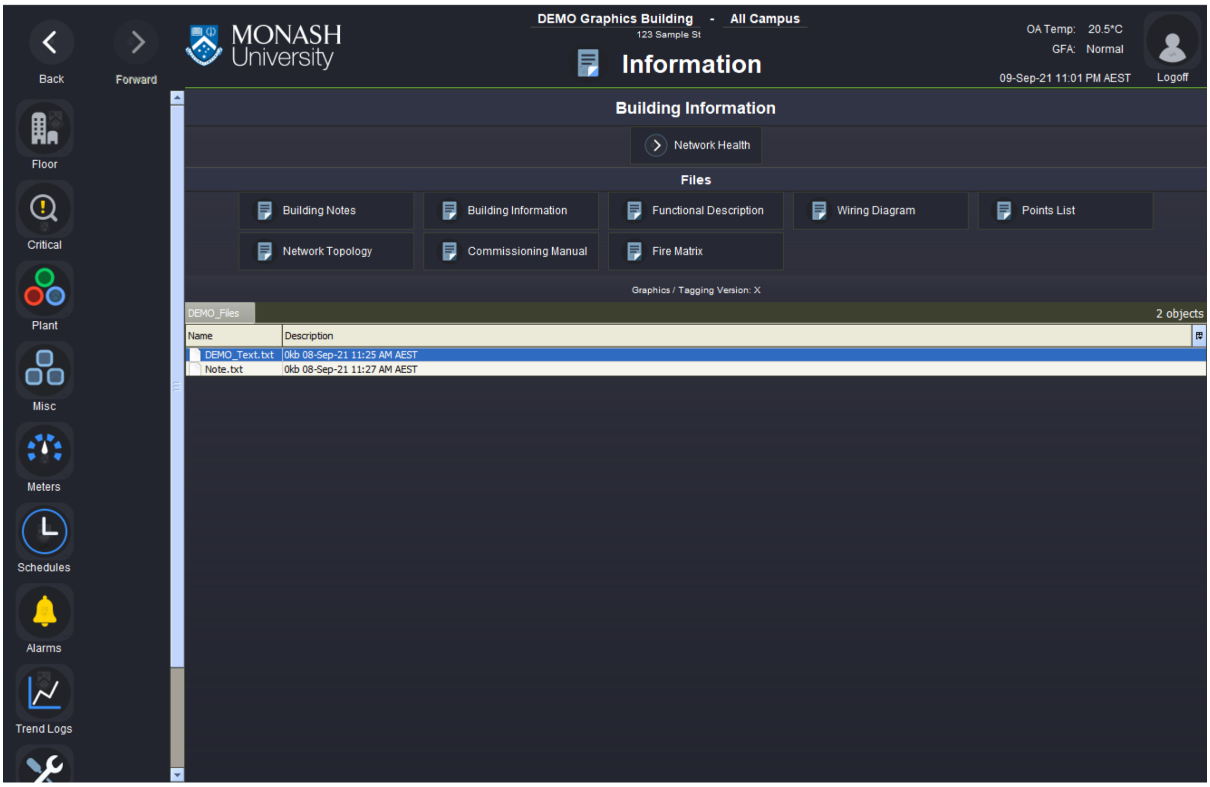
The Information page displays building information and files. The network health page is accessible from this view. The image below illustrates a main content section with a separate sub content section called ‘Files’.

Information in the main content section consists of;

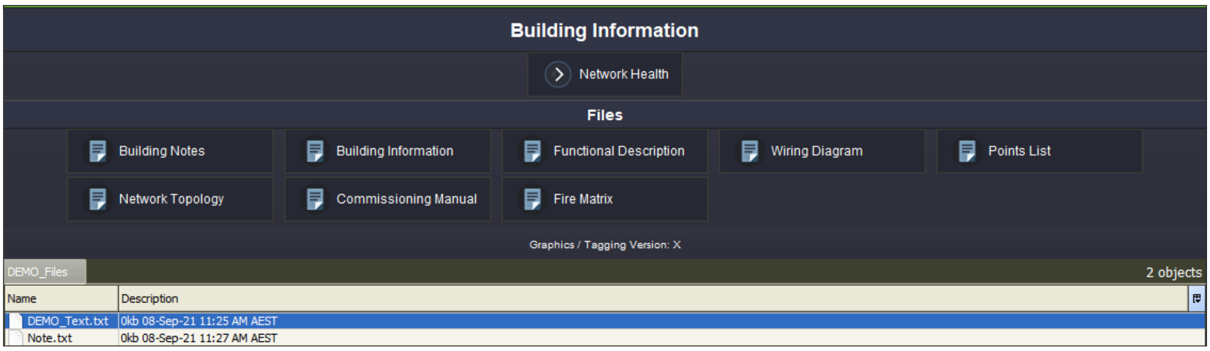
- Network health (links to page)

- Building Notes (pop up text file)
- Building information (pop up text file)
- Functional Description (links to document)
- Wiring Diagrams (links to document)
- Points List (links to document)
- Network Topology (links to document)
- Commissioning Manual (links to document)
- Fire Matrix (links to document)
- File Directory

The file directory location must be updated to an appropriate path, for example a building specific folder. Where there is no documentation available, remove the button in the Files section and update column count appropriately. Where an additional file button is required, a button may be added. Where additional non-document page links are required, add them in the section where Network Health is located.



(Information page)

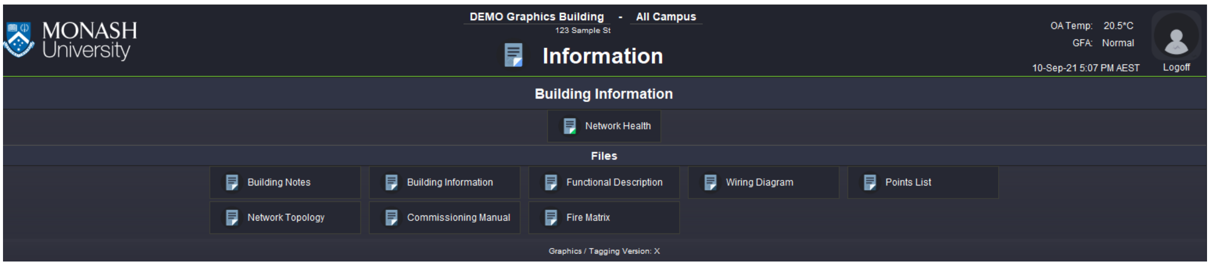


(Main content close up)

4.13 Network Health

The network health page contains data about the devices on the network. It can be navigated to by the floor selector page or the building information page. If multiple network health pages exist, such as for different priorities of devices, the links must navigate the user to the most important page first.

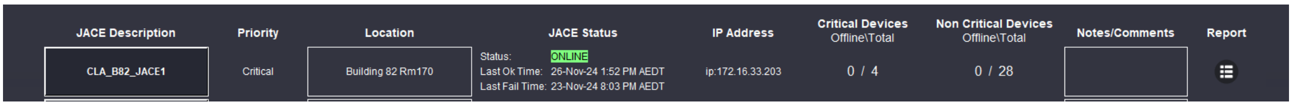
Network health information needs to be provided on the building level graphics as well as the campus level graphics.



(Example of network health navigation button)

4.13.1 Summary Row

The network health page displays a list of relevant network devices. Each row displays a set of values that identify the device as well as values that indicate how many are operational. The data is contained in a summary table and follows the same standards for layout and styling. The button in the first column contains a hyperlink to the device being displayed. For ease of setup, each row may be setup as a pxInclude that accepts a list of ords that correspond to the different label values.



JACE Description	Priority	Location	JACE Status	IP Address	Critical Devices Offline\Total	Non Critical Devices Offline\Total	Notes/Comments	Report
CLA_B82_JACE1	Critical	Building 82 Rm170	Status: ONLINE Last Ok Time: 26-Nov-24 1:52 PM AEDT Last Fail Time: 23-Nov-24 8:03 PM AEDT	ip:172.16.33.203	0 / 4	0 / 28		

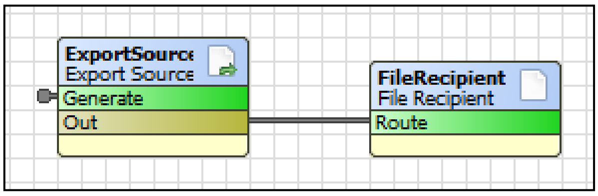
(Example of the network health page)

The information for each network device is to show JACE name, criticality, location, last ok time, last failure time, IP addresses, critical devices offline statistics, non-critical device offline statistics and an editable field to write notes. See the image above for column names and table cell layout. In addition, an icon button is used to open a report pop up page. The report pop-up displays a dynamic text file that displays information such as device name, last ok time, status, device address, device location and critical (yes/no).

4.13.2 Device Report

The popup works by retrieving data that is generated from a query and placing it into a report file that is created on the JACE. This file is imported to the supervisor. This file method is used since specific device data cannot be retrieved from the supervisor directly. The default text file view is an edit view. The icon button should open a read only version of the text file. Therefore, the button must be configured where the pop-up ord is changed from default view to the "AX Text File Viewer".

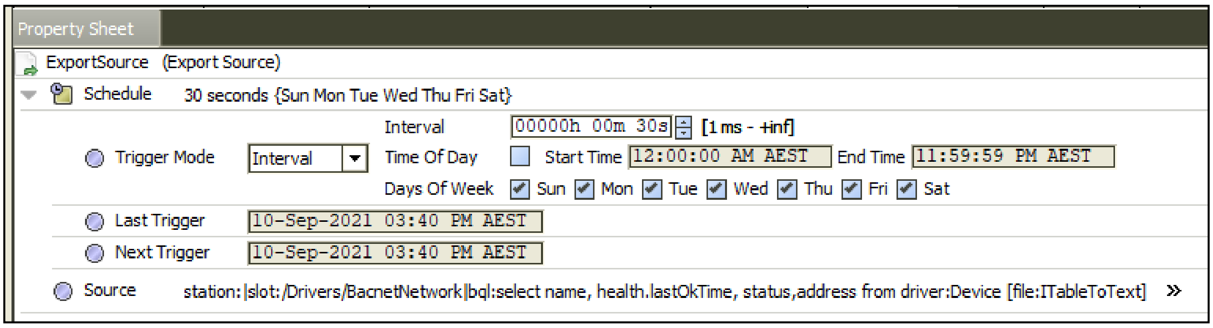
To create the JACE's network health pop up file you will need to connect to the relevant JACE. You must navigate to the JACE's report service. From here you will need add an ExportSource to gather the data and a FileRecipient to export the data.



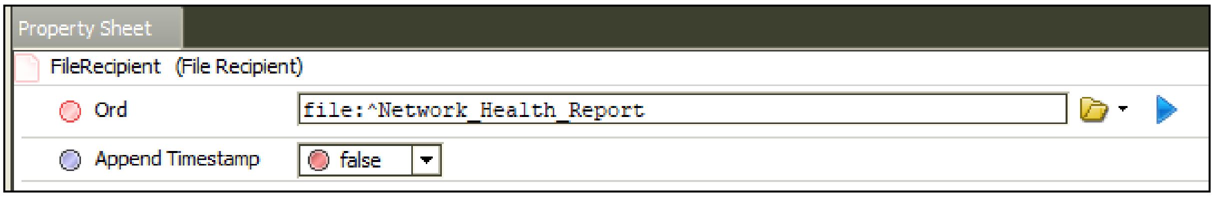
The ExportSource should have a schedule set to trigger updates at a reasonable pace, for example once or twice a minute. The source of the data comes from a bql table which should retrieve the name, ip address, and the devices status at the minimum. The data in the table must be formatted correctly here since the supervisor will not be able to adjust any data from the file, only read it. This means that column names and row orders need to done via the bql query. Additional information can be queried where devices are tagged.

For more information on BQL view the Niagara help documents:

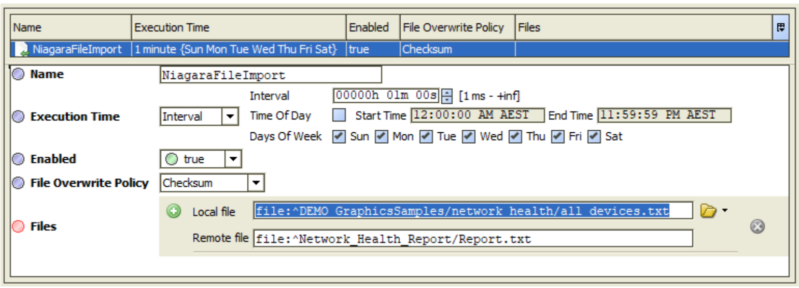
local:|***module***://***docDeveloper***/***doc***/***bql***.***html***



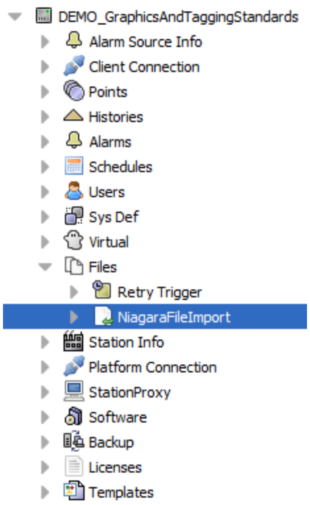
The FileRecipient must save the file to a unique file that is named according to its purpose. If the query is for all devices, then the file should be named “network_health_devices”. If an offline device report is required, a file name such as “network_health_devices_offline” should be used to differentiate it from the main report. Do not append the timestamp to the filename. The file requires a consistent file and ord name to read the data from.



To import the file on the supervisor, go to the NiagaraFileDeviceExt of the station on the supervisor and add a NiagaraFileImport object. Set the execution time to an appropriate time, such as 1 minute. Input the correct local and remote file paths ords. For the local file path, make sure to put it in the building graphics folder.



(Add NiagaraFileImport pop-up window)



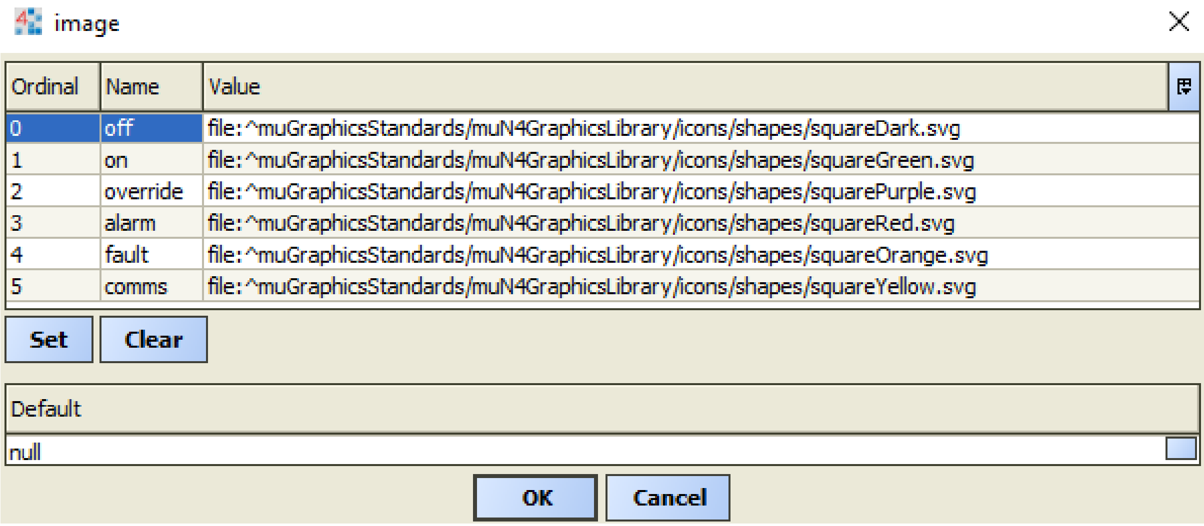
(NiagaraFileDeviceExt with added NiagaraFileImport)

Name	Last Ok Time	Status	address	Enabled	Device Location	Critical
B82_Generator_HLI	28-Apr-22 5:26 PM AEST	{unackedAlarm}	99991:172.16.33.197:47808	true	MSSB-1 (Rm421)	Yes
KNX_Dali_Gateway	28-Apr-22 5:26 PM AEST	{ok}	99991:172.16.33.33:47808	true	MSSB-1 (Rm421)	No
dev_1082147	28-Apr-22 5:26 PM AEST	{ok}	18201:2f	true	MSSB-1 (Rm421)	No
dev_1082148	28-Apr-22 5:26 PM AEST	{ok}	18201:30	true	MSSB-1 (Rm421)	No
dev_1082149	28-Apr-22 5:26 PM AEST	{ok}	18201:31	true	MSSB-1 (Rm421)	No
dev_1082150	28-Apr-22 5:26 PM AEST	{ok}	18201:32	true	DDC Panel 3 (Rm378)	Yes
dev_1082151	28-Apr-22 5:26 PM AEST	{ok}	18201:33	true	DDC Panel 3 (Rm378)	Yes
dev_1082152	28-Apr-22 5:26 PM AEST	{ok}	18201:34	true	DDC Panel 3 (Rm378)	Yes
dev_1082153	28-Apr-22 5:26 PM AEST	{ok}	18201:35	true	DDC Panel 3 (Rm378)	Yes
dev_1082154	28-Apr-22 5:26 PM AEST	{ok}	18201:36	true	DDC Panel 3 (Rm378)	Yes
dev_1082155	28-Apr-22 5:26 PM AEST	{ok}	18201:37	true	DDC Panel 3 (Rm378)	Yes
dev_1082156	28-Apr-22 5:26 PM AEST	{ok}	18201:38	true	DDC Panel 3 (Rm378)	Yes
dev_1082157	28-Apr-22 5:26 PM AEST	{ok}	18201:39	true	DDC Panel 3 (Rm378)	Yes
dev_1082158	28-Apr-22 5:26 PM AEST	{ok}	18201:3a	true	DDC Panel 3 (Rm378)	Yes
dev_1082159	28-Apr-22 5:26 PM AEST	{ok}	18201:3b	true	DDC Panel 3 (Rm378)	Yes
dev_1082160	28-Apr-22 5:26 PM AEST	{ok}	18201:3c	true	DDC Panel 3 (Rm378)	Yes
dev_1082161	28-Apr-22 5:26 PM AEST	{ok}	18201:3d	true	DDC Panel 3 (Rm378)	Yes
dev_1082162	28-Apr-22 5:26 PM AEST	{ok}	18201:3e	true	DDC Panel 3 (Rm378)	Yes
dev_1082163	28-Apr-22 5:26 PM AEST	{ok}	18201:3f	true	DDC Panel 3 (Rm378)	Yes
dev_1082164	28-Apr-22 5:26 PM AEST	{ok}	18200:40	true	Ceiling Space (RmG73)	No
dev_1082165	28-Apr-22 5:26 PM AEST	{ok}	18200:41	true	Ceiling Space (RmG73)	No
dev_1082166	28-Apr-22 5:26 PM AEST	{ok}	18200:42	true	Ceiling Space (RmG73)	No
dev_1082167	28-Apr-22 5:26 PM AEST	{ok}	18200:43	true	Ceiling Space (RmG73)	No
dev_1082168	28-Apr-22 5:26 PM AEST	{ok}	18201:44	true	On VAV-7-2	No
dev_1082169	28-Apr-22 5:26 PM AEST	{ok}	18200:45	true	On VAV-7-3	No
dev_1082170	28-Apr-22 5:26 PM AEST	{ok}	18200:46	true	On VAV-7-4	No
dev_1082171	28-Apr-22 5:26 PM AEST	{ok}	18200:47	true	On VAV-7-5	No
dev_1082172	28-Apr-22 5:26 PM AEST	{ok}	18200:48	true	On VAV-7-6	No
dev_1082177	28-Apr-22 5:26 PM AEST	{ok}	18200:4d	true	On VAV-7-7	No
dev_1082181	28-Apr-22 5:26 PM AEST	{ok}	18200:51	true	On VAV-7-8	No
dev_1082600	28-Apr-22 5:22 PM AEST	{ok}	99991:172.16.33.158:47808	true	B38 (RmPLG05)	No

(Example of a B82 JACE Report)

4.14 Icon Summary View (Legacy)

The icon summary is a previous standard used to navigation and view individual equipment pages. It is a unique page that is used to quickly determine whether a unit is running, or has points that are in override, fault, comms error or alarm. The button has a background image that is linked to an EnumWritable point that is specifically for that unit. The EnumWritable's output is animated to the Ordinal values and images shown below. The Enum out value is determined by graphics logic that is located in a 'graphicsLogic' folder structure. The logic uses components from the baja, vykonPro, kitControl and control modules, with the purpose of creating a hierarchy in which it will decide which colour value will take precedence over another. For example, if a unit is running (green), but is in alarm (red), the enum value and the background image associated with it will be shown (Ordinal 3 - Red).



image

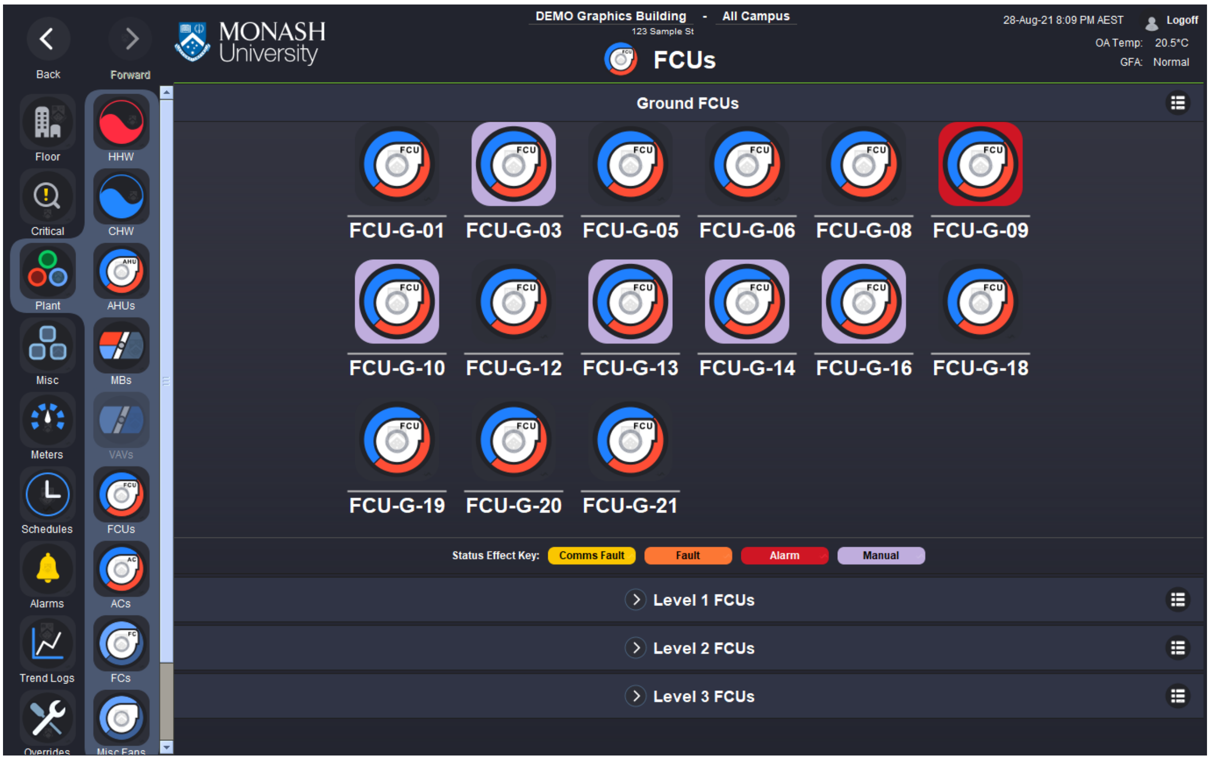
Ordinal	Name	Value
0	off	file:^muGraphicsStandards/muN4GraphicsLibrary/icons/shapes/squareDark.svg
1	on	file:^muGraphicsStandards/muN4GraphicsLibrary/icons/shapes/squareGreen.svg
2	override	file:^muGraphicsStandards/muN4GraphicsLibrary/icons/shapes/squarePurple.svg
3	alarm	file:^muGraphicsStandards/muN4GraphicsLibrary/icons/shapes/squareRed.svg
4	fault	file:^muGraphicsStandards/muN4GraphicsLibrary/icons/shapes/squareOrange.svg
5	comms	file:^muGraphicsStandards/muN4GraphicsLibrary/icons/shapes/squareYellow.svg

Set Clear

Default

null

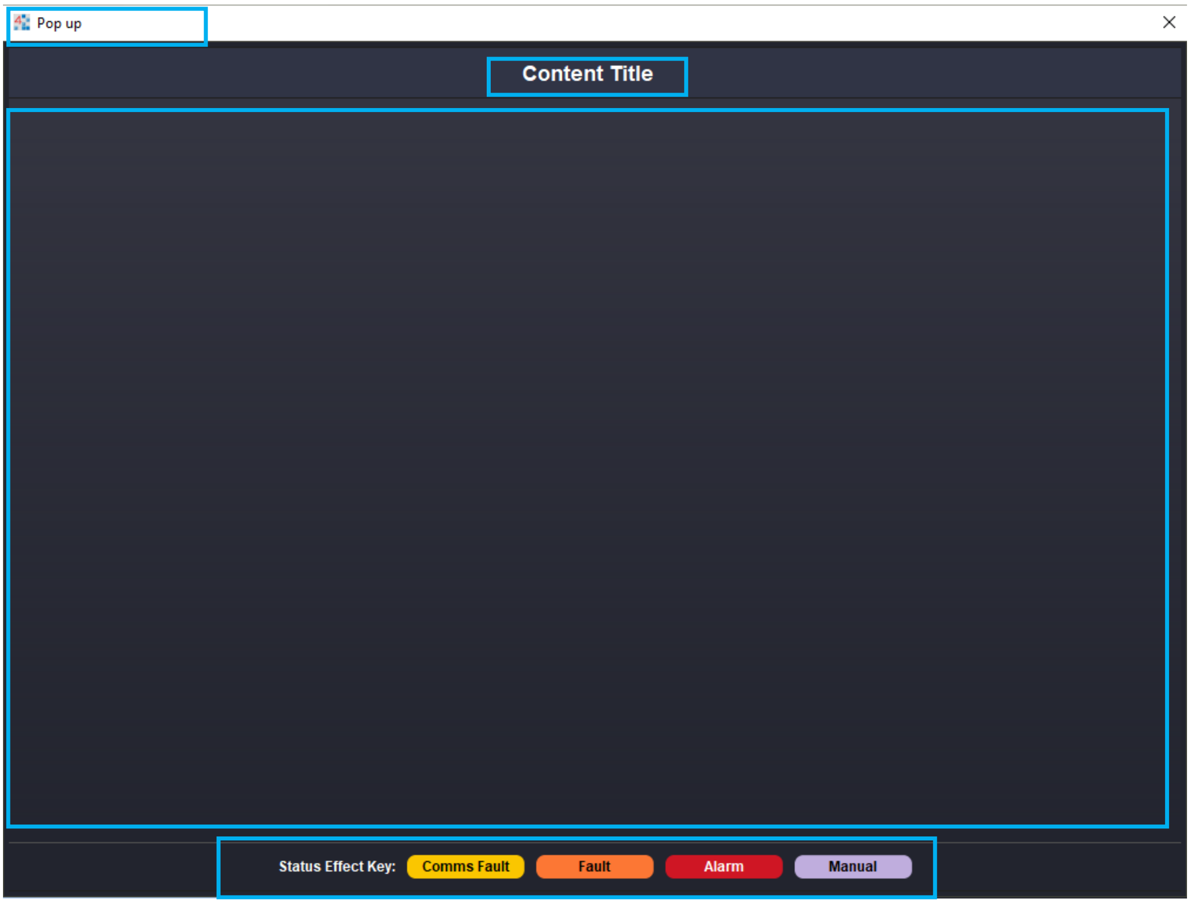
OK Cancel



5 POP-UP WINDOWS

Pop-up windows are used for displaying information such as settings, notes and charts. These pop-up windows do not require navigation menus or headers. The intent is to allow users to open relevant information and exit the pop-up.

A pop-up window must not hyperlink to other pages, but it may open another pop-up window. An example is a settings pop-up window with values and chart buttons. Clicking the chart button will open a pop-up of the relevant chart. All history chart popups may use the default popup size, 800 x 600px. All other pop-up windows are to open to a minimum size of 1024 x 768px. Ideally, the popup content does not exceed the pop-up window size. Where this occurs, ensure a scroll pane is used where appropriate. All pop-ups must have a practical pop-up name and title. A template is available, comprising of a content title, content section and status effect key, as per the image below. The image below indicates key areas of the popup window.



6 DISPLAYING VALUES: BOUNDLABELS AND BUTTONS

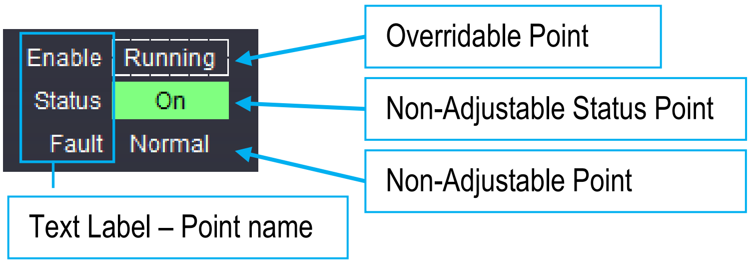
A BoundLabel is a widget that has the capacity to display images and information, as well as animate specific properties based off Ords. The BoundlLabel properties must be configured by the user in accordance with the graphic standard. The widget is found in the kitPx module. Pre-configured BoundLabel templates are available on the Monash supervisor.

A button is use to navigate to different pages and can be displayed as icons, icons and text or just text. Buttons can also be used for control, such as a manual rotate trigger in a lead-lag system. Pre-configured button templates are available on the Monash Supervisor.

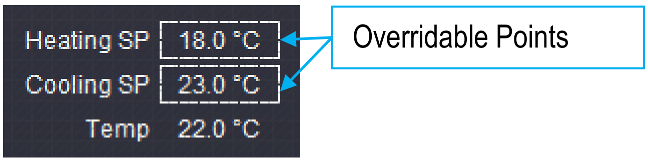
The section below outlines the specifications for BoundLabels and buttons.

6.1 Configuration for Point Types

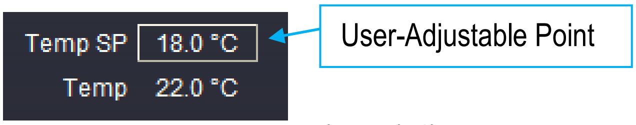
There are 4 main BoundLabel styles that need to be configured across the Monash University graphics. These are: text label, overridable, user adjustable, non-adjustable and non-adjustable status points. Below are examples identifying the different styles.



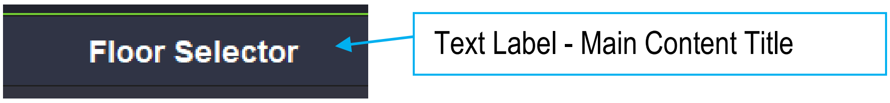
(Example 1)



(Example 2)



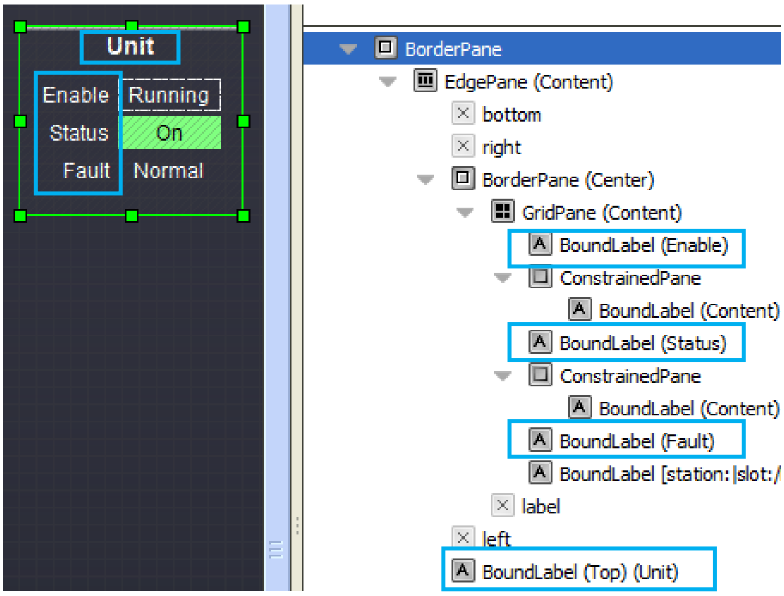
(Example 3)



(Example 4)

6.2 Text Label: Overview

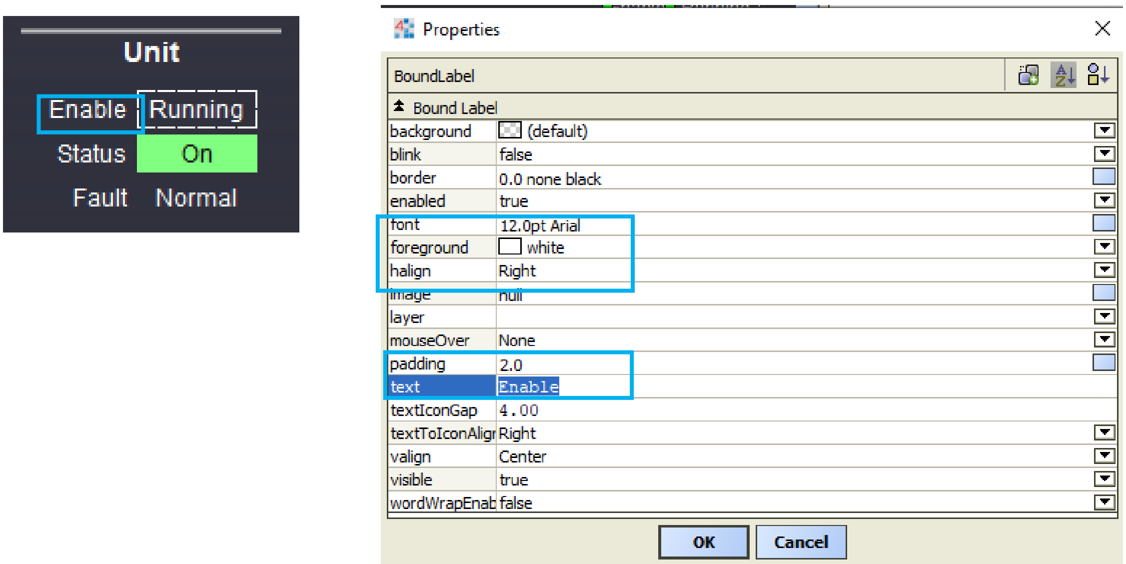
Text Labels refers to a BoundLabel that has been configured in order to show text to help clarify elements on a page. These BoundLabel's can be in the form of titles or labels next to point values. A label from the baja module can be used, but has less modifiable properties than a BoundLabel.



(Example indicating the BoundLabels configured for text only)

6.3 Text Label: Default

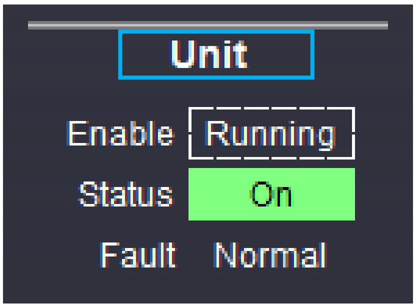
The image below illustrates the minimum requirements for a default text label. This style is to be used for all labels associated with adjacent point values unless otherwise specified. Text field is to be updated as required.

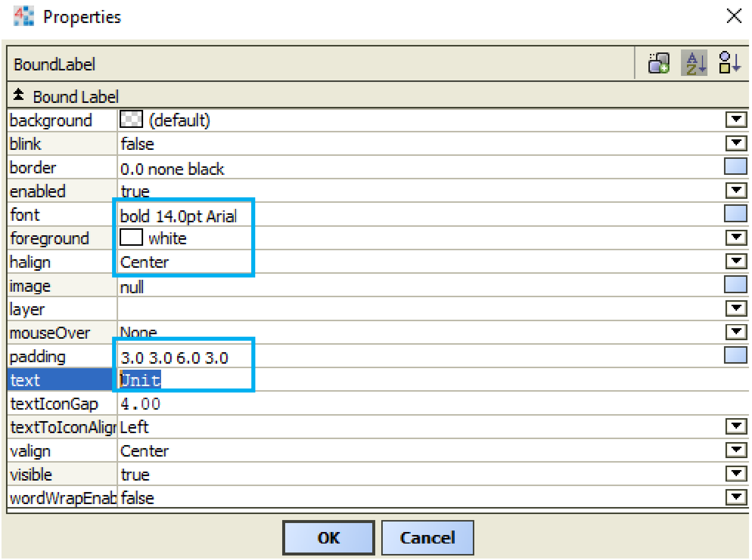


(Text only BoundLabel properties unless otherwise specified)

6.4 Text Label: Schematic GridPane Title

The image below illustrates the minimum requirements for a text label used for a schematic GridPane title. Text field is to be updated as required.



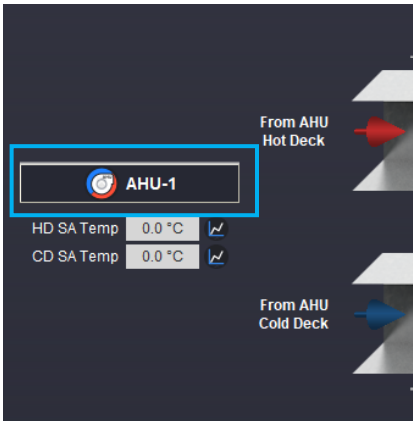


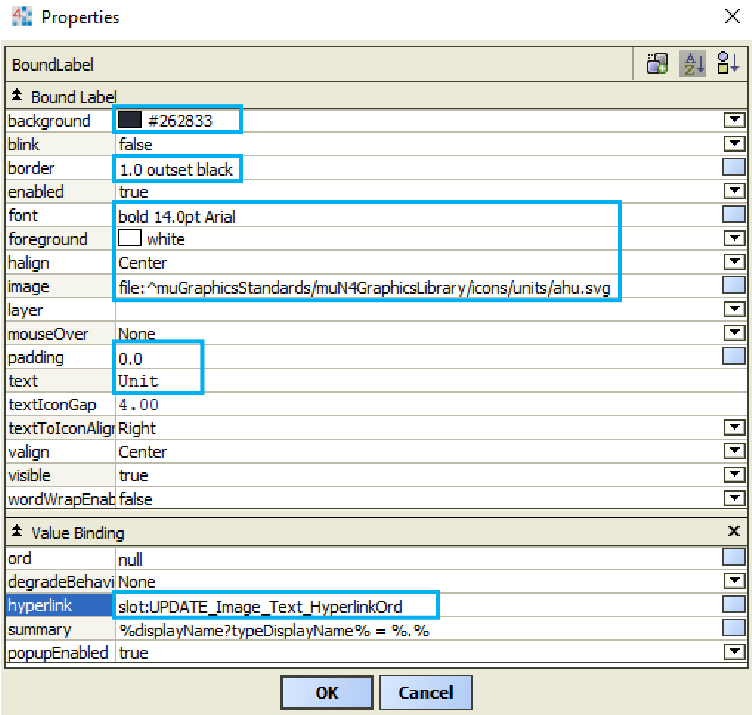
(Text only schematic GridPane title properties)

6.5 GridPane Title Button

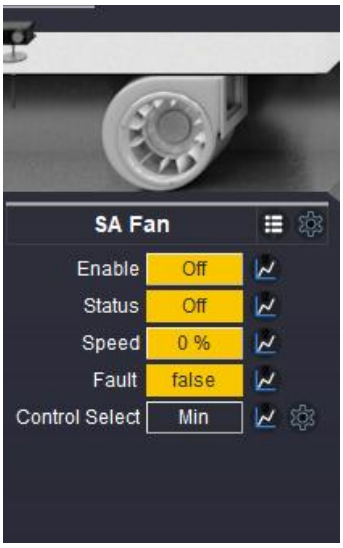
GridPane Title buttons are used to navigate to a unit that is not part of the current schematic. It must have an appropriate icon. For all equipment specific settings or high-level interface (HLI) pop-up pages, a unique title is required.

Below is an example of an AHU that is serving a unit, but is not part of the unit or located on the same floor.





(GridPane hyperlink title properties)

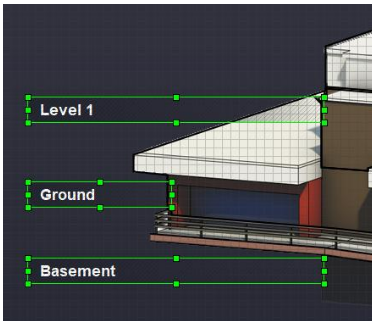


(Example button for HLI and equipment specific Settings pop-up)

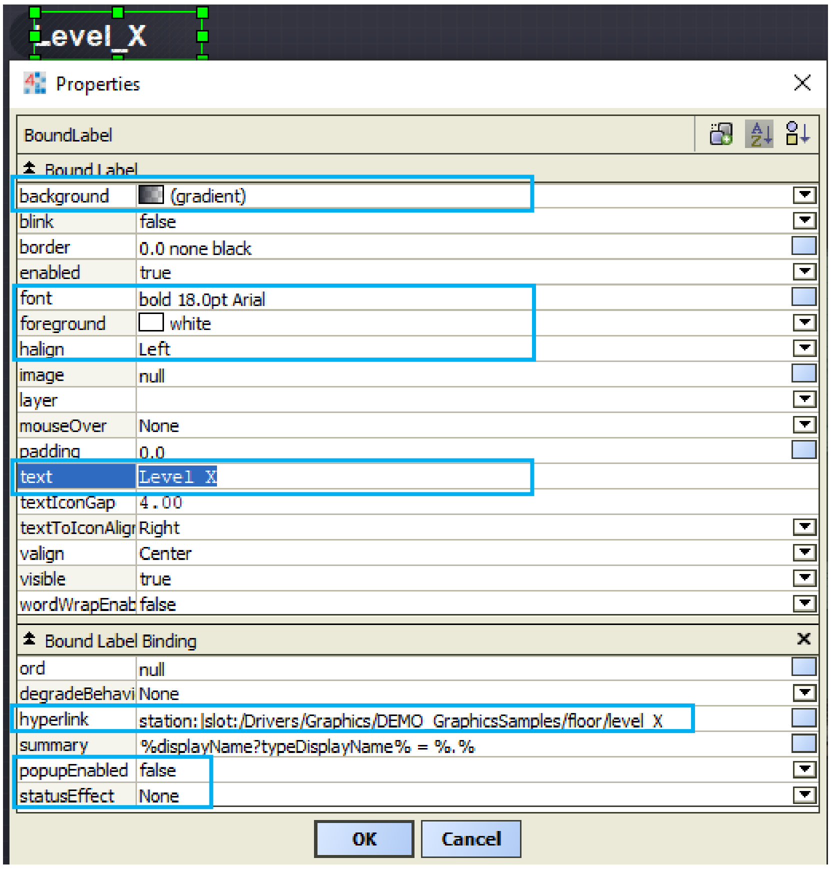
6.6 Text Label: Floor Selector

The BoundLabel is used as a way to indicate the levels of a building and navigate to available Floor pages. The building image used must be a suitable visual representation of the building in order for BoundLabels to be placed adjacent to the corresponding level. The text must indicate the page that the label navigates to. Left and a right variation are to be used as needed. It is recommended that all labels are aligned to the left of the building. Where it is more practical to have labels to the right of the building image, the correct label configuration and alignment must be implemented. The information below illustrates the minimum requirements for a text label used for the Floor Selector BoundLabel.

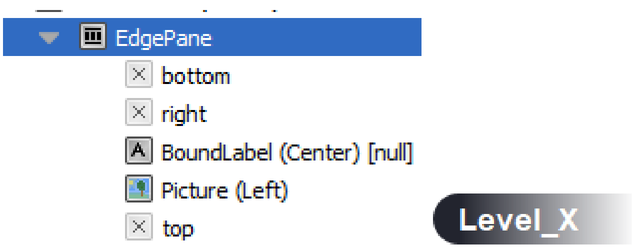
(Floor Selector Page)



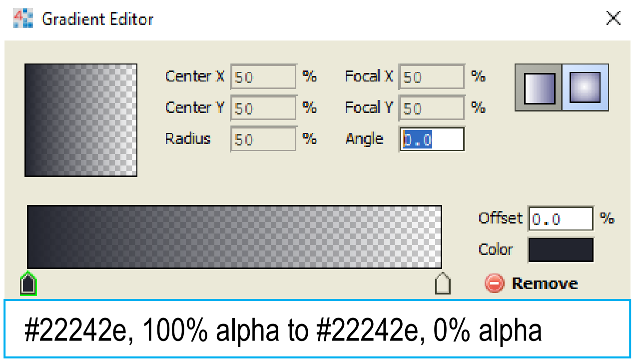
(Floor Selector BoundLabels)



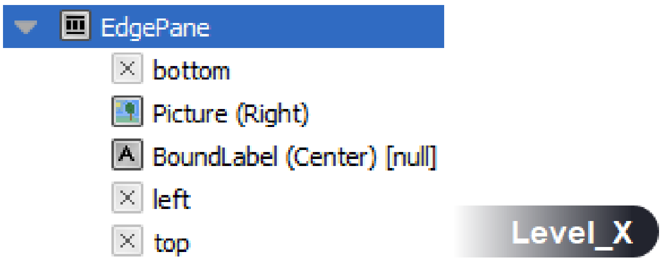
(Left Floor Selector BoundLabel properties)



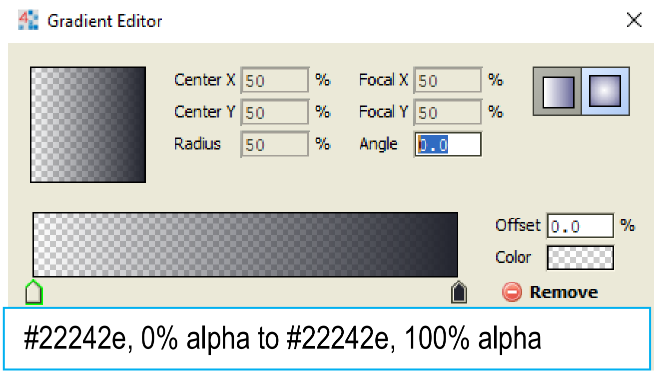
(Left widget configuration)



(Left label configuration)

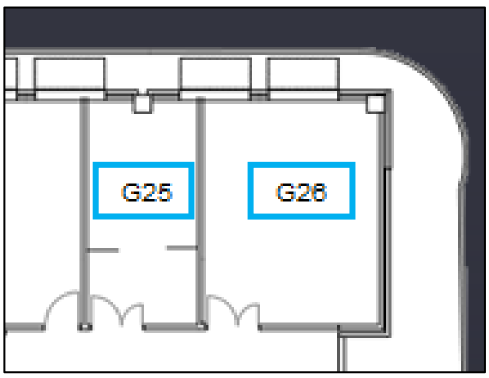


(Right widget configuration)

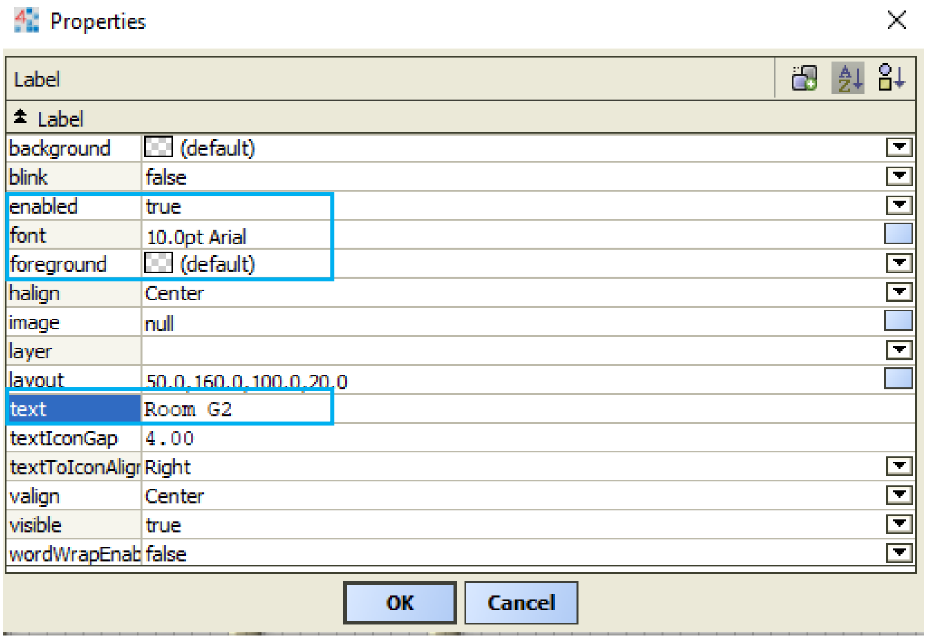


(Right label configuration)

6.7 Text Label: Room Name Label



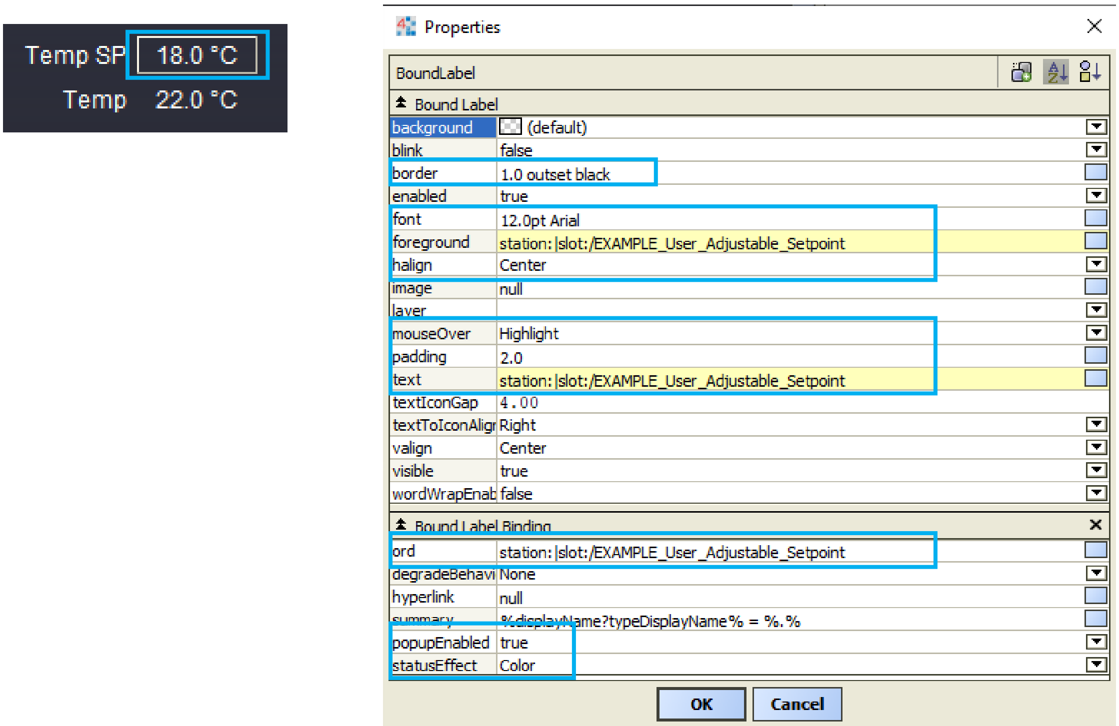
(Floorplan room labels)



(Text only floorplan properties)

6.8 Display Points: User Adjustable

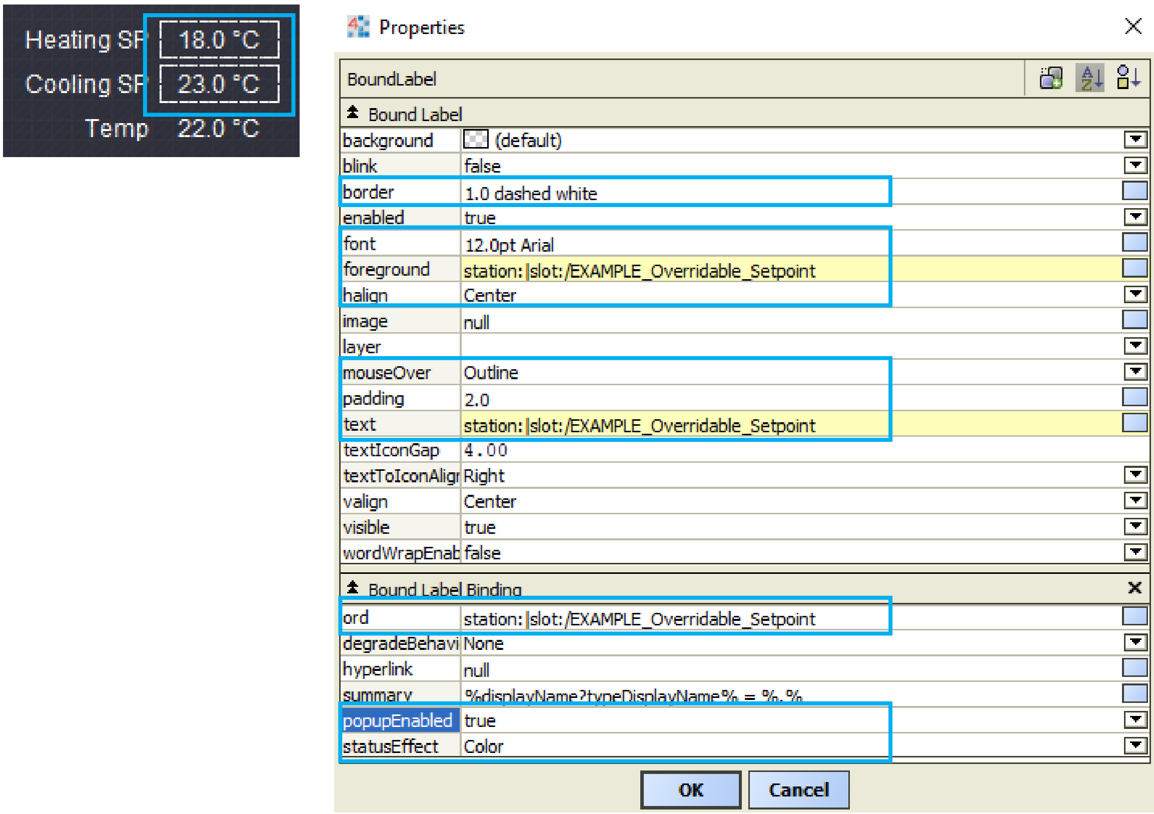
The image below illustrates the minimum requirements for a text label used for user-adjustable points. The Bound Label Binding ord is to be updated as required. In addition, the text field must be animated with the point ord, for example, %out.value%. The foreground is to be animated to Fixed Simple, value 'white' colour. This is to prevent the foreground colour reverting to default (black) after a point's status effect colour is triggered, for example if a point goes into Alarm or Fault and returns to normal.



(User adjustable BoundLabel properties)

6.9 Display Points: Overridable

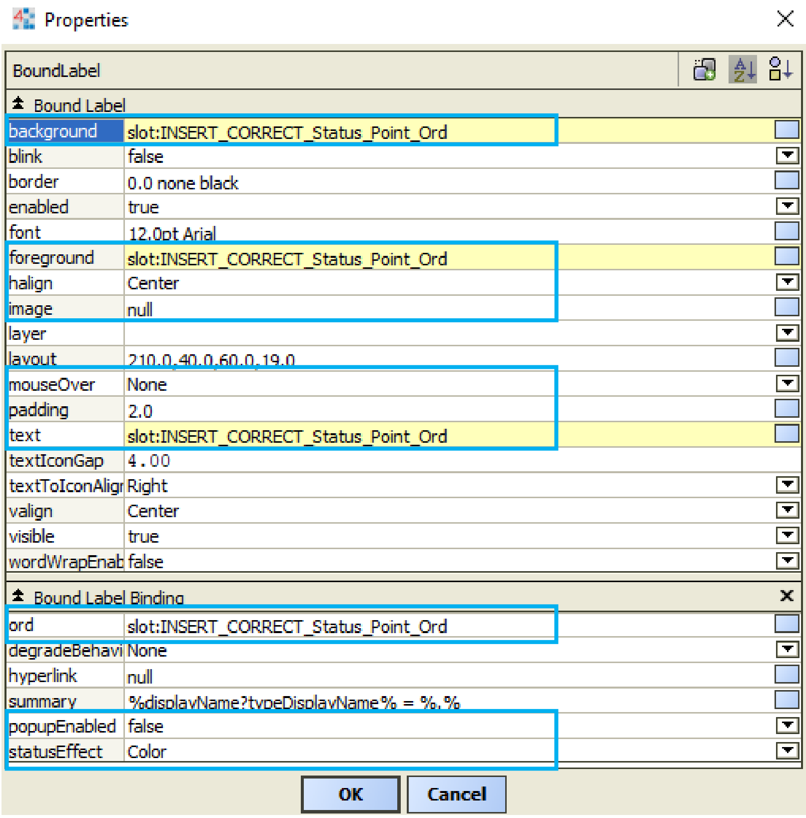
The image below illustrates the minimum requirements for a text label to be configured and used for overridable points. The Bound Label Binding ord is to be updated as required. In addition, the text field must be animated with the point ord, for example, %out.value%. The foreground is to be animated to Fixed Simple, value ‘white’ colour. This is to prevent the foreground colour reverting to default (black) after a point’s status effect colour is triggered, for example if a point goes into Alarm or Fault and returns to normal. A pre-configured BoundLabel template is available, see *Graphical Application Guide – Templates*.



(Overridable BoundLabel properties)

6.10 Display Points: Status (Green)

BoundLabels that display unit status points must animate to green (#80ff80) when the unit is running. The Bound Label Binding ord is to be updated as required. In addition, the text field must be animated with the point ord, for example, %out.value%. The foreground property must be animated using Boolean To Simple; true value ‘Black’ and false value ‘White’. This ensures text colour is displayed correctly when label background animates green. The background property must be animated using Boolean To Simple; true value ‘#80ff80’ and false value ‘null’. The images below illustrate the minimum requirements for a status label to be configured, where the background label is animated.

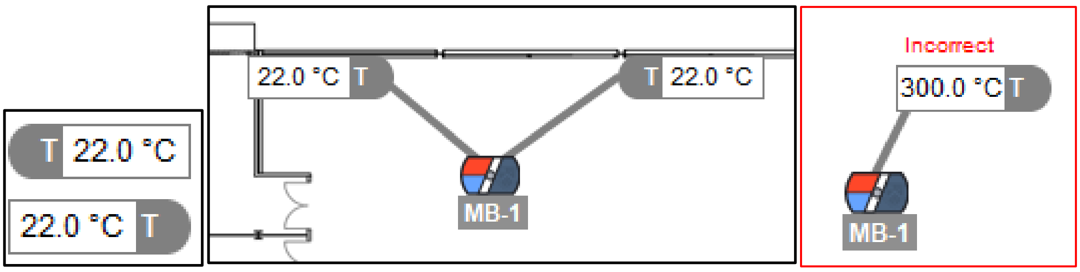


Unit Status Examples:

Unit	Point	Type	Background Animation
Chiller	Status	Boolean	True
Pumps	Pumps	Boolean	True
-	Schedule	Boolean	False
-	State	Enum	True

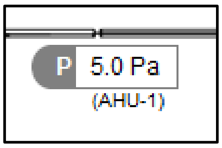
6.11 Display Points: Floor Plan Sensor

The BoundLabel is used as a way to indicate the location, sensor type and to navigate to the associated page. The text is editable and label size can be adjusted to a preferred size. Instead of ‘T’, which in the example below signifies a temperature sensor, other abbreviations can be used, for example ‘P’ for pressure sensor. Abbreviations must be clear and where it is unclear, a key must be used as per the Floor Plan standard. Left and a right variation are to be used as needed. The curved end of the label should indicate the location of the sensor. The indicators leading back to the unit should always be positions from the curved end of the label and the unit, as shown below. The information below illustrates the minimum requirements for a text label used on floorplans. BoundLabel. For more information on floor icons and label sensors, see *Floor Plan – Floor Icons and Labels* in the document.

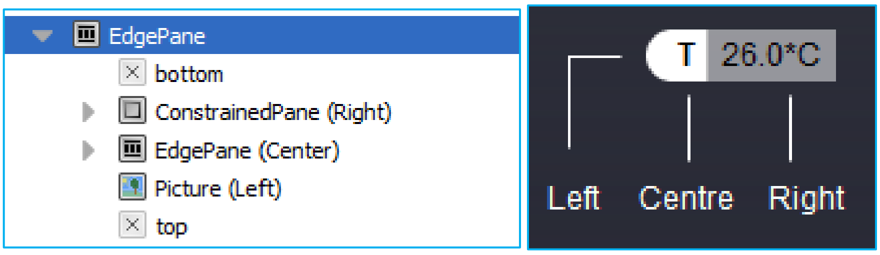


(Left and right sensor labels and an example of an incorrect use of an indicator)

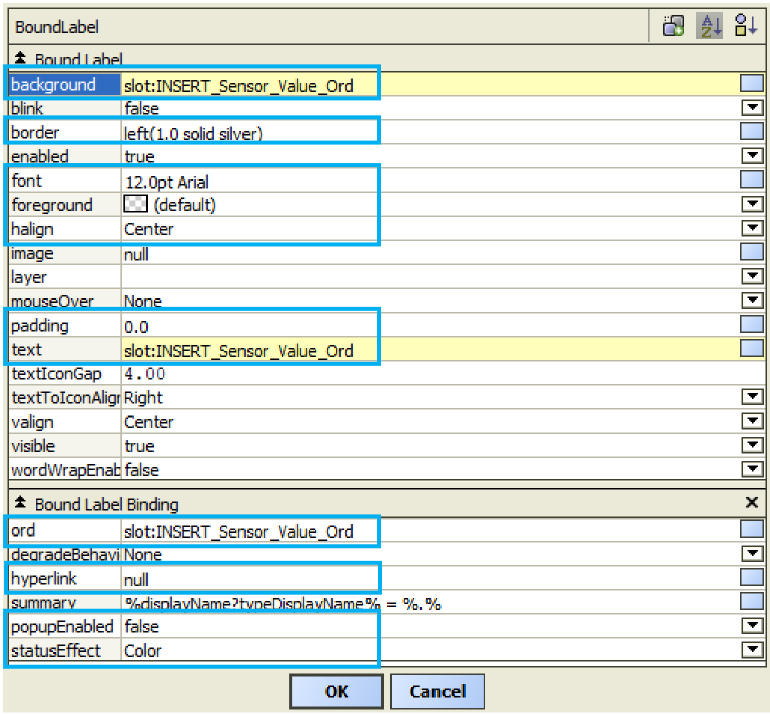
MONASH University



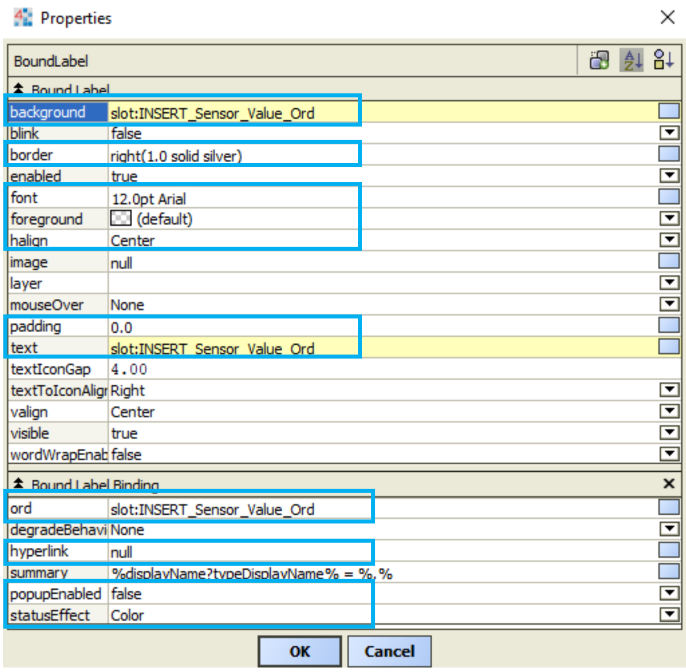
(Left pressure sensor label where unit is located on a different level – Arial 10pt - Unit name in brackets)



(Left sensor widget)

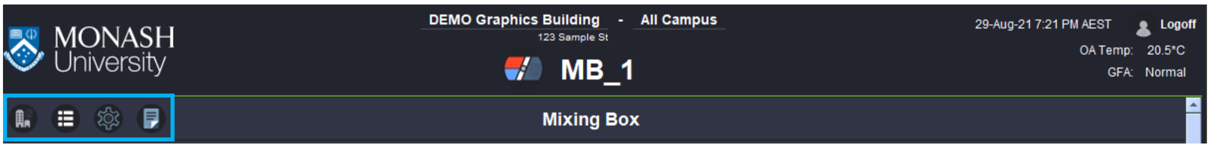


(Left sensor BoundLabel properties)



(Right sensor BoundLabel properties)

6.12 Content Title Buttons



6.12.1 Floorplan

The floorplan button is to be enabled on all equipment pages and positioned as the first button in the row. It is indicated by an icon that matches the floorplans nav menu icon. The button needs to have a tooltip (hover text) that says “Floorplan”. When the button is clicked it needs to navigate the user to the specific floorplan that the unit is located on.

6.12.2 Detailed View

The detailed view button is to be enabled on pages that contain two views for the equipment being displayed, such as schematics for meters. This button is to be indicated by an info icon positioned directly after the floorplan button. The button needs to have a tooltip (hover text) that says “Detailed View”. When the button is clicked it needs to navigate the user to another view in the same nav location named “DETAILED” that contains additional information than what is available on the standard GFX view.

6.12.3 Settings

The settings button is to be enabled to display additional control points that are not shown in the main view. The button, when used in the content title section, must be used for settings that affect the whole system. For equipment specific settings, for example a fan or damper, a button beneath the point values on the schematic diagram is to be used. See *Main Content – Pop-up Window* section. This button is indicated by a gear icon and positioned after the detailed view. The button needs to have a tooltip (hover text) that says “General Settings”. When the button is clicked it needs to navigate the user to another view in the same nav location named “SETTINGS” that contains additional writable points such as setpoints and lockouts as well as information detailing how the they relate to equipment.

MONASH University

6.12.4 Notes

The notes button is to be enabled on all equipment pages. It is to be positioned at the end of the row of buttons after the settings button. It is indicated by an icon of a report and a tooltip (hover text) that says “Notes”. When the button is clicked, the user is directed to a pop-up window displaying the notes' view. This view is in same nav location named “NOTES”. The notes view needs to allow the user to write notes about the unit into a text field that can be saved and view by anyone.

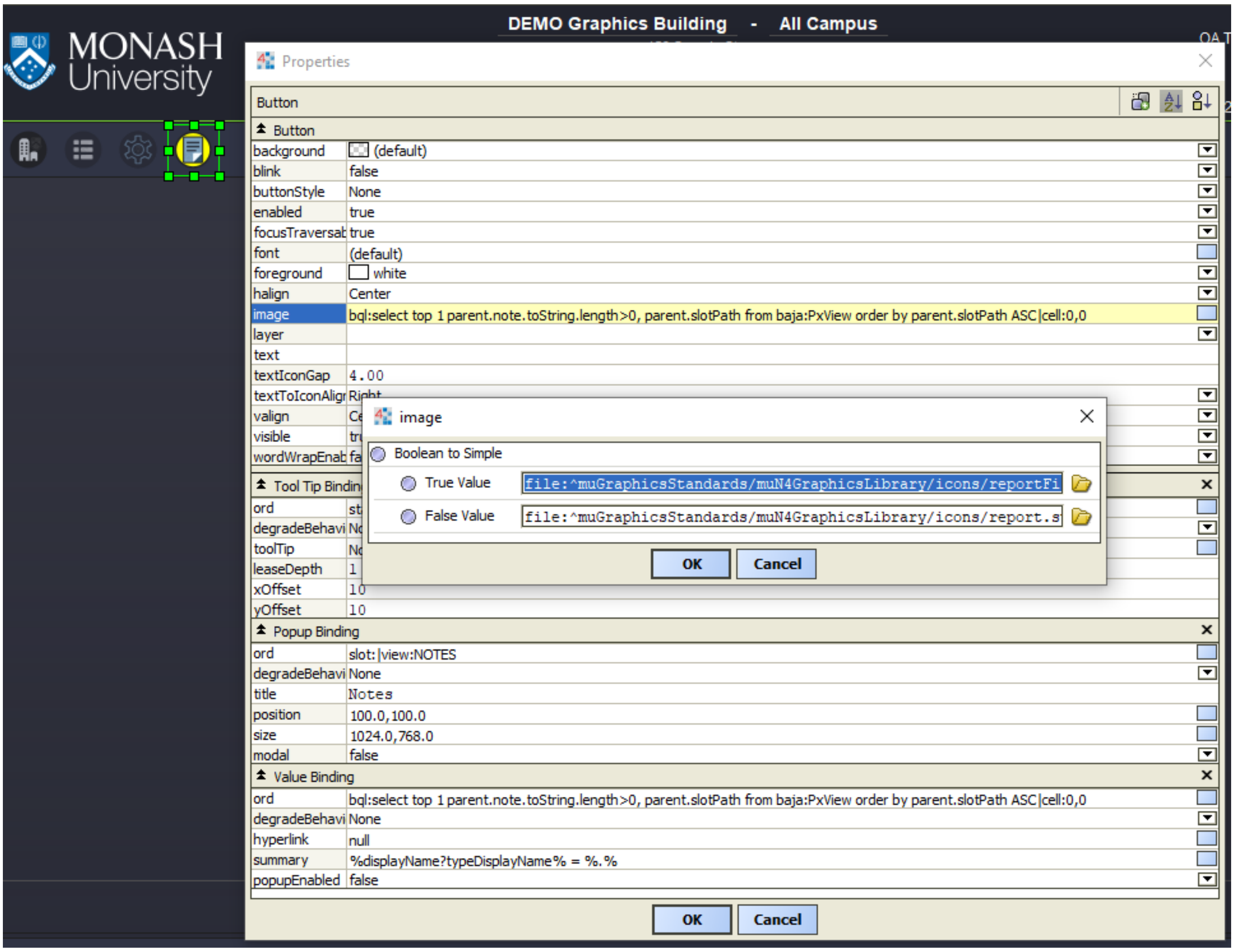
A BQL query is to be used to animate the button image using 2 report icon images. The button must display a yellow background report image when notes are present and a normal report image when no notes are present.

Example BQL:

bql:select top 1 parent.note.toString.length>0, parent.slotPath from baja:PxView order by parent.slotPath ASC|cell:0,0

Button Images:

Item	Icon
Button (Has Notes)	reportFillYellow.svg
Button (No Notes)	report.svg

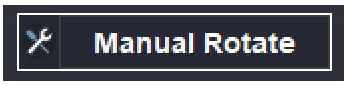


(Button with animated BQL)

6.13 Buttons

Below are some examples of buttons used across the graphics. The uses are outlined in the relevant sections of the document where the buttons will be implemented. Templates are available for all buttons.

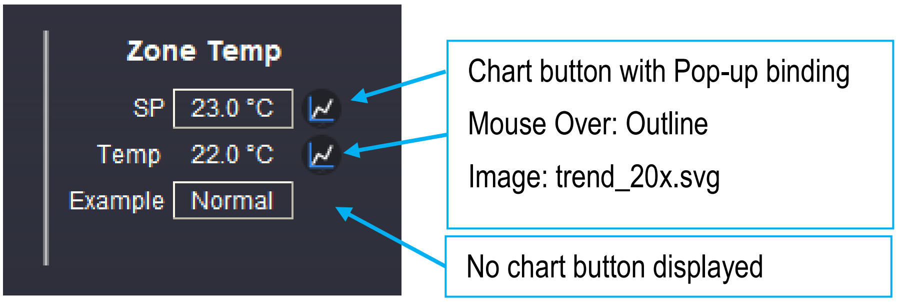
(Button with icon and hyperlink)



(Button with action binding)

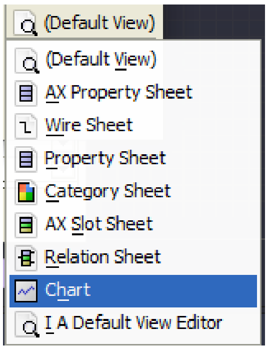
6.14 Chart Button

History charts can be viewed via the front-end graphics using a chart button or dedicated 'Trend Data' section. A chart button must be placed to the right-side of the associated point. The chart icon is 20px and a template is available. The button must open a pop-up window to the relevant history chart view. Where there is no history available or required, no chart button is to be displayed.

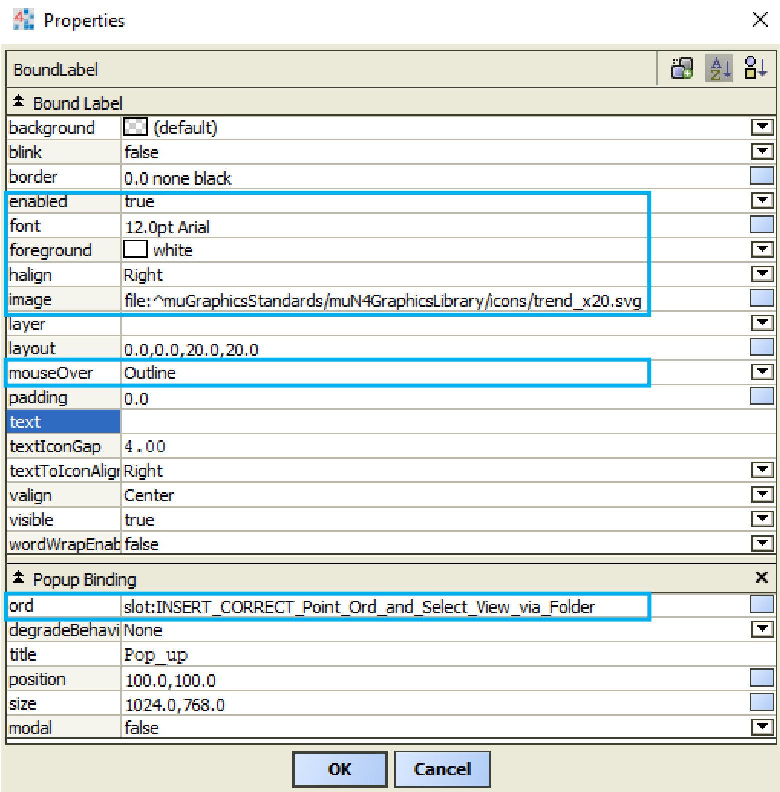


(Button examples)

6.14.1 Properties



(View select via the folder button)



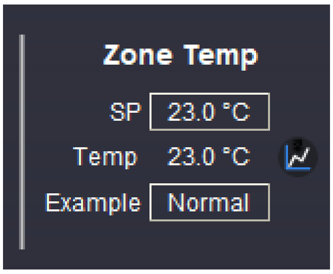
(Chart button properties)

7 TREND LOGS

Trend logs can be viewed via the front-end graphics using a chart button or dedicated 'Trend Data' section.

7.1 Chart Button

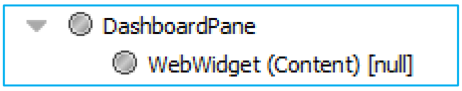
The chart button is a small icon configured to hyperlink to the Niagara Chart View of the respective point. The button is location to the right of the label displaying the point's value. For more information about the Chart Button, see the *Displaying Values: BoundLabels and Buttons* section of the document.



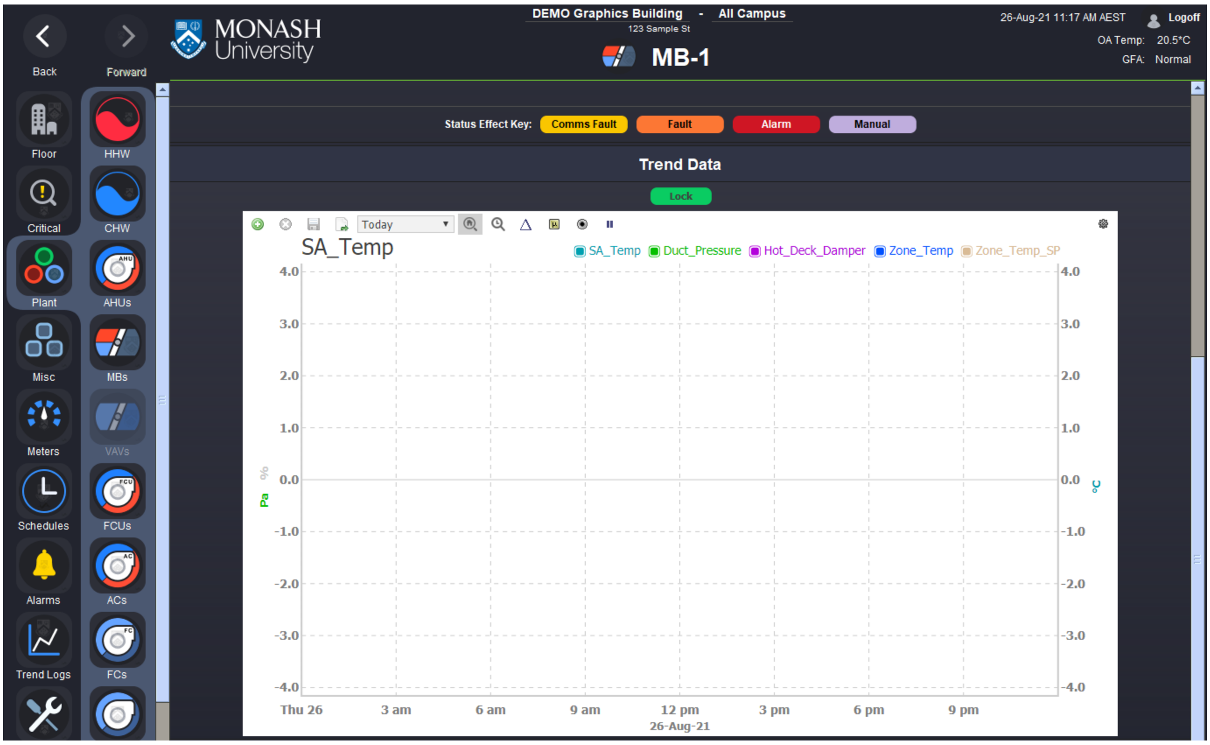
(Chart button)

7.2 Trend Data (Mandatory)

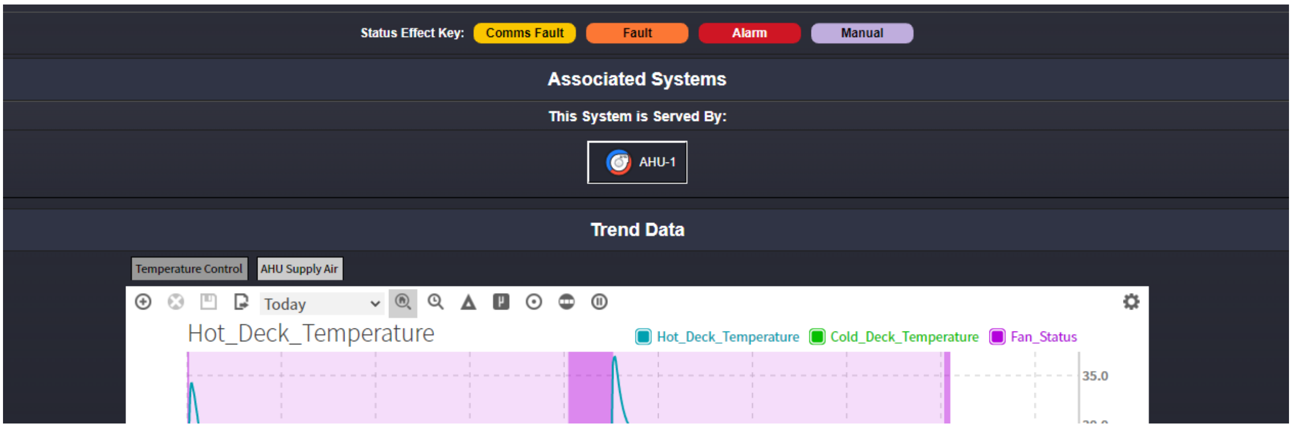
The 'Trend Data' section must be inserted below the main content and Status Effect Key in the widget hierarchy. The chart widget must be placed in a dashboard pane. A pre-configured chart with multiple trends is required, the page must have chart buttons adjacent to point values, as well as a dedicated 'Trend Data' section. This chart section must be the last section, at the bottom on the page. Pages that require additional sections such as zone group or associated units must be inserted between the status key and the trend section.



(Widget tree, displays chart inside DashboardPane)



(Trend Data section – below the MB-1 ductwork schematic diagram)



(Tabbed Pane example where multiple charts are required)

Chart widgets within dashboard panes are to be populated with categories of data relevant for the asset or system. Categories are specific to the asset or system, and some examples are below. The intent is to provide ease of access to historical information for maintenance/troubleshooting purposes, and must cover sensors involved in key control loops and their related outputs. A maximum of 20 trendlogs can be loaded within the tabbed panes – if more than 20 histories are involved in the system, then up to 20 should be provided on the tabbed panes and any additional provided as pop-ups via buttons arranged vertically to the left of the tabbed pane.

Charts should be grouped in batches of up to 5 relevant trends, allowing for users to add additional trends via the dashboard feature.

Examples provided below are indicative, and actual points used should reflect the specific equipment type:

Temperature Control (Space Temperature, Space Setpoint, Valve Positions)

Temperature Control (Supply Air Temperature, Supply Air Temperature Setpoint, Valve Positions)

Pressure Control (Static Pressure, Static Pressure Setpoint, Speed Output)

Pressure Control (Differential Pressure, Differential Pressure Setpoint, Bypass Position)

Pressure Control (Field Pressure, Field Pressure Setpoint, Pump Speed)

Damper Control (CO2 Level, Ambient Temperature, Damper Signal, Economy Mode Status)

Setpoint Control (Calculated VAV Demand, Supply Air Temperature Setpoint, Supply Air Temperature)

CH-1 (Leaving Water Temperature, Entering Water Temperature, Status, DP, Pump Speed)

CH-2 (Leaving Water Temperature, Entering Water Temperature, Status, DP, Pump Speed)

Common (Field Flow Temperature, Field Return Temperature, Bypass Valve Position)

etc.

Chart categories used should be considerate to the asset or system type and from the perspective of future maintainability of the equipment for a technical or non-technical user. For a complex system or if in doubt for the charts to present, provide proposed arrangement to the BAS Administrator for review.

Monash University

8 STATUS EFFECT KEY

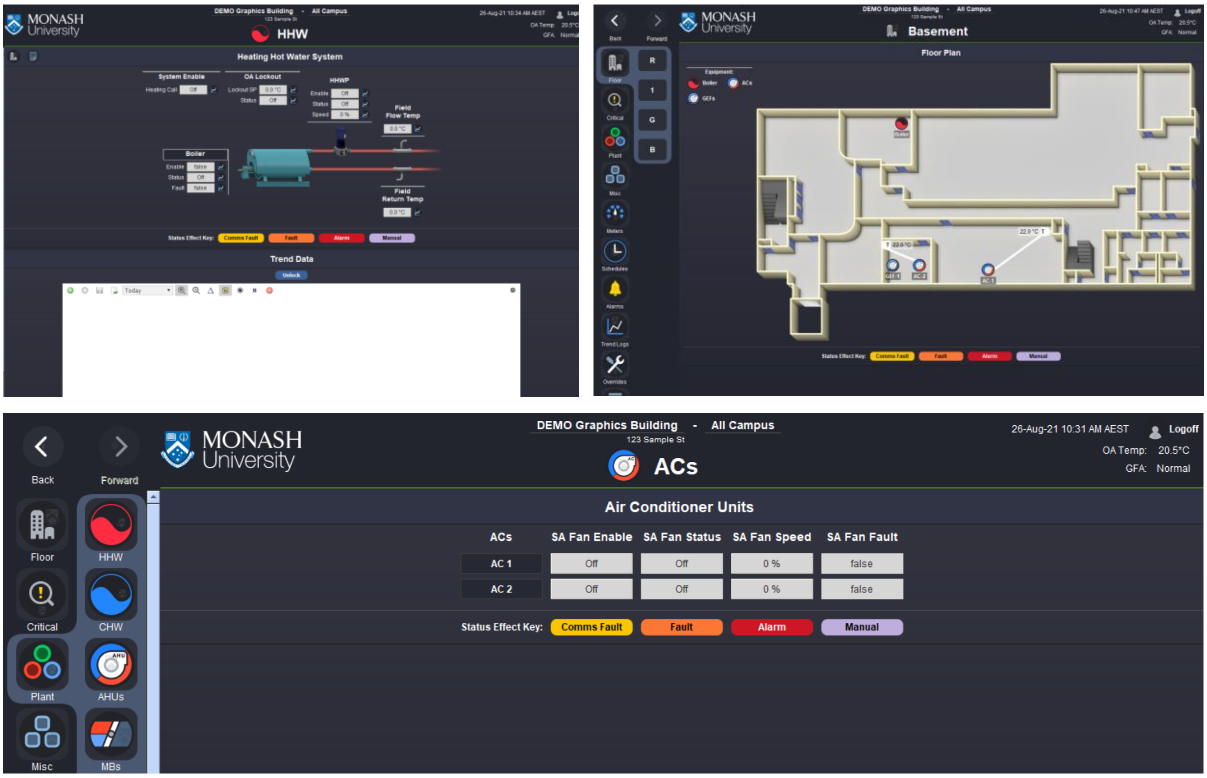
A status effect key must be positioned beneath any main content section where points are being displayed.

Standard status effect colours are;

- Comms Fault = Yellow
- Fault = Orange
- Alarm = Red
- Manual = Purple
- Enabled = Green

These are handled by Niagara.

Where a page has multiple sections inside the content area, the status effect key must be positioned beneath the first section. Only one status effect key must be shown per page as indicated in the examples below.



(Position examples – Schematic showing multiple sections separated by a content title, Floorplan Page and Summary Table)

9 ALARMS

Refer to the Monash 'BAS Alarm Standards' document for additional information to this section, which has been updated to provide additional clarity on alarm configuration.

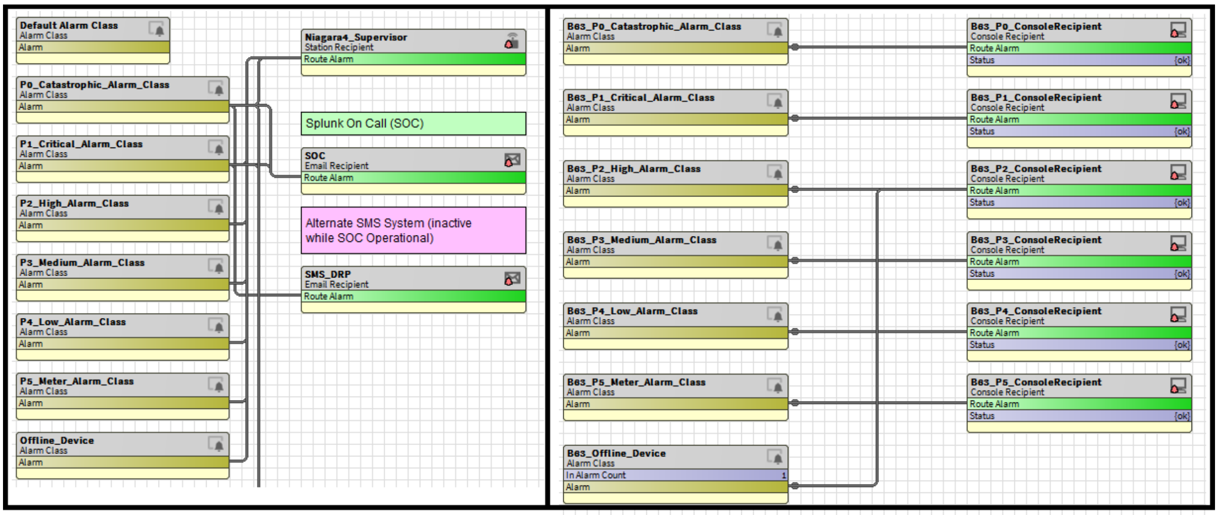
9.1 Alarm Class

The supervisor and JACEs group alarms using alarm classes, the alarm classes can be used to create a hierarchy so that they can be separated and processed based on their importance. The alarm classes you will need are as follows.

Key	Description	Colour	Title	AIMS	BPD Response Time
P0	Priority 0	Purple	Life And Safety	Yes	Immediate
P1	Priority 1	Red	Critical	Yes	1 Hour
P2	Priority 2	Orange	High	Some	4 Hours
P3	Priority 3	Blue	Medium	No	Next Business Day
P4	Priority 4	Black	Low	No	Monitored Weekly
P5	Priority 5	Pink	Metering Alarms	No	Monitored Monthly
P10	Priority 10	N/A	Client Only	Yes	N/A

The alarm classes must reside under the Alarm Service and have a have identical naming on both the supervisor and each JACE so that they are recognised to mean the same thing on each station. Folders must be used in the Alarm service to separate buildings.

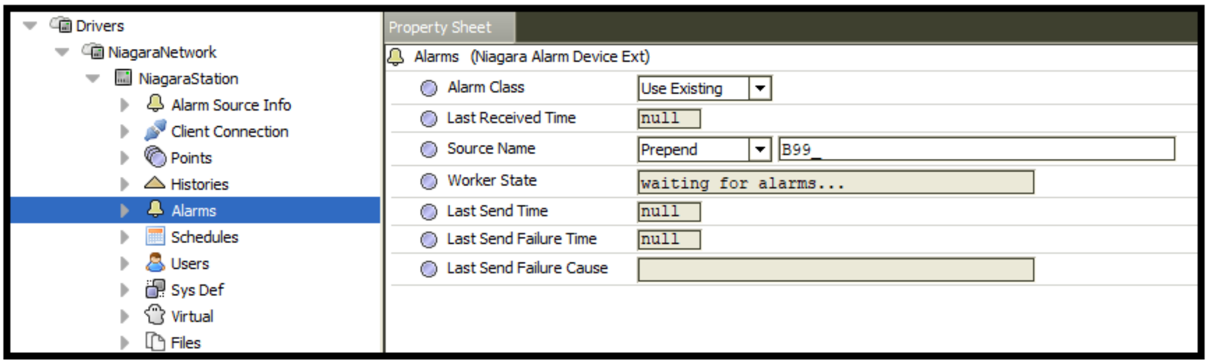
Alarm classes on the JACEs will also need to be routed to a Station Recipient component so that they can be seen on the supervisor station. Alarm classes that require an AIMS alert to be sent out must be set up on the JACE to ensure that they are sent immediately instead of on the next sync with the supervisor. Only JACE offline AIMS alarms are to be sent from the Supervisor.



JACE Alarm Service Routing

Supervisor Alarm Service Routing

Note: The alarm classes in the example image above have the building number prepended to them. This is to separate out the building's alarms from all of the other alarms in the campus. To replicate this functionality, you must do so on the supervisor under the Alarms section of the device.



(Example of setting an alarm class prefix)

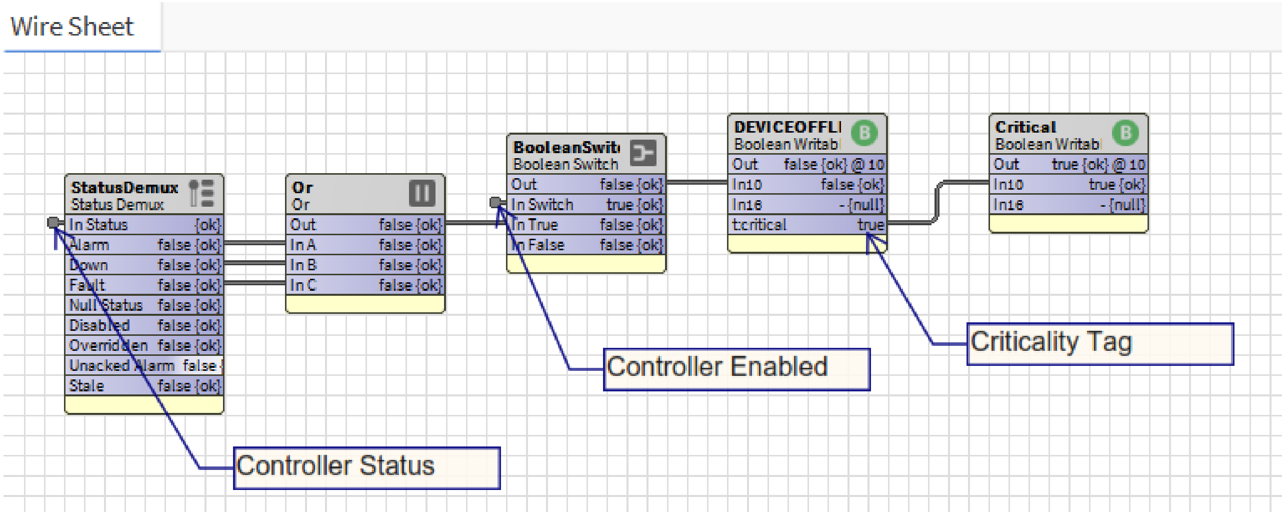
The naming convention for alarm classes on the supervisor is: [Building]_[AlarmClass]. The building name variable is mandatory, and changes how the alarms are handled. General alarm classes are no longer in use – e.g. Critical_Alarm_Class or Medium_Alarm_Class. All alarm class must be building specific, prepended with the building number and routed to the appropriate console.

A 'Building Alarm List' in the Monash approved format must be submitted to the Monash University BAS Administrator.

9.2 Offline Device Alarm

As part of the Monash standard, an AIMS alarm is to be sent when a critical JACE or 'any critical field device' is offline. An alarm for **each** field device is not required. Each device will need to additional programming in order to determine what priority alarm should be thrown when it goes offline. The following section will outline the requirements. These are the only AIMS alarms that can be, and are, required to be sent from the Supervisor.

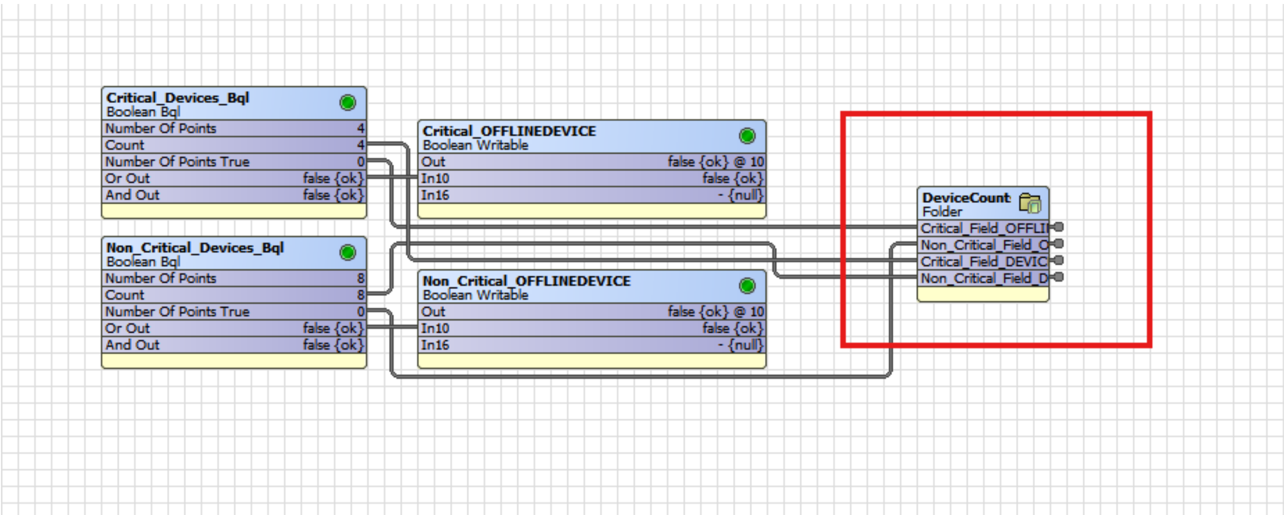
A 'StatusDemux' component must be added that links the relevant device statuses to an 'Or' component so that when a point has any of the chosen statuses it will trigger a Boolean value. The 'Or' component can then be linked to point that denotes that the device is offline. A switch may be inserted before the end point to disable the logic under certain circumstances such as the device being disabled.



The "DEVICEOFFLINE" point must be tagged with "critical" and the tag linked to a Boolean writable named "Critical".

A BooleanBQL is to be created for each JACE to query all of the "DEVICEOFFLINE" points created for each device. The BooleanBQLs query is prepended with a NEQL query to retrieve a filtered list of only critical devices.

Link the "orOut" of the BooleanBQL to a BooleanWritable Point. This point will be used to represent all devices contained in the BooleanBQLs results. Next, add an alarm extension to the BooleanWritable point with the appropriate alarm class. Finally, you can route the alarm to a recipient (alarm console, SOC, Email,etc) in the alarm service. Counts and statistics used for network health graphics are contained in the folder in red box pictured below.



9.2.1 Determining Priority of Offline Device Alarm

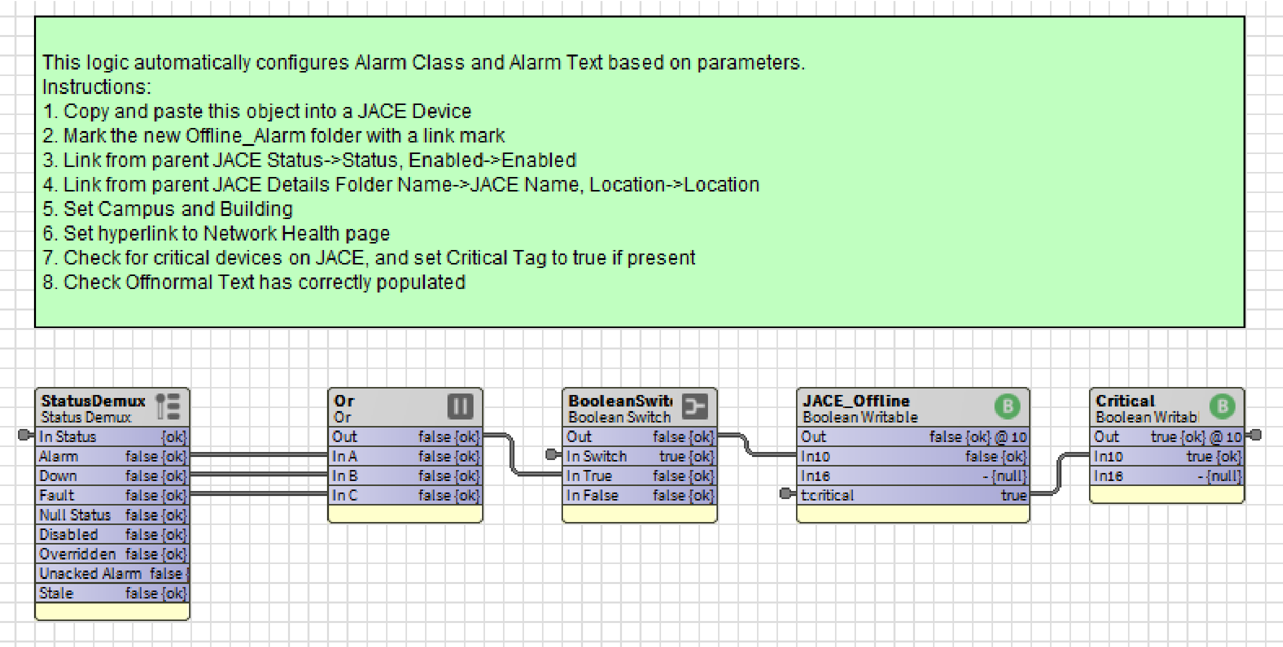
Every JACE is to provide an offline device alarm via the Supervisor. One for the JACE and the other for the 'field devices'. Software should reside in the Supervisor in a common location and be logically split up per building (i.e. not all buildings on one wiresheet).

Each JACE and field device (BAS controller, meter, HLI, etc) is to be tagged as a "device" and also tagged if 'critical' (true/false).

- If any critical JACE or device is offline, the alarm is Critical class and is sent to the console and AIMS.
- If a non-critical JACE or device is offline, it is OfflineDevice class and is to send a console alarm only.

9.2.2 JACE Offline Alarms

A folder named "Offline_Alarm" must be present as a slot directly under each JACE in the supervisor. This folder requires links added to it, but otherwise automatically configures the offline device alarm and alarm class routing based on criticality flag and other metadata.



9.3 Alarm Hyperlink

All alarm source extensions must be configured as per the 'Building Alarm List' that is submitted to Monash University Administrator for approval. In addition, the alarm source extension must use an appropriate hyperlink ord. This is beneficial as the user can quickly navigate to the relevant graphics page via the alarm console on work bench. The hyperlink ord must use "ip:tridium.bdp.monash.edu….." format.

9.4 Alarm Console

As per the graphic standard, dedicated alarm pages per building will display alarm consoles. Note that the building alarms must also be received by the campus wide consoles. As well as any other console recipients outlined by the Monash BAS Administrator.

Refer to the Monash BAS Alarm Standard (link) for format of console alarm.

9.5 Alarm Enable

For equipment alarms such as freezers, incubators, etc. an enable alarm checkbox is to be displayed as well as alarm parameters (setpoint and delay). Below is an example of an alarm enable GridPane. The checkbox may appear in a summary or a schematic review. Alarm requirements to be approved by Monash University BAS Administrator. The checkbox must be ordered below a bound label with "Set" action required to change the state of the checkbox – the checkbox itself is a visual aid only. The bound label does not need to display any information.

(Alarm Enable check box)

9.6 Controlled Environments Minimum Graphics Content

For each controlled environment within the Tridium BAS, the following information is presented for the user within the associated table row: Room Number (as a button navigable to the relevant floor plan), Port Number (settable), MU Asset Number (settable), Alarm Description (settable), Alarm Status (with trend popup adjacent), Alarm Enable (as a label requiring a set action to adjust, and trend popup adjacent), AIMS Team assigned to the alarm, Alarm On Delay, Alarm High Limit (N/A if a Boolean alarm), Alarm Low Limit (N/A if a Boolean alarm), Comments/Instructions Popup (highlights icon yellow when notes are present, but not when alarm instructions are present).

Room No.	Port No.	Asset No.	Alarm Description	Status	Alarm Enable	AIMS Team	Alarm On Delay	Alarm High Limit	Alarm Low Limit	Comments/ Instructions
Rm1U46	1-23	MU036287	-30 Freezer	Normal	Enabled	ACBD-B247ARmU46Freezer	300 s	N/A	N/A	
Rm1U46	1-24	MU036288	-30 Freezer	Normal	Enabled	ACBD-B247ARmU46Freezer	300 s	N/A	N/A	
Rm1U46	1-30	MU028684	+4 Fridge	2.7 °C	Enabled	ACBD-B247ARmU46Freezer	1800 s	8.0 °C	-1.0 °C	

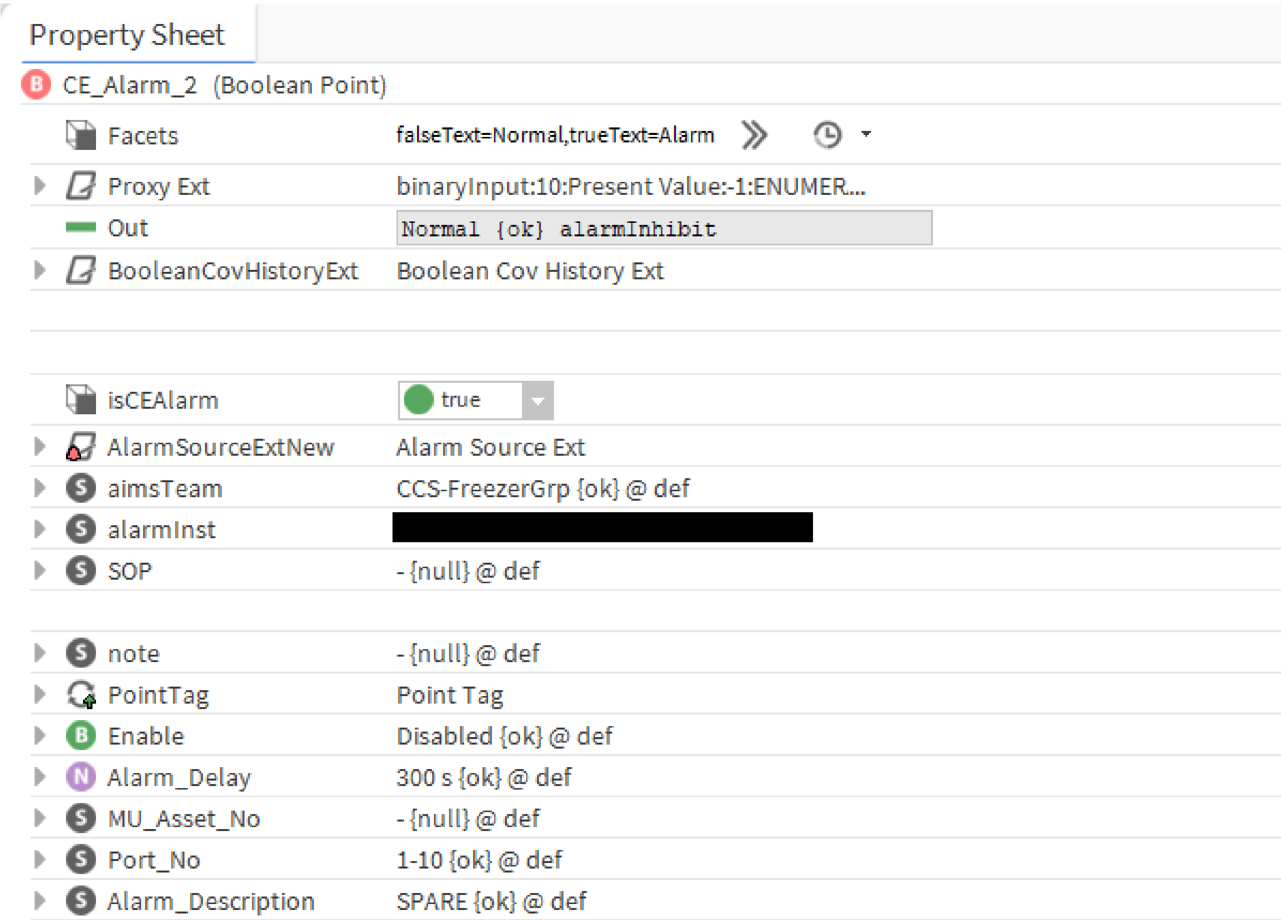
(Controlled Environment Summary Table)

The Comments/Instructions popup consists of two field editors. The Alarm Instructions field is settable, and provides alarm response information that is ingested by AIMS and presented as an annotation on the alarm. The Notes field is settable, and provides information relevant to the alarm but is not sent with the alarm to AIMS.

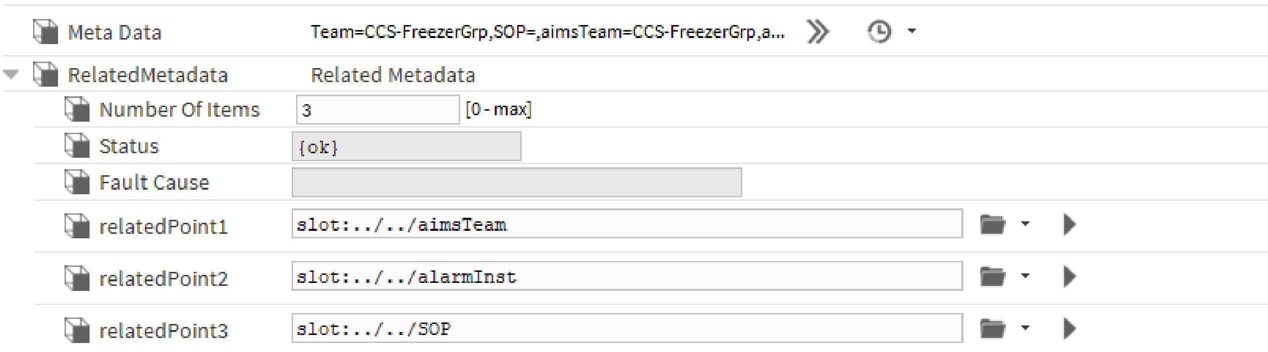
(Controlled Environment Note Fields Popup)

To achieve the functionality above, slots on alarm points should be configured per the below screenshot (slots under the point for aimsTeam, alarmInst, SOP, note, Enable, AlarmDelay, Alarm Limits (if applicable), MU_Asset_No, Port_No, Alarm Description.

To attach the relevant metadata to the alarm extension, use the Vykon Pro “RelatedMetadata” component with relative ords matching to the aimsTeam, alarmInst and SOP slots (example pictured below).



(Slots on a Boolean freezer alarm point)

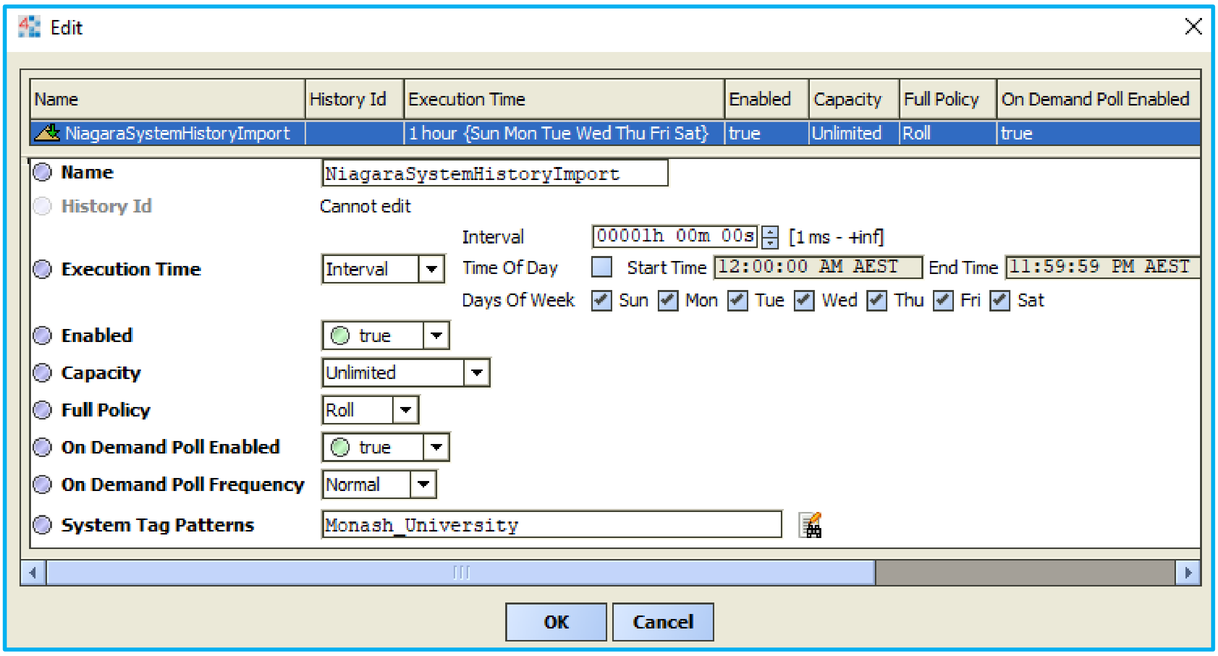


(Usage of RelatedMetadata component to attach fields to the alarm extension)

10 HISTORY

10.1 Import

Each JACE controller will be connected to the Supervisor and all trend logs setup on the JACE shall be imported into the Supervisor via the Tridium NiagaraN4 protocol. Histories in the JACE are to use the System Tag, "Monash_University". The Monash University supervisor is to import histories based on this tag.



(Supervisor - System history import tag properties example)

10.2 Grouping

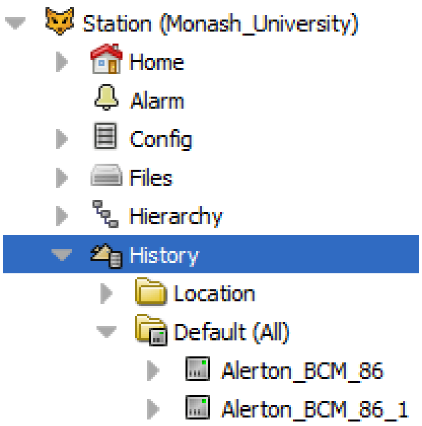
History grouping allows you to organise and view imported histories in folders in an alternate structure to 'Default (All)'. The 'Default (All)' history structure on the Monash Supervisor is organised by JACE Station. As per the Monash Graphics Standard, history groups are defined using the History Group Manager with the following history properties; Campus, Building, System, Unit. Each history that is imported to the supervisor must be configured with these property names. The tag used for each property must be logical and consistent with any existing campus, building, system or equipment tag. For example, if building 96 uses 'B96' for the 'Building' property, then this should be perpetuated. A key benefit to grouping is that users can locate histories and build charts more easily regardless of the point's location due to the standardised history grouping structure used across the campuses and buildings.

For more information on history grouping and implementation, see Niagara documentation:

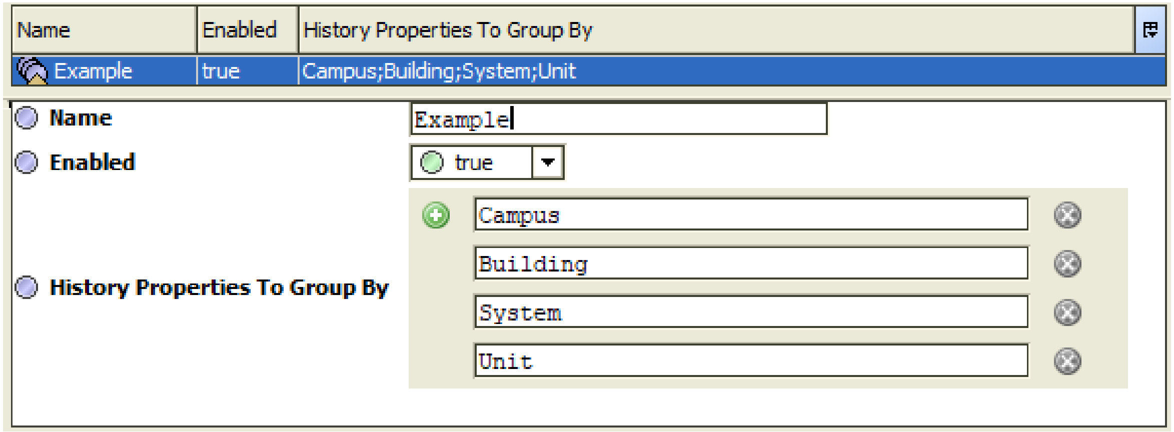
local:|module://docHistories/doc

History tag examples:

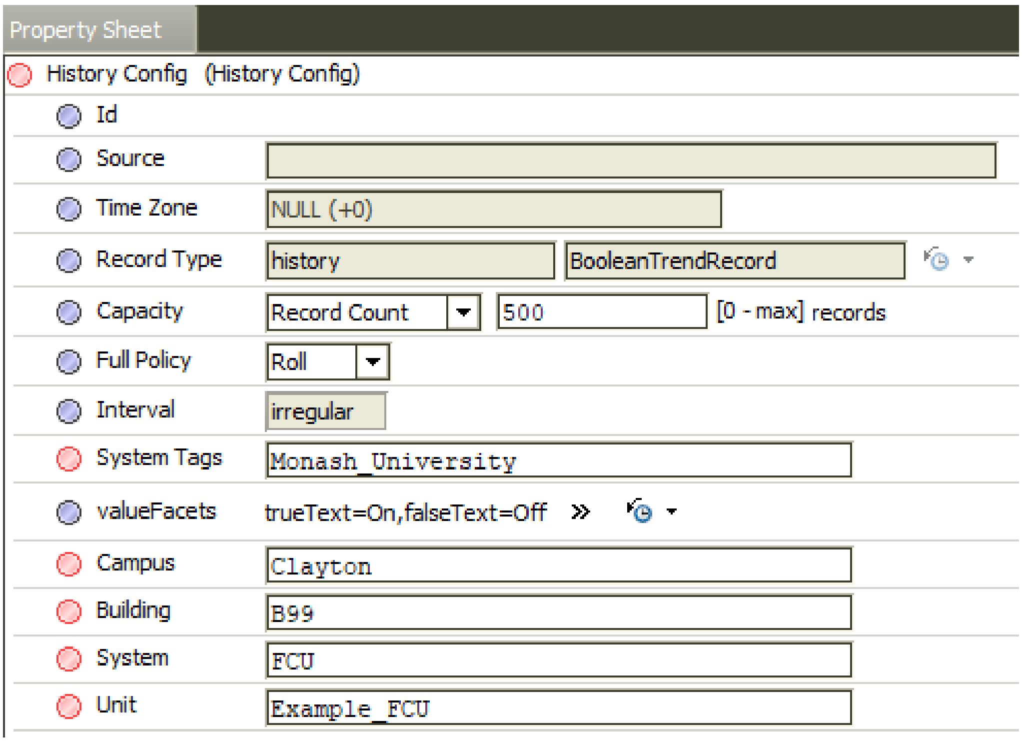
Campus	Building	System	Unit Name
Clayton	B96	AHU	AHU-2
Caulfield	B53	FCU	FCU-1-1
Alfred Hospital	AMREP	Freezer Room	Freezer Alarms
Clayton	DEMO_GraphicsSample	Metering	DB_B_1_Power



(History group folder location on Supervisor)



(Example, History Group Manager, New History Group Popup)



(Example, Point History Extension - History Config, string slots)

10.3 RdbmsNetwork

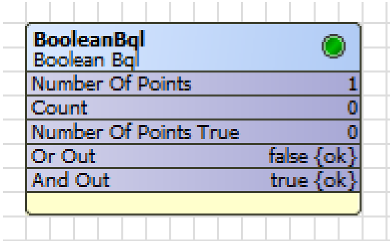
All histories must be exported to the Rdbms database. The Sql Server Database, BPD_SQL_PROD1, is to be used and is located on the Monash Supervisor. Note, System Tags cannot be utilised when importing histories.

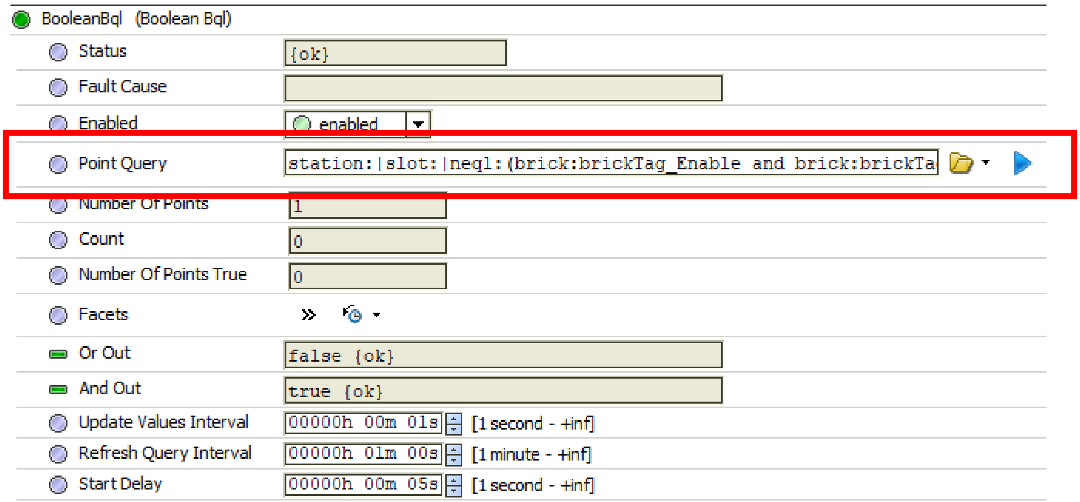
For more information:

local:|module://docRdbms/doc/ExportingHistoryDataToAnRdbmsDataba-7C3B7D13.html

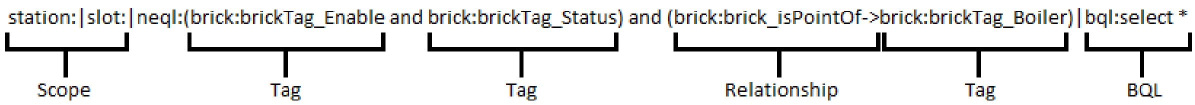
11 TAGGING

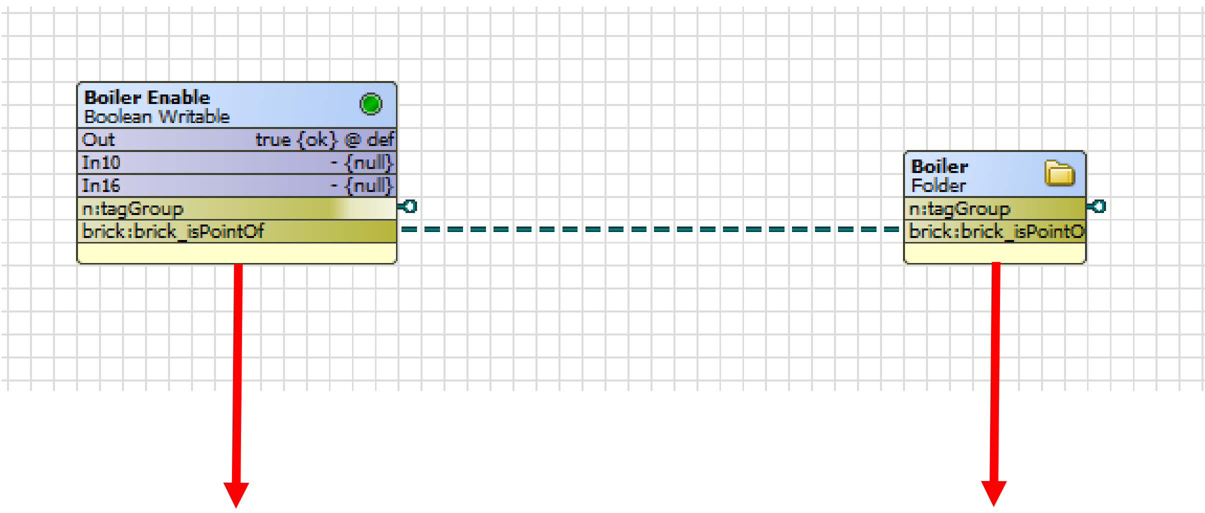
NEQL is a tag query syntax built into Niagara. NEQL can be used in conjunction with BooleanBQL blocks to count points and report summaries. Refer to the Monash Tagging Standards for more information on tagging requirements.

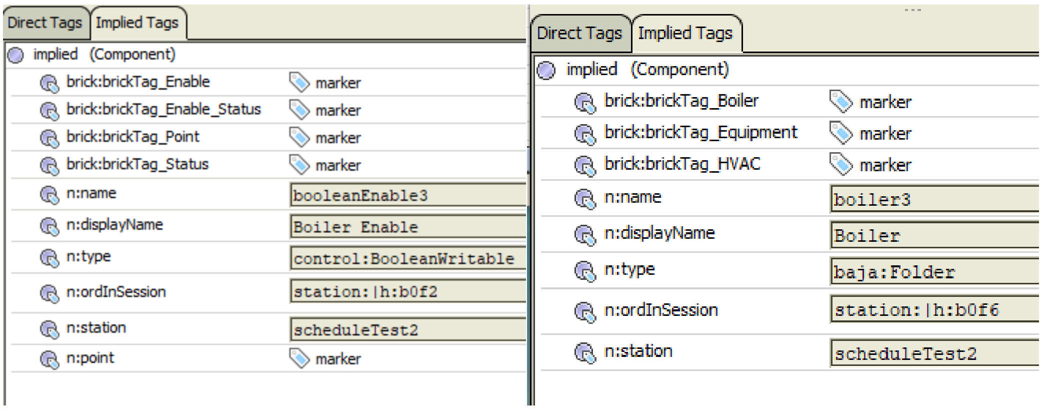




The structure of a point query using NEQL is as follows







The example above queries all enable points for boilers. You can see the “isPointOf” relationship from the boolean point to the boiler entity, and their respective tags used to make this query.

You can use any combination of tags and relationship ids to build queries. Please refer to the station’s tag dictionary service for a list of currently installed tag dictionaries.

For more information view the Niagara help documents: module://docDeveloper/doc/neql.html

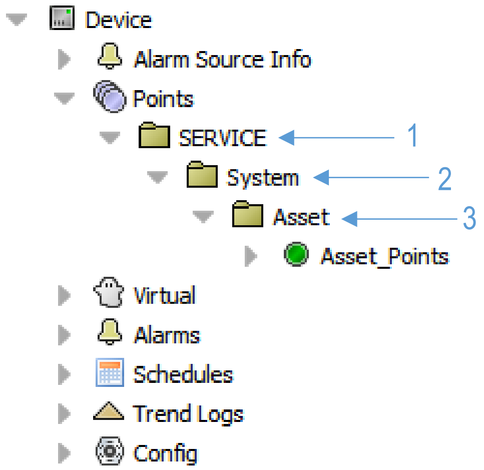
12 BAS FOLDER / FILE STANDARD

12.1 Component Space - Points Folder Structure

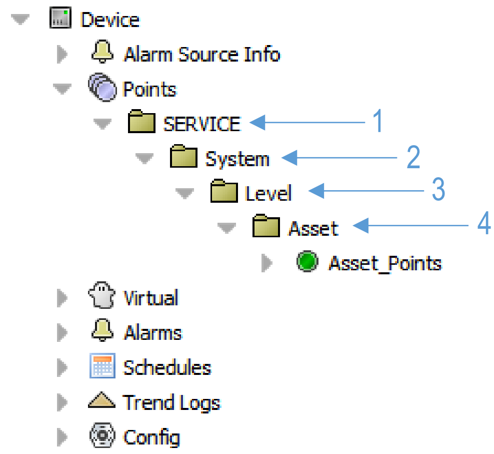
To ensure consistency and enhance engineering efficiency within the Tridium BAS system, it is crucial to follow the specified folder structure. This structure is designed to align with the graphics structure, promoting familiarity and coherence while navigating through either the graphics or folders.

Commencing from a device points directory, the choice between two folder structures is contingent on whether the ultimate asset functions as a central plant serving an entire building or multiple floors, or as a single unit catering to a specific room or a few rooms.

The folder structure for central plant:

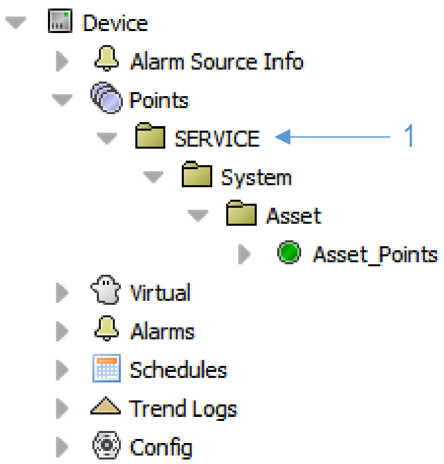


The folder structure for other assets:



In case of any ambiguity or for systems, asset types, etc., not explicitly covered here, kindly propose the folder structure to the BAS Administrator for approval.

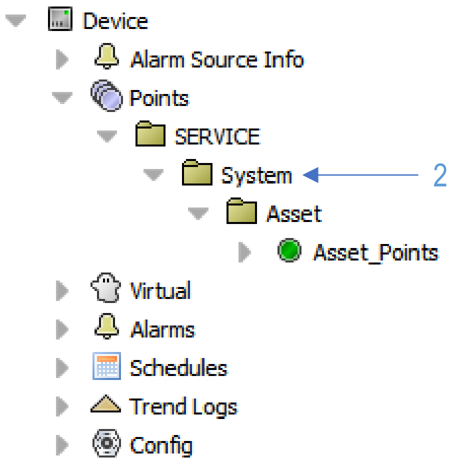
12.1.1 Folder 1 – SERVICE



There are four service folders corresponding to four of the primary graphics menu items.

Name	Child Systems
CRITIAL	Controlled environments, Gas detection, Critical equipment monitoring etc.
HVAC	All plant, AHU's, FCU's, AC's, CHW, HHW etc.
MISC	All other monitored/interfaced equipment not considered critical, i.e. lighting interface, Hydraulics etc.
METERING	All metering devices

12.1.2 Folder 2 – System



This folder should be named with the overarching term referring to a particular system. For instance, HHW would include assets such as pumps, boilers, heat exchangers, bypass valves, and more. CHW would cover assets like pumps, chillers, cooling towers, bypass valves, and so forth.

Here are some typical examples for each service:

CRITICAL

- Gas_Detection
- CE
- Generator
- Safety_Showers
- UPS

HVAC

- HHW
- CHW
- DHW
- AHU
- FCU
- AC
- VRV
- Fan
- VAV

MISC

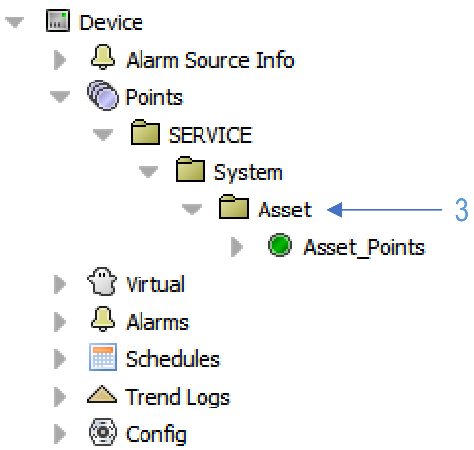
- Hydraulic
- Lighting
- Window
- Blind

METERING

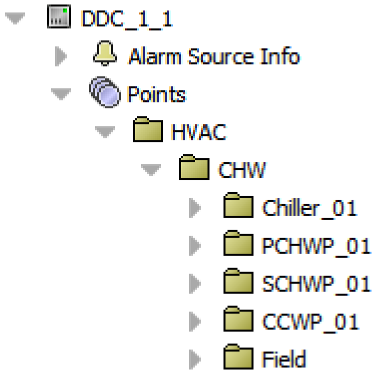
- Electrical
- Solar
- Water
- Gas
- Energy

12.1.3 Folder 3

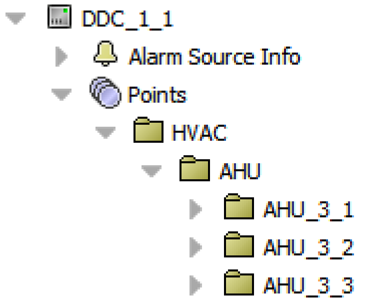
12.1.3.1 Folder 3 – Asset (Central Plant)



The Asset folder should adopt the distinct asset label as its name. For example, Under CHW you would have all assets relevant to the CHW System, for example:



Or under AHU you might have:



12.1.3.2 Folder 3 – Level (Other Assets)

This is fairly straightforward and should indicate the floor for which the asset serves the space.

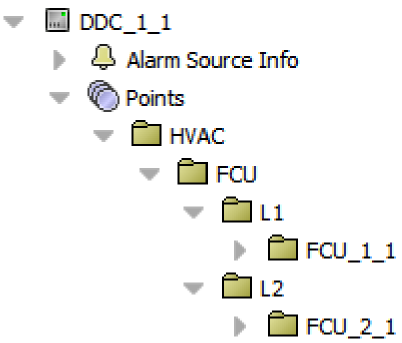
For example, an FCU may be physically located on the roof, but servers a space below on level 3 then that FCU would reside in the L3 folder.

12.1.4 Folder 4 – Asset (Non-Central plant)

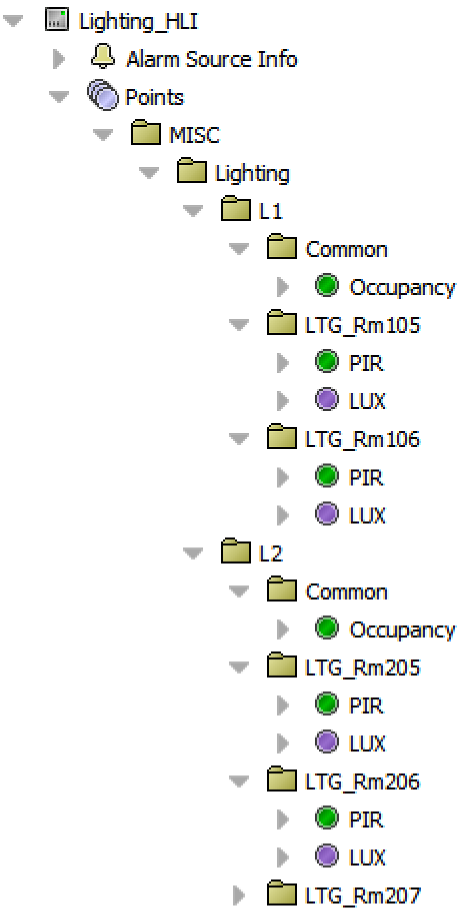
The asset folder should be named after its unique asset label. Alternatively, for assets more aptly identified by their location, such as room numbers, the folder should reflect this. For instance, folders containing FCUs, VAVs, and ACs should bear their distinct asset labels like FCU_1_1, VAV_3_2, AC_1_3. On the other hand, assets like gas detection and lighting should be named according to the room they serve; for example, GD_Rm301 and LTG_Rm205. See section 1.5 for structure examples.

12.1.5 Example folder structure

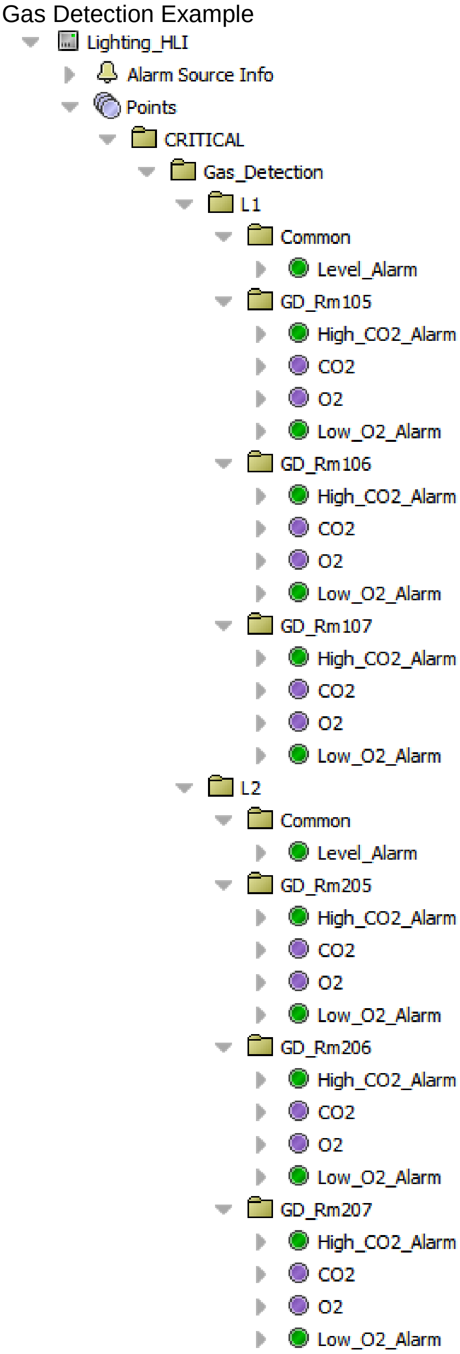
FCU Example



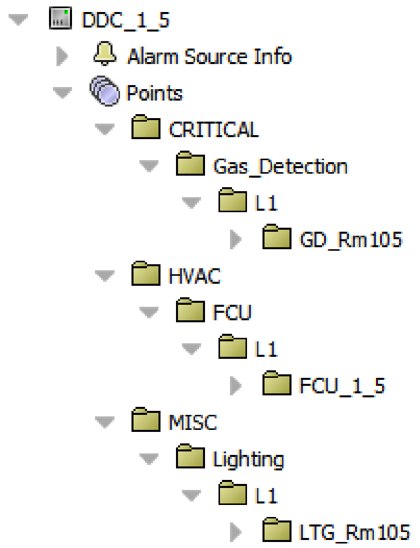
Lighting Example



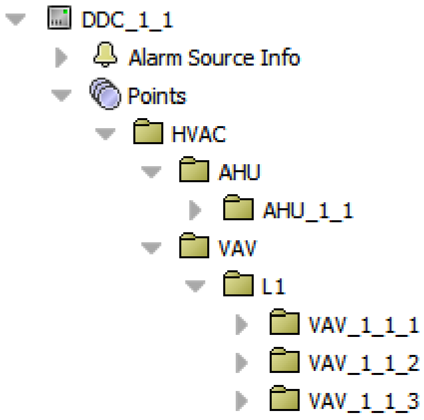
Gas Detection Example



Mixed



Mixed

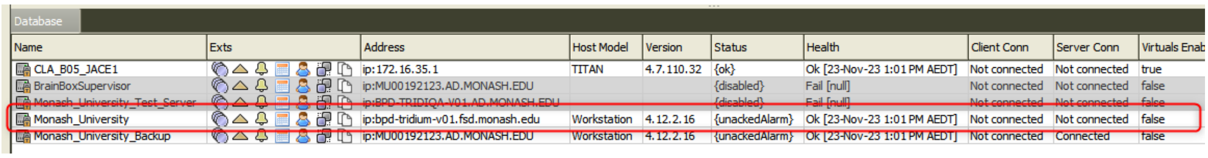


12.2 Exporting points to Supervisor

To continue the consistency at a server level, export tags should be used to synchronise folder structure and points with the supervisor.

12.2.1 Adding and configuring Export Tag Network

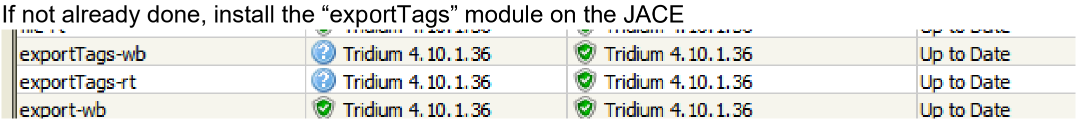
1. Ensure the supervisor has been added to the Niagara network of the JACE and has the correct credentials etc.



Database

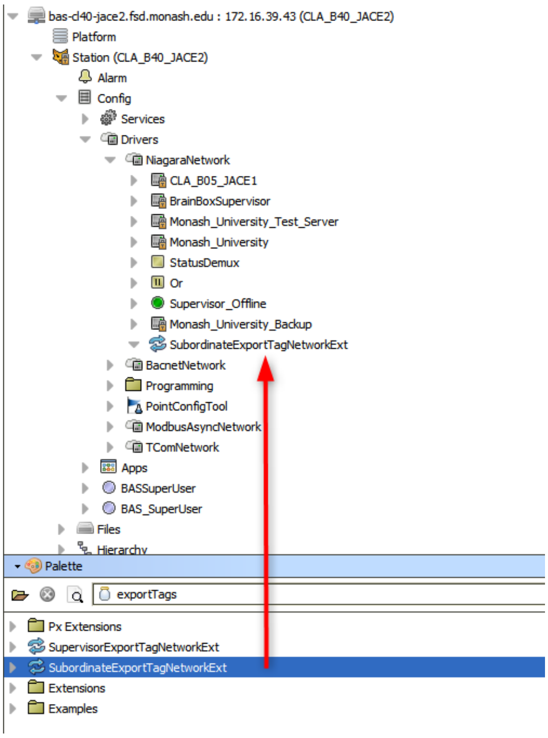
Name	Exts	Address	Host Model	Version	Status	Health	Client Conn	Server Conn	Virtuals Enab
CLA_B05_JACE1		ip:172.16.35.1	TITAN	4.7.110.32	{ok}	Ok [23-Nov-23 1:01 PM AEDT]	Not connected	Not connected	true
BrainBoxSupervisor		ip:MU00192123.AD.MONASH.EDU			{disabled}	Fail [null]	Not connected	Not connected	false
~~Monash_University_Test_Server~~		~~ip:BPD-TRIDIQA-V01.AD.MONASH.EDU~~			~~{disabled}~~	~~Fail [null]~~	~~Not connected~~	~~Not connected~~	~~false~~
Monash_University		ip:bpd-tridium-v01.fsd.monash.edu	Workstation	4.12.2.16	{unackedAlarm}	Ok [23-Nov-23 1:01 PM AEDT]	Not connected	Not connected	false
Monash_University_Backup		ip:MU00192123.AD.MONASH.EDU	Workstation	4.12.2.16	{unackedAlarm}	Ok [23-Nov-23 1:01 PM AEDT]	Not connected	Connected	false

2. If not already done, install the “exportTags” module on the JACE

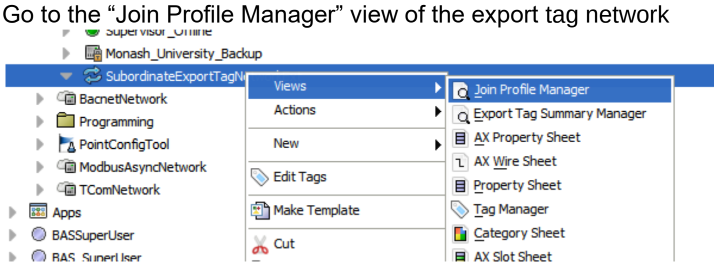


exportTags-wb	Tridium 4.10.1.36	Tridium 4.10.1.36	Up to Date
exportTags-rt	Tridium 4.10.1.36	Tridium 4.10.1.36	Up to Date
export-wb	Tridium 4.10.1.36	Tridium 4.10.1.36	Up to Date

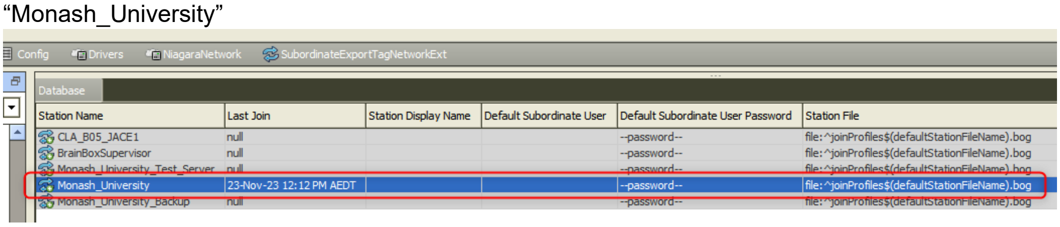
3. Open the palette in workbench and copy the “SubordinateExportTagNetworkExt” to the “NiagaraNetwork” of the JACE



4. Go to the “Join Profile Manager” view of the export tag network

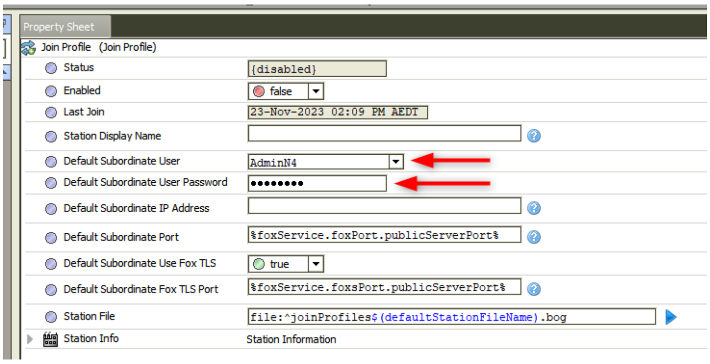


5. In the window that opens, double click on the server you want to sync with, in this case it would be “Monash_University”

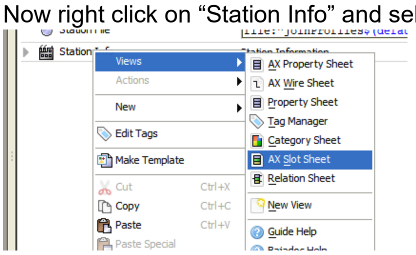


Noting: that the last join box will also say “null” the first time you open it

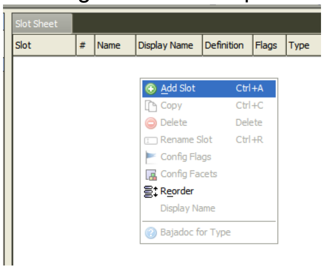
6. Now fill in the user and password details for the join profile, using the “AdminN4” credentials



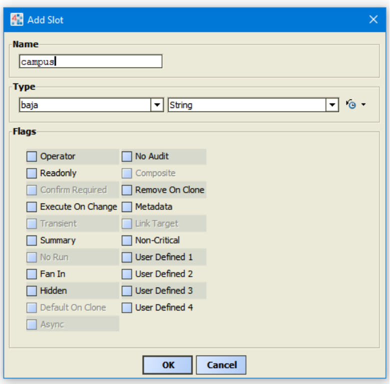
7. Now right click on "Station Info" and select the "AX Slot Sheet"



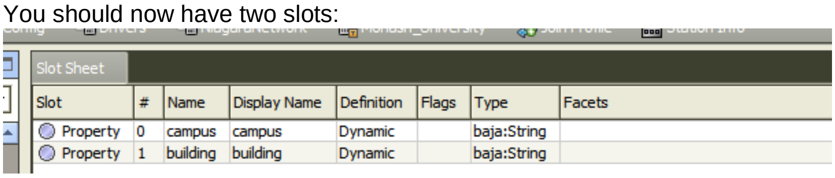
8. Then right click on the pane and select "Add Slot"



9. Add two slots of type baja:String, one named "campus" and the other named "building"

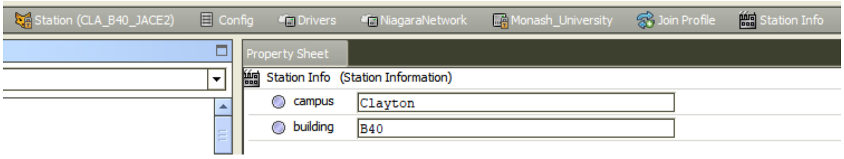


10. You should now have two slots:



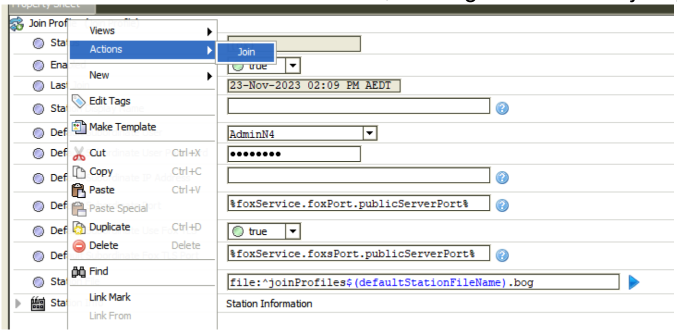
11. Now go to the "AX Property Sheet" of "Station Info" and fill in the relevant details

MONASH University

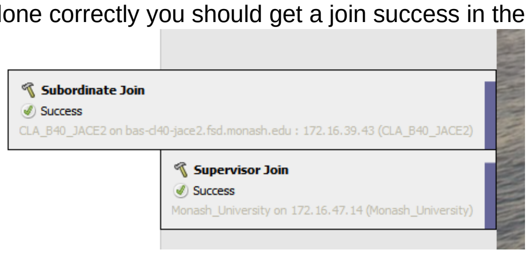


Note: the building tag will be the folder the JACE is added to on the supervisor, so follow the same naming convention, “B” followed by the building number.

12. You can now enable the Join Profile, then right click on the join profile and select the action “Join”



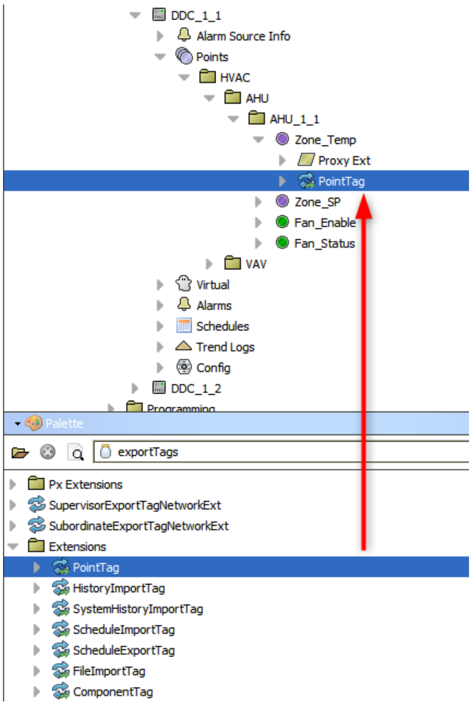
13. If all is done correctly you should get a join success in the bottom right corner of workbench



12.2.2 Adding point tags for export

To export points to the supervisor, a “PointTag” extension needs to be added to each point you want exported.

1. Copy the “PointTag” extension from the pallet to the first point you want to export

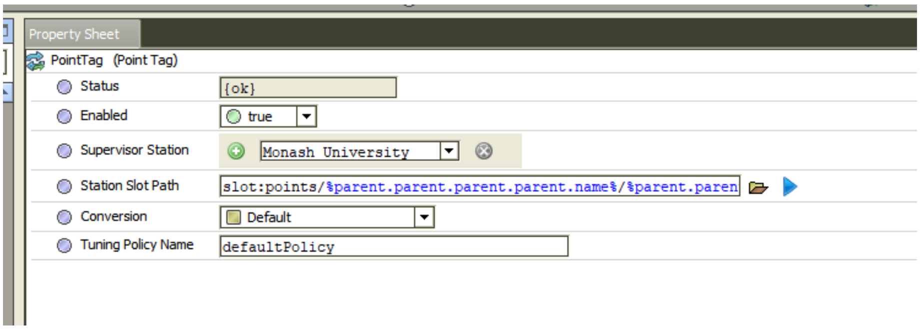


2. Now double click on the “PointTag” extension under the point to open it’s “AX Property Sheet” and fill in the details:
    - Supervisor Station: select “Monash_University”
    - Station Slot Path:
        i. For Central plant (no level folder) enter:

slot:points/%parent.parent.parent.parent.name%/%parent.parent.parent.name%/%parent.parent.name%

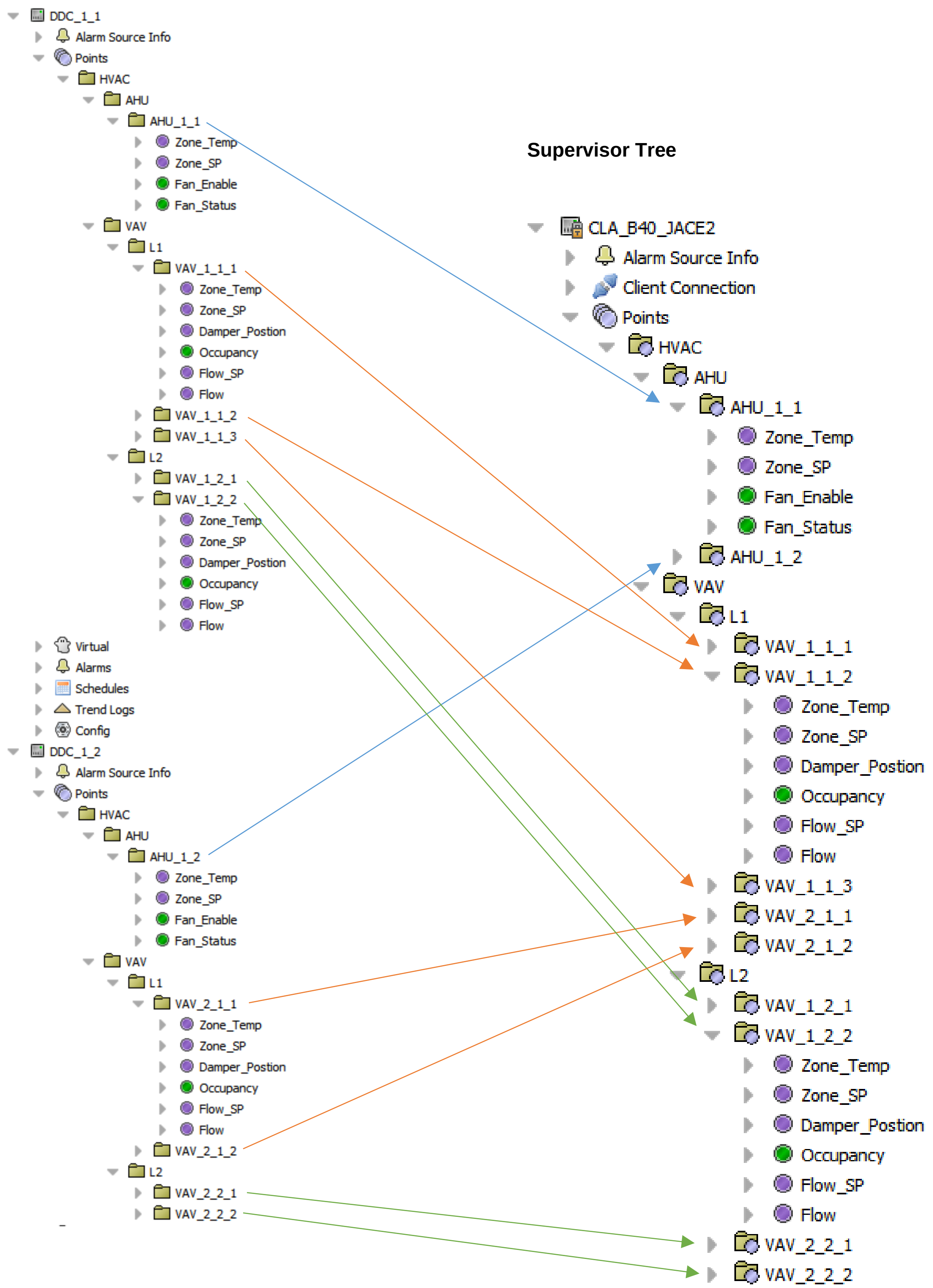
ii. For other assets (level folder) enter:

slot:points/%parent.parent.parent.parent.parent.name%/%parent.parent.parent.parent.name%/%parent.parent.parent.name%/%parent.parent.name%

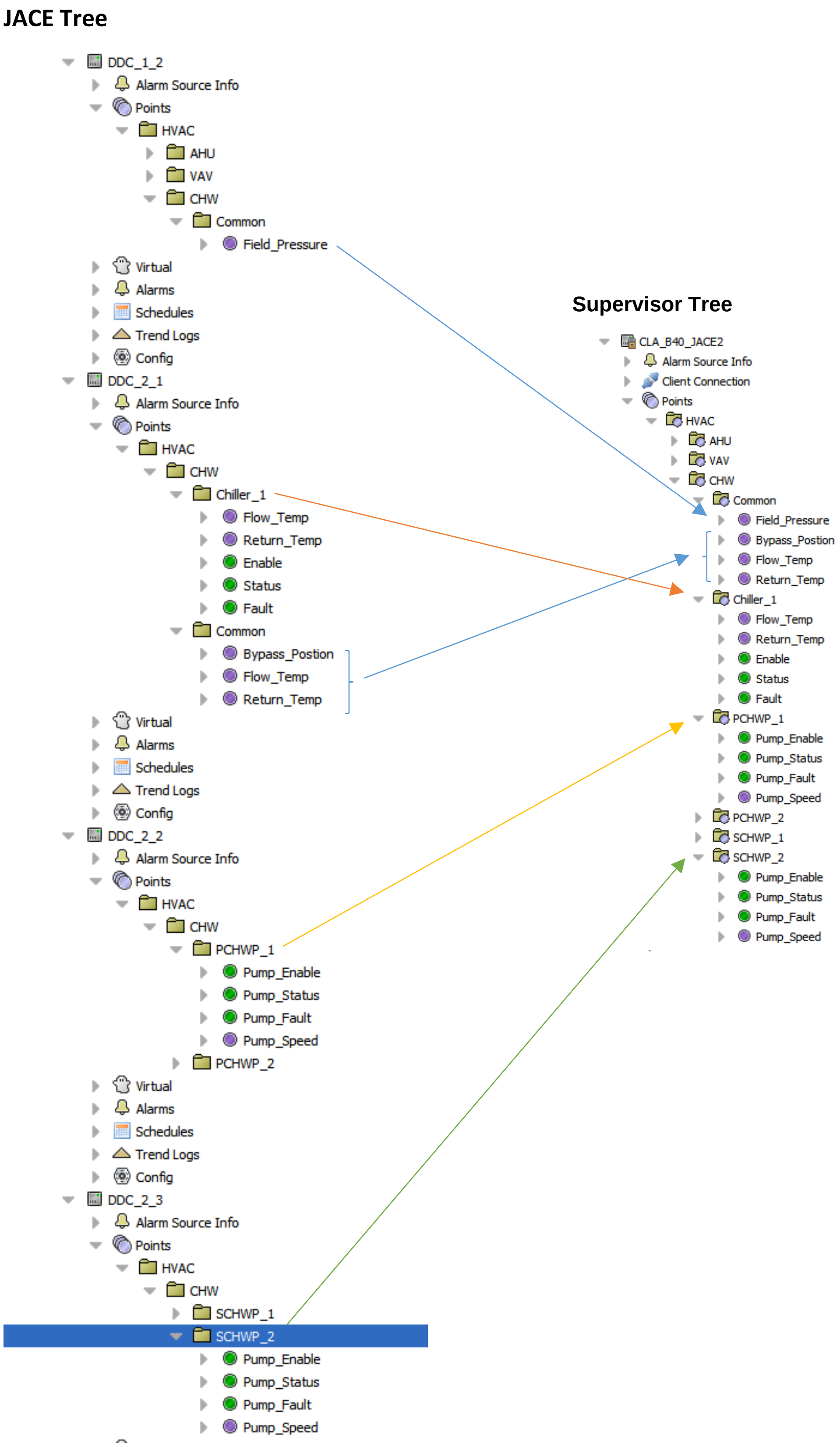


3. Once you have created a “PointTag” extension for both central plant (no level folder) and the other assets (level folder) you can copy and paste the extensions to all the relevant points in the JACE.
4. Once you are happy that all extensions are added, or you are at a point where you want to start working with points on the supervisor, then you can initiate the join action again. This should add all the points with extensions to the points folder of the JACE on the supervisor under their relevant folders.

JACE Tree



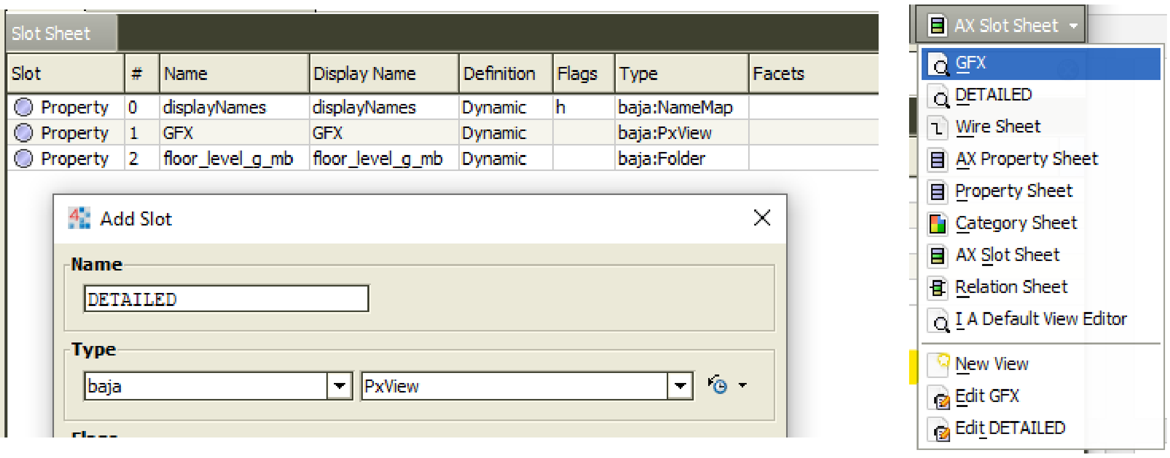
As you can see the folder structure is automatically created, and even if points/units etc. reside on different controllers on the JACE they will be consolidated in the JACE at the supervisor level. See the chilled water system example bellow.



As you can see even though the points are spread across four separate controllers, they are all consolidated at a supervisor level because a common folder structure is used.

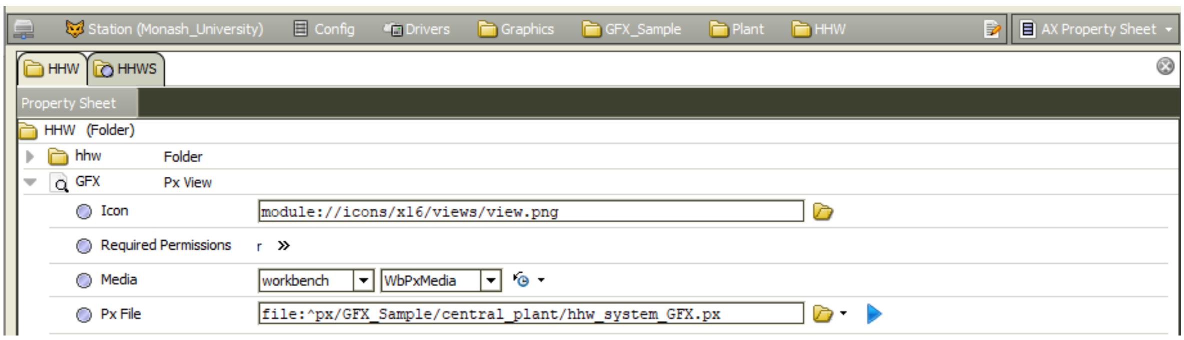
12.3 Component Space - Structure

A graphics folders is a baja:Folder the resides under the stations component space (slot:/) that has at least one view assigned to it. A graphics folder can be a container for points, programming logic, folders etc. Using Graphics Folders allows you to assign a Px File to a View without affecting existing hyperlink ords. A view is a slot that can be added to a folder using the view selector dropdown or by using the AX Slot Sheet view.



(Example showing the GFX and DETAILED views on Folder)

There are 4 views that can be assigned to a graphics folder although not all views are needed on every folder. The views are GFX, DETAILED, SETTINGS and NOTES. The GFX view is required to be present on all folders that are to be used in the graphics, it must also be the default view of all graphics folders. Each of the other views are to be added based on the content of the GFX view.



(Example of a configured view)

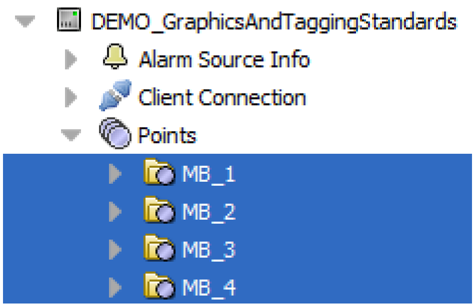
Some views may require additional points to be present on the parent folder to function correctly. These points usually store a Boolean value to toggle the state of a views widget or a String that contains additional information specific to the content being displayed. The table below contains a list of the standard points associated with views, although extra points may be added for additional functionality where necessary.

Name	Type	View	Note/Comments
equip_visibility	baja:boolean	GFX	Used on floorplans pages
duct_visibility	baja:boolean	GFX	Used on floorplans pages
note	baja:String	NOTES	Used on equipment pages, Facet: multiLine = true
serves	baja:String	GFX	Used on equipment/plant pages, is optional
location	baja:String	GFX	Used on equipment/plant pages, is optional

When a graphic is to display the points of a single asset, the PxView must be placed on the Parent folder of the points being displayed. This allows the existing points folder on the supervisor to be used instead of creating an individual graphics folder. In addition, Slot (relativised ord) and target-based referencing may be used. Tag based graphics are not to be used.

When the graphics are displaying a central system then the graphics should be placed on the system folder If the points for a unit are spread amongst multiple JACE's, pick the one with the most importance to the system, this is usually the one with the most points. For example a CHW system may have the chillers and pumps in the basement, connected to a basement JACE and cooling towers on the roof connected to a roof JACE, the most appropriate folder for the graphics would be the CHW system folder on basement JACE. Where there is minimal benefit to assigning a system's graphic page to a unit's points folder, a graphics folder may be created and used instead. For example, a page showing miscellaneous pumps in the building.

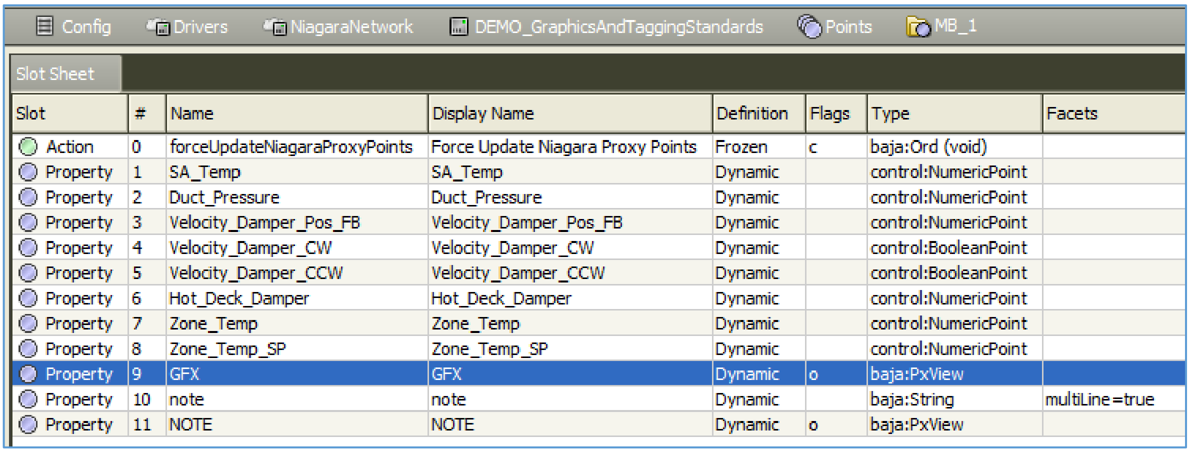
Graphic pages such as VAVs, Mixing Boxes or VSD HLIs pop-ups the PxView must be placed on the parent Points Folder, where a templated page should be used. This can be helpful for units such as VAVs or Mixing Boxes where there are many units on one floor displaying the same points. In instances where there are minor variations with units, a new page template can be made and assigned to that point folder. Degrade hide behaviour can also be implemented on the point and kitPxGraphics image. Templates may need to be made per unit type and level. For example, a template for level 1 and 2 may not be shared due to the content title bar's 'Floorplan' navigation button.



(Mixing Box points folder)

For VSD HLIs a 'GFX' view can be added to a VSD HLI points folder. As this page may be a pop-up, the page will not have a header or nav menu. The pop-up standard must be followed. See the *Pop-up Windows* section in this document.

If the graphic is not representative of a unit, such as a floorplan or a summary page, the PxView is to be placed logically in a folder under the slot:/Drivers/Graphics/<Building_Name> directory.

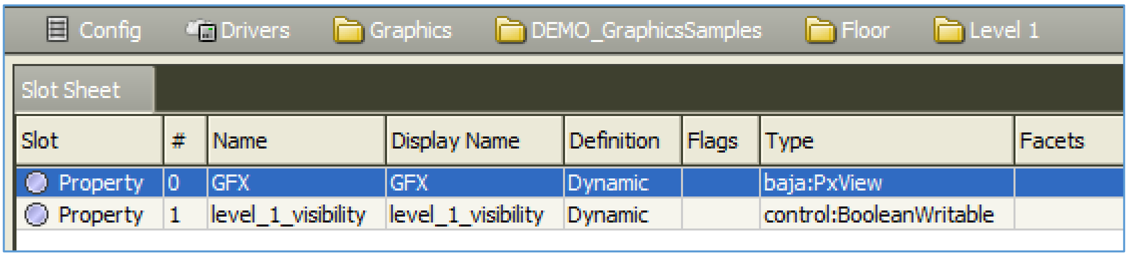


Config | Drivers | NiagaraNetwork | DEMO_GraphicsAndTaggingStandards | Points | MB_1

Slot Sheet

Slot	#	Name	Display Name	Definition	Flags	Type	Facets
Action	0	forceUpdateNiagaraProxyPoints	Force Update Niagara Proxy Points	Frozen	c	baja:Ord (void)	
Property	1	SA_Temp	SA_Temp	Dynamic		control:NumericPoint	
Property	2	Duct_Pressure	Duct_Pressure	Dynamic		control:NumericPoint	
Property	3	Velocity_Damper_Pos_FB	Velocity_Damper_Pos_FB	Dynamic		control:NumericPoint	
Property	4	Velocity_Damper_CW	Velocity_Damper_CW	Dynamic		control:BooleanPoint	
Property	5	Velocity_Damper_CCW	Velocity_Damper_CCW	Dynamic		control:BooleanPoint	
Property	6	Hot_Deck_Damper	Hot_Deck_Damper	Dynamic		control:NumericPoint	
Property	7	Zone_Temp	Zone_Temp	Dynamic		control:NumericPoint	
Property	8	Zone_Temp_SP	Zone_Temp_SP	Dynamic		control:NumericPoint	
Property	9	GFX	GFX	Dynamic	o	baja:PxView	
Property	10	note	note	Dynamic		baja:String	multiLine=true
Property	11	NOTE	NOTE	Dynamic	o	baja:PxView	

(Mixing box points folder slot sheet – GFX PxView is used for displaying graphics page)



Slot	#	Name	Display Name	Definition	Flags	Type	Facets
Property	0	GFX	GFX	Dynamic		baja:PxView	
Property	1	level_1_visibility	level_1_visibility	Dynamic		control:BooleanWritable	

(GFX PxView on a graphics folder for displaying a floor plan)

12.4 Component Space - Naming

Graphics Folder naming conventions consist of names and display names. You can view these using the AX Slot Sheet View.

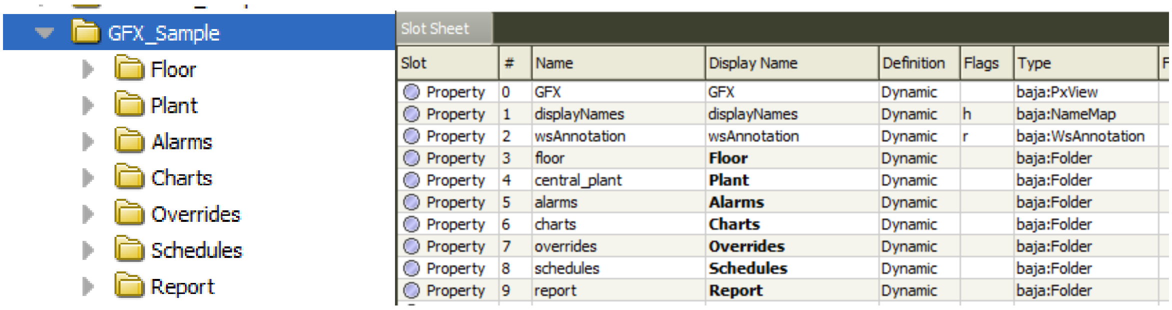
Name:

- All lower case
- Only underscores (no other special characters)

Display Name:

- Spaces allowed
- Special characters allowed

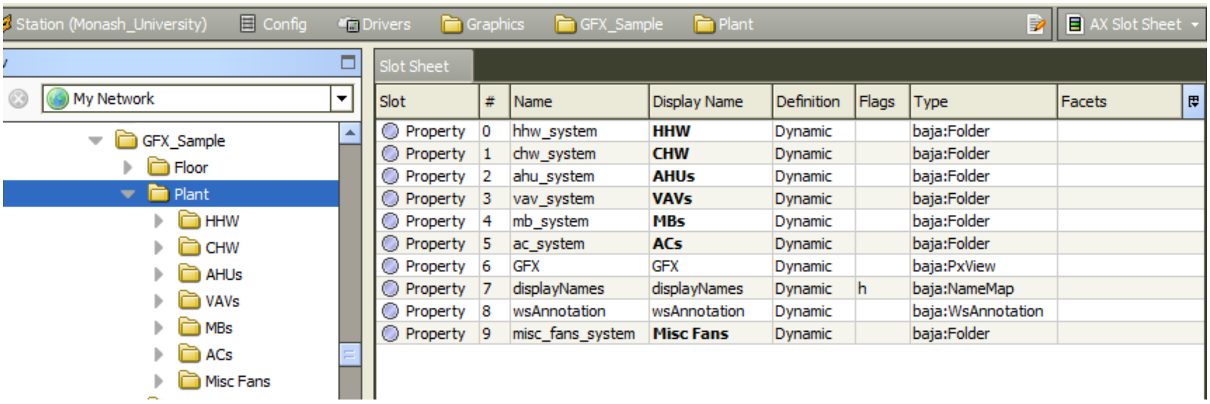
The benefit of having the name standardised are ords that do not convert special characters to ASCII code. The benefit of Display Names is being able to have a name that is easier to read, editable without changing folder structures, as well as the use of spaces.



Slot	#	Name	Display Name	Definition	Flags	Type
Property	0	GFX	GFX	Dynamic		baja:PxView
Property	1	displayNames	displayNames	Dynamic	h	baja:NameMap
Property	2	wsAnnotation	wsAnnotation	Dynamic	r	baja:WsAnnotation
Property	3	floor	**Floor**	Dynamic		baja:Folder
Property	4	central_plant	**Plant**	Dynamic		baja:Folder
Property	5	alarms	**Alarms**	Dynamic		baja:Folder
Property	6	charts	**Charts**	Dynamic		baja:Folder
Property	7	overrides	**Overrides**	Dynamic		baja:Folder
Property	8	schedules	**Schedules**	Dynamic		baja:Folder
Property	9	report	**Report**	Dynamic		baja:Folder

A basic building Graphic Folder (Root) in order.

Also shows the difference between Name and Display Name via the AX Slot Sheet View.



Slot	#	Name	Display Name	Definition	Flags	Type	Facets
Property	0	hhw_system	**HHW**	Dynamic		baja:Folder	
Property	1	chw_system	**CHW**	Dynamic		baja:Folder	
Property	2	ahu_system	**AHUs**	Dynamic		baja:Folder	
Property	3	vav_system	**VAVs**	Dynamic		baja:Folder	
Property	4	mb_system	**MBs**	Dynamic		baja:Folder	
Property	5	ac_system	**ACs**	Dynamic		baja:Folder	
Property	6	GFX	GFX	Dynamic		baja:PxView	
Property	7	displayNames	displayNames	Dynamic	h	baja:NameMap	
Property	8	wsAnnotation	wsAnnotation	Dynamic		baja:WsAnnotation	
Property	9	misc_fans_system	**Misc Fans**	Dynamic		baja:Folder	

A building's central plant Graphic Folder structure.

Also shows the difference between Name and Display Name via the AX Slot Sheet View.

In the graphics ord example, ‘GFX_Sample’ is where the Building Name would be.

station:|slot:/Drivers/Graphics/**GFX_Sample**/central_plant/hhw_system

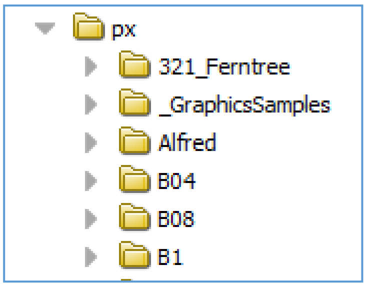
station:|slot:/Drivers/Graphics/***building name***/central_plant/hhw_system

station:|slot:/Drivers/Graphics/**B31/**central_plant/hhw_system

Graphic folder names / display names examples:

Name	Name
central_plant	Plant
misc_plant	Misc
hhw_system	HHW
chw_system	CHW
ahu_system	AHUs
vav_system	VAVs
mb_system	MBs
ac_system	ACs
misc_fans_system	Misc Fans
metering_system	Meters
power_metering_system	Power
water_metering_system	Water
gas_metering_system	Gas
solar_metering_system	Solar
lighting_system	Lighting
weather_station	Weather
level_b	Basement
level_g	Ground
level_1	Level 1

12.5 File Space - Structure



The graphics files for a building should be placed under the px directory (file:^px) in a folder that is named after the building they represent. This building folder will then be used as the root directory for all files related to buildings graphics to reside.

Under the buildings folder there should be a px file for each main menu landing page. For example: floor_GFX.px, shedules_GFX.px.

Each of the px files that have a submenu need to have a folder created that matches the base of the main menus px file name. So, the floorplan landing page “floor_GFX.px” needs to have a folder added to the building folder named “floor”.

Inside the buildings folder is an additional folder name includes. This folder contains the file that has the main menu px file named “navmenu.px”. Also contained in the include folder is another folder with the name “template”. The template folder contains all of the template files that will become a pxInclude. This can be equipment templates or page structure templates such at the page headers.

MONASH University

include

Name	Type	Size	Modified
template	Directory		27-Aug-21 4:09 PM AEST
navmenu.px	PxFile	8 KB	28-Aug-21 9:49 PM AEST
navmenu1.px	PxFile	8 KB	20-Aug-21 9:31 AM AEST

(Example of main menus include folder)

template

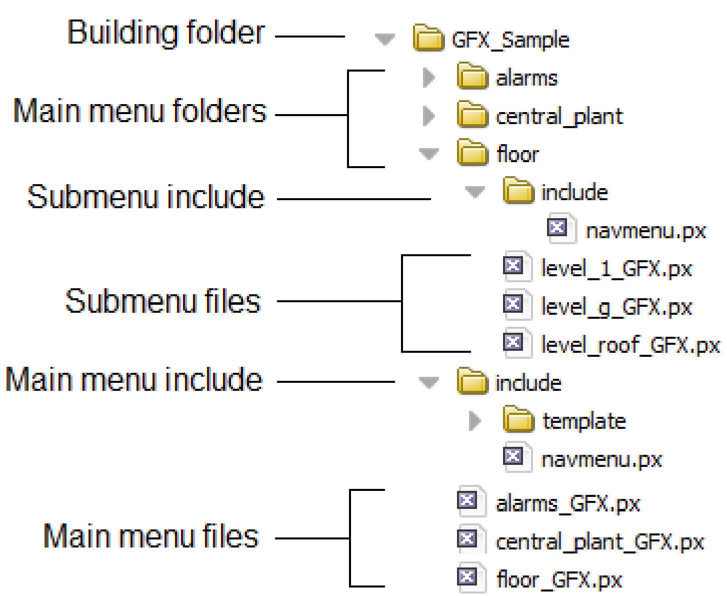
Name	Type	Size	Modified
INCLUDE_Alarm_OutOfRange_HighLimit.px	PxFile	2 KB	08-Feb-21 12:37 AM AEDT
ac.px	PxFile	4 KB	29-Jan-21 2:55 PM AEDT
ac.unit.px	PxFile	1 KB	20-Aug-21 9:31 AM AEST

(Example of main include/template folder)

In each of the folders representing a menu item, there is also an include folder, in this folder there is only a single px file called “navmenu.px”. While the navmenu px file in the under the buildings include directory contained the main menu. The navmenu file in this directory contains the submenu. This structure allows up to create submenus that change based on what main menu item is selected.

Also contained in the menu folder is the px files that represent each of the submenu pages in a similar manner to the main menu pages in the building folder.

This structure of px files and that represent the pages of the parent folder continues downward for each subdirectory with the exception menu items in the include folders.



12.6 File Space – Naming

The file space folders and files differ from the stations component space in that they only have a single name instead of a real name and an optional display name. The names allow for all characters that are supported by the operating system for use in file naming. However, the convention for all files inside the building folder are as follows:

- All file and folder names must be lowercase
- All file and folder names must not use special characters

- All files must use underscores as their separator character
- Px Files that are used in a Px View must have the view name append in uppercase (Example: floor_GFX.px)

These conventions make sure that all files are cross compatible with all operating systems as well as making them as simple and consistent as possible.

13 MULTI-BUILDING CONTROL AND MONITORING SYSTEMS

Where a system affects multiple buildings, a button must be placed in an appropriate location on the front-end graphics, for example the home or campus home menus. The following pages must be updated if new installations or upgrades work are within these systems.

- HTHW System (Control)
- External Lighting (Control)
- Microgrid (Control)
- Network Health (Monitoring)

(Future)

- Rain Water (Monitoring)
- Generator (Monitoring)
- Solar (Monitoring)
- Boilers (Monitoring)
- Chillers (Monitoring)

APPENDIX A – TEMPLATES

Parent Folder	Palette Folder	Name	Usage
GUI Template	Alarms	alarm_boolean	Alarm class chooser widgets
GUI Template	Alarms	alarm_outOfRange	Alarm class chooser widgets
GUI Template	Alarms	alarm_outOfRange_high	Alarm class chooser widgets
GUI Template	Display Points/Buttons	gridPane_title	Gridpane with title for schematics
GUI Template	Display Points/Buttons	gridPane_title_buttonLabel	The button for Gridpane title - eg. HLI hyperlink
GUI Template	Display Points/Buttons	gridPane_content_only	The inner GridPane to display points only - no title.
GUI Template	Display Points/Buttons	boundLabel_text	Text Only - to accompany point values
GUI Template	Display Points/Buttons	boundLabel_constrainedPane	To be used to define the width of the point column in a schematic GridPane
GUI Template	Display Points/Buttons	boundLabel_write_set	Writable points - To be used as per standard
GUI Template	Display Points/Buttons	boundLabel_write_override	Writable points - To be used as per standard
GUI Template	Display Points/Buttons	boundLabel_read	Read only points - To be used as per standard
GUI Template	Display Points/Buttons	boundLabel_status_green	Read only status points, where label animates green - To be used as per standard
GUI Template	Display Points/Buttons	button	Button with icon
GUI Template	Display Points/Buttons	button_actionBinding_icon	Button with action binding - eg. Manual Rotate
GUI Template	Display Points/Buttons	button_chart	Chart button - 20x20 image
GUI Template	Display Points/Buttons	checkBox	Check box toggle
GUI Template	Menu	navmenu	Use as a base for constructing building nav menu
GUI Template	Menu	submenu_icon	Sub menu button (Icon)
GUI Template	Menu	submenu_text	Sub menu button (Text only)
GUI Template	Menu	submenu_base	Sub menu highlight
GUI Template	Menu	submenu_bottom	Sub menu highlight
GUI Template	Menu	submenu_innerBottom	Sub menu highlight
GUI Template	Menu	submenu_innerTop	Sub menu highlight
GUI Template	Menu	submenu_mid	Sub menu highlight
GUI Template	Menu	submenu_noPage	Sub menu highlight
GUI Template	Menu	submenu_top	Sub menu highlight
GUI Template	Floor	label_button_gradient_L	Floorselector Label
GUI Template	Floor	label_button_gradient_R	Floorselector Label
GUI Template	Floor	label_sensor_L	Floor plan sensor label

Parent Folder	Palette Folder	Name	Usage
GUI Template	Floor	label_sensor_R	Floor plan sensor label
GUI Template	Floor	floor_icon_20x	Floor plan unit icon
GUI Template	Floor	floor_icon_32x	Floor plan unit icon
GUI Template	Floor	floor_icon_32_20x	Floor plan unit icon
GUI Template	Floor	floor_icon_24_16x	Floor plan unit icon
GUI Template	Floor	indicator	Floor plan line
GUI Template	Floor	jace	JACE location dot
GUI Template	Floor	zone	Path with spectrum bindings
GUI Template	Landing Page	icon_summary_schedule	Widget for schedule landing page
GUI Template	Landing Page	icon_summary_small	Widget for legacy icon summary view
GUI Template	Landing Page	icon_summary_large	Widget for landing page
GUI Template	Misc	NavContainerView	File directory container
Layout Template	Page Layout	page_base	Page template (inc nav menu, header, status effect key and main content)
Layout Template	Page Layout	page_navmenu_container	Standalone page nav menu template
Layout Template	Page Layout	popup_base	Page template for pop-up information/ settings window
Layout Template	Page Layout	content_main_blank	Blank main content
Layout Template	Page Layout	content_root	Main content container only
Layout Template	Page Layout	content_sub_blank	Blank sub content
Layout Template	Page Layout	content_icon_small	Legacy icon summary content
Layout Template	Page Layout	content_trends	Blank trend data section (inc chart)

APPENDIX B – TEXT AND COLOUR PALETTE

A.1 Text

Description	Font	Size	Colour
Main Content Title	Arial, Bold	28	White
Main Content Subtitle	Arial, Bold	14	White
Schematic GridPane Title	Arial, Bold	14	White
Default Text	Arial	12	White

A.2 Colour

Many of the graphics elements are templated and found in the muGraphicStandards folder on the Monash Supervisor. Below are some notable colours values:

Description	Value	Use
Main Background	#22242e	Page background colours applied to the root borderpane of a page.
Content Background	#333440	Colour used in content section
squareDark.svg	#ff303445	Image is used behind icons in the navigation menu and landing pages.
Content Gradient	linearGradient (angle(90.0) stop(0.0% #1a2d2e3b) stop(95.0% #333440)) 	
GridPane Line	(2.0 solid 'silver')	Used on schematic pages to indicate associated equipment.
Floorplan Indicator	(3.0 'grey')	Used to indicate locations and unit. Stroke weight to be determined as per what is appropriate for the floorplan image.
Zone Alarm Brush (Normal)	#8044c185	Used in the spectrum binding - mid colour.
Zone Alarm Brush (Out of Range)	#800055ff	Used in the spectrum binding – low colour.
Zone Alarm Brush (Out of Range)	#80ff0000	Used in the spectrum binding – high colour.
Zone Select Stroke	#ffbec8ff	Used for Floor selector and minimap
Zone Select Fill	#598c9eff	Used for Floor selector and minimap

APPENDIX C – POINT CONFIGURATION GUIDE

A.3 Point Action / Display Name Standard

All writable points must have action with suitable display names.

All overrideable boolean points should only have Active/Inactive/Auto exposed. For example, Manual On/Manual Off/Auto for fans and Manual Open/Manual Close/Auto for 2 position actuator.

No point to have emergency override exposed.

All analog outputs and calculated setpoints (SPs) should only have Override/Auto exposed.

Equipment that operates using internal controls such as chillers, boilers start commands facets must be "Enabled / Off" to give an indication that the system *could* operate but may not due to internal control conditions. Systems that can be directly operated….

Point Name	Category	Type	Facet	History Type	Override Facets	Override	Set	Description Format	Accepted Descriptions	Note
Alarm	Plant	Boolean Point	Normal/Alarm	BooleanCOV		False	False	[X]_Alarm	TBD	
Zone_Temp	Plant	Numeric Point	°C, Precision 1	NumericInterval		False	False			
Avg_Zone_Temp	Plant	Numeric Point	°C, Precision 1	NumericInterval		False	False			
Controlling_Zone_Temp	Plant	Numeric Point	°C, Precision 1	NumericInterval		False	False			
Zone_Temp_SP	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
Zone_Temp_Min_SP	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
Zone_Temp_Max_SP	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
Heating_SP	Plant	Numeric Writeable	°C, Precision 1	BooleanCOV		True	False			
Cooling_SP	Plant	Numeric Writeable	°C, Precision 1	BooleanCOV		True	False			
Zone_Pressure	Plant	Numeric Point	Pa, Precision 1	NumericInterval		False	False			
Zone_Pressure_SP	Plant	Numeric Writeable	Pa, Precision 1	NumericCOV		True	False			
Filter_DP_SP	Plant	Numeric Writeable	Pa, Precision 0	NumericCOV		True	False			
Filter_DP	Plant	Numeric Point	Pa, Precision 0	NumericInterval		False	False			
Filter_Status	Plant	Boolean Point	Normal/Dirty	BooleanCOV		False	False			
HEAP_Filter_DP_SP	Plant	Numeric Writeable	Pa, Precision 0	NumericCOV		True	False			
HEPA_Filter_DP	Plant	Numeric Point	Pa, Precision 0	NumericInterval		False	False			
HEPA_Filter_Status	Plant	Boolean Point	Normal/Dirty	BooleanCOV		False	False			
Fault	Plant	Boolean Point	Normal/Fault	BooleanCOV		False	False	[X]_Fault	TBD	
SA_Temp	Plant	Numeric Point	°C, Precision 1	NumericInterval		False	False			
SA_Temp_SP	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
SA_Temp_Min_SP	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
SA_Temp_Max_SP	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
SA_Pressure	Plant	Numeric Point	Pa, Precision 0	NumericInterval		False	False			
SA_Pressure_SP	Plant	Numeric Writeable	Pa, Precision 0	NumericCOV		True	False			
Fan_Enable	Plant	Boolean Writeable	On/Off	BooleanCOV	Manual On / Manual Off / Auto	True	False	[X]_Fan_Enable	SA,RA,OA,Spill_Air,EA,CT	
Fan_Status	Plant	Boolean Point	Running/Stopped	BooleanCOV		False	False	[X]_Fan_Status	SA,RA,OA,Spill_Air,EA,CT	
Fan_Speed	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False	[X]_Fan_Speed	SA,RA,OA,Spill_Air,EA,CT	
Fan_Alarm	Plant	Boolean Point	Normal/Alarm	BooleanCOV		False	False	[X]_Fan_Alarm	SA,RA,OA,Spill_Air,EA,CT	
HHWV	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			

Point Name	Category	Type	Facet	History Type	Override Facets	Override	Set	Description Format	Accepted Descriptions	Note
CHWV	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			
Enable	Plant	Boolean Writeable	Enabled/Off	BooleanCOV		True	False	[X]_Enable	Boiler, Chiller	
Status	Plant	Boolean Point	Running/Stopped	BooleanCOV		False	False	[X]_Status	Boiler, Chiller	
Fault	Plant	Boolean Point	Normal/Fault	BooleanCOV		False	False	[X]_Fault	Boiler, Chiller	
Flow_Temp	Plant	Numeric Point	°C, Precision 1	NumericInterval		False	False	[X]_Flow_Temp	CHW,CW,HHW,DHW,DCW, Boiler, Chiller, CT	
Return_Temp	Plant	Numeric Point	°C, Precision 1	NumericInterval		False	False	[X]_Return_Temp	CHW,CW,HHW,DHW,DCW, Boiler, Chiller, CT	
Pump_Enable	Plant	Boolean Writeable	On/Off	BooleanCOV		True	False	[X]_Pump_Enable	CHW,CW,HHW,DHW,DCW	
Pump_Status	Plant	Boolean Point	Running/Stopped	BooleanCOV		False	False	[X]_Pump_Status	CHW,CW,HHW,DHW,DCW	
Pump_Speed	Plant	Numeric Writeable	%, Precision 1	NumericInterval		True	False	[X]_Pump_Speed	CHW,CW,HHW,DHW,DCW	
Pump_Alarm	Plant	Boolean Point	Normal/Alarm	BooleanCOV		False	False	[X]_Pump_Alarm	CHW,CW,HHW,DHW,DCW	
DP	Plant	Numeric Point	Pa, Precision 0	NumericInterval		False	False			Lab pressures are to be set to Precision 1.
DP_SP	Plant	Numeric Writeable	Pa, Precision 0	NumericCOV		True	False			Lab pressures are to be set to Precision 1.
Blacklist	Plant	Boolean Writeable	Use/Ignore	BooleanCOV		True	True			
Air_Flow	Plant	Numeric Point	L/s, Precision 0	NumericInterval		False	False			
Requested_Flow	Plant	Numeric Writeable	L/s, Precision 0	NumericInterval		True	False			Precision 1 if setpoint is under 50L/s
Damper_Position	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			Precision 1 if setpoint is under 50L/s
Damper_Position	Plant	Boolean Writeable	Open/Closed	BooleanCOV	Manual Open / Manual Close / Auto	True	False			
Min_Flow_SP	Plant	Numeric Writeable	L/s, Precision 0	NumericCOV		True	False			Precision 1 if setpoint is under 50L/s
Max_Flow_SP	Plant	Numeric Writeable	L/s, Precision 0	NumericCOV		True	False			Precision 1 if setpoint is under 50L/s
Heating_Flow_SP	Plant	Numeric Writeable	L/s, Precision 0	NumericCOV		True	False			Precision 1 if setpoint is under 50L/s
K_Factor	Plant	Numeric Writeable	Precision 2	NumericCOV		True	True			Including any P or I control variables
Mode	Plant	Enum Writable	Program Dependent	EnumCOV		True	False			
Occupancy	Plant	Boolean Writeable	Occupied/Unoccupied	BooleanCOV		True	False			
Zone_Humidity	Plant	Numeric Point	%RH, Precision 1	NumericInterval		False	False			
Zone_Humidity_SP	Plant	Numeric Writeable	%RH, Precision 1	NumericCOV		True	False			
Humidifier_Status	Plant	Boolean Point	Running/Stopped	BooleanCOV		False	False			
Humidifier_Output	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			
Humidifier_Enable	Plant	Boolean Writeable	On/Off	BooleanCOV		True	False			
Time_Schedule	Plant	Boolean Writeable	On/Off	BooleanCOV		True	False			
Cooling_Call	Plant	Boolean Writeable	On/Off	BooleanCOV		True	False			
Heating_Call	Plant	Boolean Writeable	On/Off	BooleanCOV		True	False			
Cooling_Demand	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			
Heating_Demand	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			
OA_Temp	Misc	Numeric Point	°C, Precision 1	NumericInterval		False	False			
OA_Humidity	Misc	Numeric Point	%RH, Precision 1	NumericInterval		False	False			
GFA	Misc	Boolean Point	Normal/Alarm	BooleanCOV		False	False			

Point Name	Category	Type	Facet	History Type	Override Facets	Override	Set	Description Format	Accepted Descriptions	Note
Status	VSD/HLI	Boolean Point	Running/Stopped	BooleanCOV		False	False			
Fault	VSD/HLI	Boolean Point	Normal/Fault	BooleanCOV		False	False			
Alarm	VSD/HLI	Boolean Point	Normal/Alarm	BooleanCOV		False	False			
Speed	VSD/HLI	Numeric Point	%, Precision 0	NumericInterval		False	False			
Frequency	VSD/HLI	Numeric Point	Hz, Precision 1	NumericInterval		False	False			
Current	VSD/HLI	Numeric Point	A, Precision 1	NumericInterval		False	False			
Power	VSD/HLI	Numeric Point	kW, Precision 1	NumericInterval		False	False			
kWh	VSD/HLI	Numeric Point	kWh, Precision 1	NumericInterval		False	False			
Run_Hours	VSD/HLI	Numeric Point	Hrs, Precision 1	NumericInterval		False	False			
Night_Purge	Plant	Boolean Point	On/Off	BooleanCOV		False	False			
Night_Purge_Allow	Plant	Boolean Writeable	Enabled/Disabled	BooleanCOV		True	False			
Deadband	Plant	Numeric Writeable	°C, Precision 1	NumericCOV		True	False			
Bypass_Valve	Plant	Numeric Writeable	%, Precision 0	NumericInterval		True	False			
Lead_Pump	Plant	Enum Writable	Program Dependent	EnumCOV		True	False			
Lead_Boiler	Plant	Enum Writable	Program Dependent	EnumCOV		True	False			
Lead_Chiller	Plant	Enum Writable	Program Dependent	EnumCOV		True	False			
AH_Timer	Plant	Numeric Point	mins, Precision 1	NumericInterval		False	False			
AH_Timer_SP	Plant	Numeric Writeable	mins, Precision 1	NumericCOV		True	False			
AH_Call	Plant	Boolean Point	On/Off	BooleanCOV		False	False			
Opt_Start	Plant	Boolean Point	On/Off	BooleanCOV		False	False			
Opt_Start_Allow	Plant	Boolean Writeable	Enabled/Disabled	BooleanCOV		True	False			
CO2	Plant	Numeric Point	ppm, Precision 0	NumericInterval		False	False			
CO2_SP	Plant	Numeric Writeable	ppm, Precision 0	NumericCOV		True	False			
CO	Plant	Numeric Point	ppm, Precision 1	NumericInterval		False	False			
CO_SP	Plant	Numeric Writeable	ppm, Precision 1	NumericCOV		True	False			
AH_Push_Button	Plant	Boolean Point	On/Off	BooleanCOV		False	False			
Economy_Mode	Plant	Boolean Point	On/Off	BooleanCOV		False	False			
Economy_Mode_Allow	Plant	Boolean Point	Enabled/Disabled	BooleanCOV		False	False			
P_1_Voltage	Metering	Numeric Point	V, Precision 1	NumericInterval		False	False			
P_1_Current	Metering	Numeric Point	A, Precision 1	NumericInterval		False	False			
P_1_Active_Power	Metering	Numeric Point	kW, Precision 1	NumericInterval		False	False			
P_1_Reactive_Power	Metering	Numeric Point	kVar, Precision 1	NumericInterval		False	False			
P_1_Power_Factor	Metering	Numeric Point	pf, Precision 2	NumericInterval		False	False			
P_1_App_Power	Metering	Numeric Point	kVA, Precision 1	NumericInterval		False	False			
P_2_Voltage	Metering	Numeric Point	V, Precision 1	NumericInterval		False	False			
P_2_Current	Metering	Numeric Point	A, Precision 1	NumericInterval		False	False			
P_2_Active_Power	Metering	Numeric Point	kW, Precision 1	NumericInterval		False	False			
P_2_Reactive_Power	Metering	Numeric Point	kVar, Precision 1	NumericInterval		False	False			

Point Name	Category	Type	Facet	History Type	Override Facets	Override	Set	Description Format	Accepted Descriptions	Note
P_2_Power_Factor	Metering	Numeric Point	pf, Precision 2	NumericInterval		False	False			
P_2_App_Power	Metering	Numeric Point	kVA, Precision 1	NumericInterval		False	False			
P_3_Voltage	Metering	Numeric Point	V, Precision 1	NumericInterval		False	False			
P_3_Current	Metering	Numeric Point	A, Precision 1	NumericInterval		False	False			
P_3_Active_Power	Metering	Numeric Point	kW, Precision 1	NumericInterval		False	False			
P_3_Reactive_Power	Metering	Numeric Point	kVar, Precision 1	NumericInterval		False	False			
P_3_Power_Factor	Metering	Numeric Point	pf, Precision 2	NumericInterval		False	False			
P_3_App_Power	Metering	Numeric Point	kVA, Precision 1	NumericInterval		False	False			
Total_Active_Power	Metering	Numeric Point	kW, Precision 1	NumericInterval		False	False			
Total_Reactive_Power	Metering	Numeric Point	kVar, Precision 1	NumericInterval		False	False			
Total_App_Power	Metering	Numeric Point	kVA, Precision 1	NumericInterval		False	False			
Total_Power_Factor	Metering	Numeric Point	pf, Precision 2	NumericInterval		False	False			
Frequency	Metering	Numeric Point	Hz, Precision 1	NumericInterval		False	False			
P_1_2_Voltage	Metering	Numeric Point	V, Precision 1	NumericInterval		False	False			
P_2_3_Voltage	Metering	Numeric Point	V, Precision 1	NumericInterval		False	False			
P_3_1_Voltage	Metering	Numeric Point	V, Precision 1	NumericInterval		False	False			
P_1_Voltage_THD	Metering	Numeric Point	%, Precision 1	NumericInterval		False	False			
P_2_Voltage_THD	Metering	Numeric Point	%, Precision 1	NumericInterval		False	False			
P_3_Voltage_THD	Metering	Numeric Point	%, Precision 1	NumericInterval		False	False			
P_1_Current_THD	Metering	Numeric Point	%, Precision 1	NumericInterval		False	False			
P_2_Current_THD	Metering	Numeric Point	%, Precision 1	NumericInterval		False	False			
P_3_Current_THD	Metering	Numeric Point	%, Precision 1	NumericInterval		False	False			
Total_Active_Energy	Metering	Numeric Point	kWh, Precision 1	NumericInterval		False	False			
Total_Reactive_Energy	Metering	Numeric Point	kVarh, Precision 1	NumericInterval		True	False			

APPENDIX D – ABBREVIATIONS EXAMPLES

Acronym	Full Name	Description
AH	After_Hours	
B	Basement	
B1	Basement 1	
CD	Cold Deck	
CHW	Chilled Water	
CHWV	Chilled Water Valve	
CO	Carbon Monoxide	
CO2	Carbon Dioxide	
CT	Cooling Tower	
CW	Condenser Water	
DCW	Domestic Cold Water	
DHW	Domestic Hot Water	
DP	Differential Pressure	
EA	Exhaust Air	
G	Ground	
HD	Hot Deck	
HHW	Heating Hot Water	
HHWV	Heating Hot Water Valve	
L1	Level 1	
L2	Level 2	
LB	Lower Basement	
LG	Lower Ground	
M	Mezzanine	
O2	Oxygen	
OA	Outside Air	
P_1	Phase_1	Also used for Phase A
P_2	Phase_2	Also used for Phase B
P_3	Phase_3	Also used for Phase C
RA	Return Air	
SA	Supply Air	
SP	Setpoint	A user/technician controllable value, Setpoint should be used for programming controlled setpoints
THD	Total Harmonic Distortion	
AH	After_Hours	

APPENDIX E – ICONS

Usage	Icon	Floorplan Icon	Location	Category
AC	ac.svg		Icons/Units	Plant
ACU	acu.svg		Icons/Units	Plant
AHU	ahu.svg		Icons/Units	Plant
CCWP	ccwp.svg		Icons/Units	Plant
CHWP	chwp.svg		Icons/Units	Plant
CPEF	cpef.svg		Icons/Units	Misc Fans
CRAC	crac.svg		Icons/Units	Plant
EAF	eaf.svg		Icons/Units	Misc Fans
EF	ef.svg		Icons/Units	Misc Fans
Evaporative Cooling	evap.svg		Icons/Units	Plant
FC	fc.svg		Icons/Units	Plant
FCU	fcu.svg		Icons/Units	Plant
GEF	gef.svg		Icons/Units	Misc Fans
HHWP	hhwp.svg		Icons/Units	Plant
KEF	kef.svg		Icons/Units	Misc Fans
MB	mb.svg	mbSml.svg	Icons/Units	Plant
Misc Fans	misc.svg		Icons/Units	Misc Fans
OAF	oaf.svg		Icons/Units	Misc Fans
PAC	pac.svg		Icons/Units	Plant
Sump Pumps	pumpGrey.svg		Icons/Units	Misc Plant
RAF	raf.svg		Icons/Units	Misc Fans
Reheat VAVs	reheat.svg	reheatSml.svg	Icons/Units	Plant
SA Fan	saf.svg		Icons/Units	Plant
SPF	spf.svg		Icons/Units	Misc Fans
TEF	tef.svg		Icons/Units	Misc Fans
VAV	vav.svg	vavSml.svg	Icons/Units	Plant
Alarms	alarm.svg		Icons	Core
Alarms	alarm_x20.svg		Icons	Navigation
Alarms - High Priority	alarmFill.svg		Icons	Core
Alarms - High Priority	alarmFill_x20.svg		Icons	Navigation
Alarms - Critical	alarmStroke.svg		Icons	Core
Alarms - Critical	alarmStroke_x20.svg		Icons	Navigation
Alarms / Notifications	alertLook.svg		Icons	Misc
Alarms / Notifications	alertLook_x20.svg		Icons	Navigation
Analytics/ Analysis	analytics.svg		Icons	Misc
Navigation	arrowDown.svg		Icons	Misc
Navigation	arrowDown_x20.svg		Icons	Misc
Navigation with Status	arrowDownGreen.svg		Icons	Misc

Navigation with Status	arrowDownGreen_x20.svg	Icons	Misc
Navigation	arrowL.svg	Icons	Misc
Navigation	arrowL_x20.svg	Icons	Misc
Navigation with Status	arrowLGreen.svg	Icons	Misc
Navigation with Status	arrowLGreen_x20.svg	Icons	Misc
Navigation	arrowR.svg	Icons	Misc
Navigation	arrowR_x20.svg	Icons	Misc
Navigation with Status	arrowRGreen.svg	Icons	Misc
Navigation with Status	arrowRGreen_x20.svg	Icons	Misc
Navigation	arrowUp.svg	Icons	Misc
Navigation	arrowUp_x20.svg	Icons	Misc
Navigation with Status	arrowUpGreen.svg	Icons	Misc
Navigation with Status	arrowUpGreen_x20.svg	Icons	Misc
Navigate Previous	back.svg	Icons	Core
Navigate Previous	back_x64.svg	Icons	Core
Power Storage	battery.svg	Icons	Misc
Boilers/ HHWS	boiler.svg	Icons	Plant
Floorplan	building.svg	Icons	Core
Chiller / CHWS	chiller.svg	Icons	Plant
PLCs/ Controls	circuit.svg	Icons	Misc
Schedules	clockBlue.svg	Icons	Core
Schedules	clockBlue_x20.svg	Icons	Navigation
Schedules	clockBlueDark.svg	Icons	Core
Schedules	clockGreen.svg	Icons	Core
Schedules	clockSplit.svg	Icons	Misc
Settings / Configuration	cog.svg	Icons	Core
Settings / Configuration	cogGrey.svg	Icons	Misc
Settings / Configuration	cogGrey_x20.svg	Icons	Navigation
IT / Server / Device Status	computer.svg	Icons	Misc
Condenser Systems	condenser.svg	Icons	Plant
Cooling Systems	cool.svg	Icons	Misc
Cooling Tower	coolingTower.svg	Icons	Plant
Errors / Unknown	cross.svg	Icons	Misc
Observation	eyeGreen.svg	Icons	Misc
Observation	eyeGrey.svg	Icons	Misc
Misc Unit	fanBlue.svg	Icons	Misc Plant
Misc Unit with Heating Element	fanBlueRed.svg	Icons	Misc Plant
Fire Alarms	fire.svg	Icons	Misc / Fire
Navigate Forward	forward.svg	Icons	Core
Navigate Forward	forward_x64.svg	Icons	Core

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